FINALTERM EXAMINATION
Fall 2009
MTH301- Calculus II

Time: 120 min
Marks: 80

Question No: 1 (Marks: 1) - Please choose one

π is an example of -----------

► Irrational numbers

► Rational numbers

► Integers

► Natural numbers

Question No: 2 (Marks: 1) - Please choose one

Straight line is a special kind of ----------------

► Surface

► Curve

► Plane

► Parabola

Question No: 3 (Marks: 1) - Please choose one

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An ordered triple corresponds to ----------- in three dimensional space.

► A unique point

► A point in each octant

► Three points

► Infinite number of points

Question No: 4 (Marks: 1) - Please choose one

The angles which a line makes with positive x ,y and z-axis are known as ------------

► Direction cosines

► Direction ratios

► Direction angles

Question No: 5 (Marks: 1) - Please choose one

Is the function $f(x, y)$ continuous at origin? If not, why?

$f(x, y) = 4xy + \sin 3x^2 y$

► $f(x, y)$ is continuous at origin

► $f(0, 0)$ is not defined

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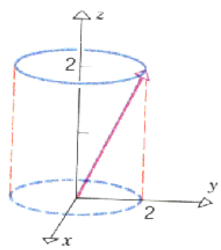
► $f(0,0)$ is defined but $\lim_{(x,y)\to(0,0)} f(x,y)$ does not exist

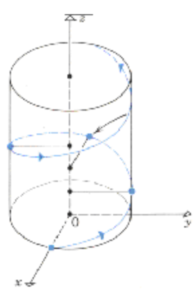
► $f(0,0)$ is defined and $\lim_{(x,y)\to(0,0)} f(x,y)$ exists but these two numbers are not equal.

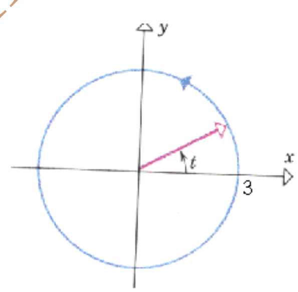
Question No: 6 (Marks: 1) - Please choose one

Match the following vector-valued function with its graph.

$r(t) = 3\cos t\,\hat{i} + 3\sin t\,\hat{j}$ and $0 \le t \le 2\pi$

► 

► 

► 

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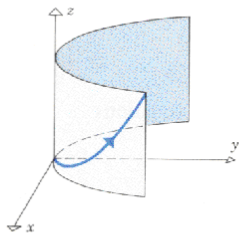
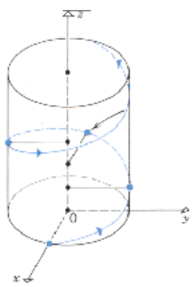
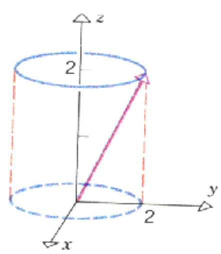
▶ 

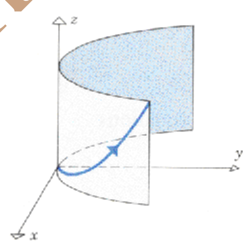
Question No: 7 (Marks: 1) - Please choose one

Match the following vector-valued function with its graph.

$r(t) = t\hat{i} + t^2\hat{j} + t^3\hat{k}$ and $t \geq 0$

▶ 

▶ 

▶ 

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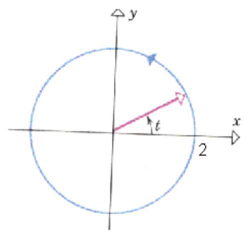
▶ 

Question No: 8 (Marks: 1) - Please choose one

What are the parametric equations that correspond to the following vector equation?

$\vec{r}(t) = \sin^2 t\,\hat{i} + (1-\cos 2t)\,\hat{j}$

▶ $x=\sin^2 t \quad , \quad y = 1-\cos 2t \quad , \quad z=0$

▶ $y=\sin^2 t \quad , \quad x = 1-\cos 2t \quad , \quad z=0$

▶ $x=\sin^2 t \quad , \quad y = 1-\cos 2t \quad , \quad z=1$

▶ $x=\sin^2 t \quad , \quad y = \cos 2t \quad , \quad z=1$

Question No: 9 (Marks: 1) - Please choose one

Is the following vector-valued function $\vec{r}(t)$ continuous at $t=0$? If not, why?

$\vec{r}(t) = (4\cos t, \sqrt{t}\ , 4\sin t)$

▶ $\vec{r}(0)$ is not defined

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▶ $\vec{r}(0)$ is defined but $\lim\limits_{t\to 0}\vec{r}(t)$ does not exist

▶ $\vec{r}(0)$ is defined and $\lim\limits_{t\to 0}\vec{r}(t)$ exists but these two numbers are not equal.

▶ $\vec{r}(t)$ is continuous at $t=0$

Question No: 10 (Marks: 1) - Please choose one

What is the derivative of following vector-valued function?

$\vec{r}(t) = (\cos 5t,\ \tan t\ ,\ 6\sin t)$

▶ $\vec{r}'(t) = \left(\dfrac{\sin 5t}{5},\ \sec t\ ,\ 6\cos t\right)$

▶ $\vec{r}'(t) = (\dfrac{-\sin 5t}{5},\ \sec t\ ,\ 6\cos t)$

▶ $\vec{r}'(t) = (-5\sin 5t,\ \sec^2 t\ ,\ 6\cos t)$

▶ $\vec{r}'(t) = (\sin 5t,\ \sec^2 t\ ,\ -6\cos t)$

Question No: 11 (Marks: 1) - Please choose one

The following differential is exact

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$dz = (3x^2 y + 2)\ dx + (x^3 + y)\ dy$

► True

► False

Question No: 12 (Marks: 1) - Please choose one

The following differential is exact
$dz = (3x^2 + 4xy)\ dx + (2x^2 + 2y)\ dy$

► True

► False

Question No: 13 (Marks: 1) - Please choose one

Which one of the following is correct Wallis Sine formula when n is odd and $n \geq 3$?

► $\int_0^{\frac{\pi}{2}} \sin^n x\ dx = \frac{\pi}{2} \frac{(n-1)}{n} \frac{(n-3)}{(n-2)} \frac{(n-5)}{(n-4)} ------- \frac{5}{6} \frac{3}{4} \frac{1}{2}$

► $\int_0^{\frac{\pi}{2}} \sin^n x\ dx = \frac{\pi}{2} \frac{(n)}{(n-1)} \frac{(n-2)}{(n-3)} \frac{(n-4)}{(n-5)} ------- \frac{6}{5} \frac{4}{3} \frac{2}{1}$

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$$\int_0^{\frac{\pi}{2}} \sin^n x \, dx = \frac{(n-1)}{n}\frac{(n-3)}{(n-2)}\frac{(n-5)}{(n-4)}-------\frac{6}{7}\frac{4}{5}\frac{2}{3}$$

▶

$$\int_0^{\frac{\pi}{2}} \sin^n x \, dx = \frac{(n)}{(n-1)}\frac{(n-2)}{(n-3)}\frac{(n-4)}{(n-5)}-------\frac{6}{5}\frac{4}{3}\frac{2}{1}$$

▶

Question No: 14 (Marks: 1) - Please choose one

Which of the following is correct?

$$\int_0^{\frac{\pi}{2}} \sin^4 x \, dx = \frac{3}{16}$$

▶

$$\int_0^{\frac{\pi}{2}} \sin^4 x \, dx = \frac{3\pi}{16}$$

▶

$$\int_0^{\frac{\pi}{2}} \sin^4 x \, dx = \frac{3}{8}$$

▶

$$\int_0^{\frac{\pi}{2}} \sin^4 x \, dx = \frac{2\pi}{3}$$

▶

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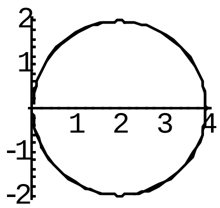
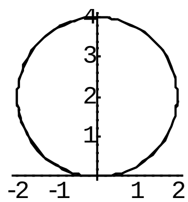
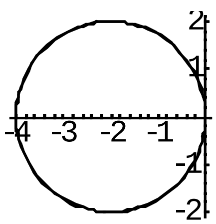
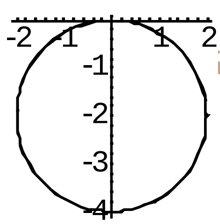
Question No: 15 (Marks: 1) - Please choose one

Match the following equation in polar co-ordinates with its graph.
$r = 4\sin\theta$

► 

► 

► 

► 

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Question No: 16 (Marks: 1) - Please choose one

If the equation of a curve, in polar co-ordinates, remains unchanged after replacing (r, θ) by $(r, \pi - \theta)$ then the curve is said to be symmetric about which of the following?

► Initial line

► y-axis

► Pole

Question No: 17 (Marks: 1) - Please choose one

What is the period of a periodic function defined by $f(x)=\sin\frac{x}{2}$?

► $\frac{\pi}{2}$

► π

► $\frac{3\pi}{2}$

► 4π

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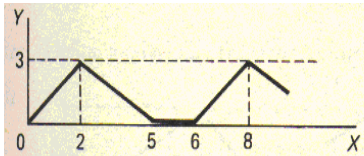
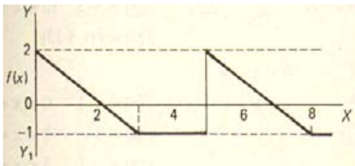
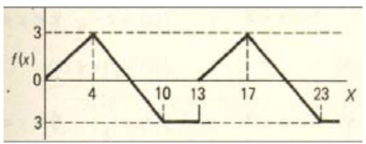
Question No: 18 (Marks: 1) - Please choose one

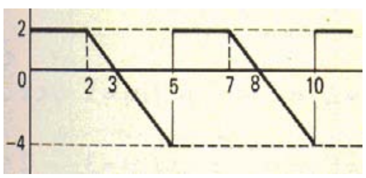
Match the following periodic function with its graph.

$$f(x)=\begin{cases}\frac{3}{4}x & 0<x<4\\ 7-x & 4<x<10\\ -3 & 10<x<13\end{cases}$$

► 

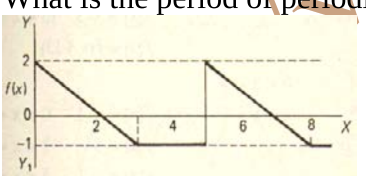
► 

► 

► 

Question No: 19 (Marks: 1) - Please choose one

What is the period of periodic function whose graph is as below?



► 2

► 3

► 4

► 5

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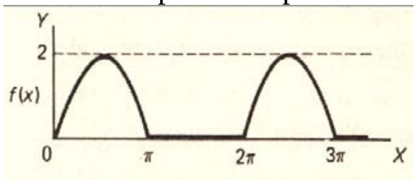
Question No: 20 (Marks: 1) - Please choose one

What is the period of periodic function whose graph is as below?



- ► 0
- ► 2
- ► π
- ► 2π

Question No: 21 (Marks: 1) - Please choose one

Polar co-ordinates of a point are $\left(-2, \frac{-3\pi}{2}\right)$. Which of the following is another possible polar co-ordinates representation of this point?

- ► $\left(2, \frac{-\pi}{4}\right)$
- ► $\left(2, \frac{-\pi}{2}\right)$
- ► $\left(2, \frac{-\pi}{3}\right)$

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$\left(2, \frac{3\pi}{4}\right)$

►

Question No: 22 (Marks: 1) - Please choose one

The function $f(x) = x^3 e^x$ is -------------

► Even function

► Odd function

► Neither even nor odd

Question No: 23 (Marks: 1) - Please choose one

The graph of an even function is symmetrical about ---------------

► x-axis

► y-axis

► origin

Question No: 24 (Marks: 1) - Please choose one

At which point the vertex of parabola, represented by the equation $y = x^2 - 4x + 3$, occurs?

► (0, 3)

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- ► $(2, -1)$
- ► $(-2, 15)$
- ► $(1, 0)$

Question No: 25 (Marks: 1) - Please choose one

The equation $y = x^2 - 4x + 2$ represents a parabola. Find a point at which the vertex of given parabola occurs?

- ► $(2, -2)$
- ► $(-4, 34)$
- ► $(0, 0)$
- ► $(-2, 14)$

Question No: 26 (Marks: 1) - Please choose one

Is the function $f(x, y)$ continuous at origin? If not, why?

$f(x, y) = \frac{xy}{x^2 + y^2}$

- ► $f(x, y)$ is continuous at origin
- ► $\lim_{(x,y)\to(0,0)} f(x, y)$ does not exist

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► $f(0, 0)$ is defined and $\lim_{(x,y)\to(0,0)} f(x, y)$ exists but these two numbers are not equal.

Question No: 27 (Marks: 1) - Please choose one

Sign of line integral is reversed when -----------

► path of integration is divided into parts.

► path of integration is parallel to y-axis.

► direction of path of integration is reversed.

► path of integration is parallel to x-axis.

Question No: 28 (Marks: 1) - Please choose one

What is Laplace transform of a function F(t)?

(s is a constant)

► $\int_0^s e^{-st} F(t)\, dt$

► $\int_0^\infty e^{st} F(t)\, dt$

► $\int_{-\infty}^\infty e^{-st} F(t)\, dt$

► $\int_0^\infty e^{-st} F(t)\, dt$

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Question No: 29 (Marks: 1) - Please choose one

What is the value of $L\{e^{5t}\}$ if L denotes laplace transform?

- ▶ $L\{e^{5t}\} = \dfrac{1}{s-5}$
- ▶ $L\{e^{5t}\} = \dfrac{s}{s^2+25}$
- ▶ $L\{e^{5t}\} = \dfrac{5}{s^2+25}$
- ▶ $L\{e^{5t}\} = \dfrac{5!}{s^6}$

Question No: 30 (Marks: 1) - Please choose one

What is the Laplace Inverse Transform of $\dfrac{1}{s+1}$

- ▶ $L^{-1}\left\{\dfrac{1}{s+1}\right\} = t+1$
- ▶ $L^{-1}\left\{\dfrac{1}{s+1}\right\} = e^{-t}+e^{t}$
- ▶ $L^{-1}\left\{\dfrac{1}{s+1}\right\} = e^{t}$

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▶ $L^{-1}\left\{\frac{1}{s+1}\right\}=e^{-t}$

Question No: 31 (Marks: 1) - Please choose one

What is Laplace Inverse Transform of $\frac{5}{s^2+25}$

▶ $L^{-1}\left\{\frac{5}{s^2+25}\right\}=\sin 5t$

▶ $L^{-1}\left\{\frac{5}{s^2+25}\right\}=\cos 5t$

▶ $L^{-1}\left\{\frac{5}{s^2+25}\right\}=\sin 25t$

▶ $L^{-1}\left\{\frac{5}{s^2+25}\right\}=\cos 25t$

Question No: 32 (Marks: 1) - Please choose one

What is $L\{-6\}$ if L denotes Laplace Transform?

▶ $L\{-6\}=\frac{1}{s+6}$

▶ $L\{-6\}=\frac{-6}{s}$

▶ $L\{-6\}=\frac{s}{s^2+36}$

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► $L\{-6\} = \frac{-6}{s^2+36}$

Question No: 33 (Marks: 1) - Please choose one

Evaluate the line integral $\int_C (3x+2y)\,dx + (2x-y)\,dy$ where C is the line segment from (0, 0) to (2, 0).

► 6

► -6

► 0

► Do not exist

Question No: 34 (Marks: 1) - Please choose one

Evaluate the line integral $\int_C (2x+y)\,dx + (x^2-y)\,dy$ where C is the line segment from (0, 0) to (0, 2).

► -4

► -2

► 0

► 2

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Question No: 35 (Marks: 1) - Please choose one

Plane is an example of ---------------------

► Curve

► Surface

► Sphere

► Cone

Question No: 36 (Marks: 1) - Please choose one

$If\ R=\{(x,y)/0\le x\le 2\, and\, -1\le y\le 1\}, then$
$\iint_R (x+2y^2)dA=$

$\int_{-1}^{1}\int_{0}^{2}(x+2y^2)dydx$

►

$\int_{0}^{2}\int_{1}^{-1}(x+2y^2)dxdy$

►

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► $\int_{-1}^{1}\int_{0}^{2}(x+2y^2)dxdy$

► $\int_{1}^{2}\int_{-1}^{0}(x+2y^2)dxdy$

Question No: 37 (Marks: 1) - Please choose one

To evaluate the line integral, the integrand is expressed in terms of x, y, z with

► $dr = dx\hat{i} + dy\,\hat{j}$

► $dr = dx\hat{i} + dy\,\hat{j} + dy\,\hat{k}$

► $dr = dx + dy + dz$

► $dr = dx + dy$

Question No: 38 (Marks: 1) - Please choose one

Match the following equation in polar co-ordinates with its graph.
$r = a$
where a is an arbitrary constant.

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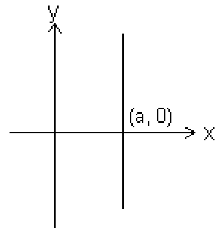
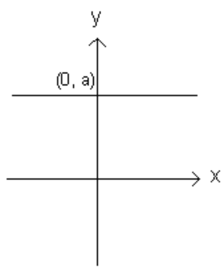
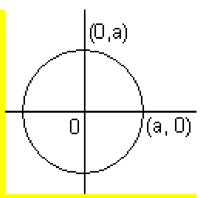
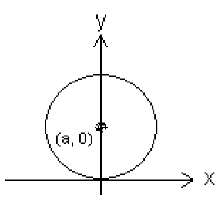
► 

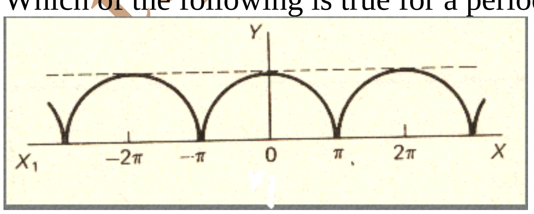
► 

► 

► 

Question No: 39 (Marks: 1) - Please choose one

Which of the following is true for a periodic function whose graph is as below?



► Even function

► Odd function

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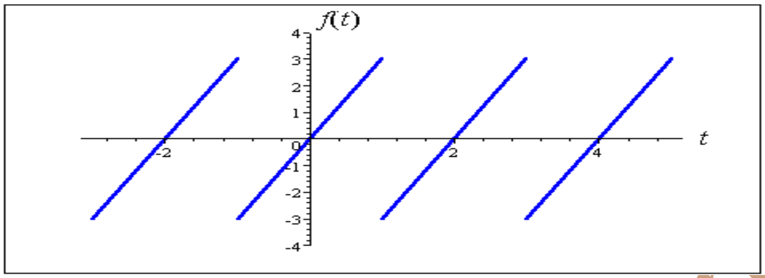
► Neither even nor odd function

Question No: 40 (Marks: 1) - Please choose one



The graph of "saw tooth wave" given above is -------------

► An odd function

► An even function

► Neither even nor odd

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Question No: 1 (Marks: 2) - Please choose one

Laplace transform of 't ' is

► $\frac{1}{s}$

- $\frac{1}{s^2}$
- e^{-s}
- s

Question No: 2 (Marks: 2) - Please choose one

Symmetric equation for the line through (1,3,5) and (2,-2,3) is

- $x-2=-\frac{y+2}{3}=-\frac{z-3}{5}$
- $x+2=-\frac{y+3}{5}=-\frac{z+5}{2}$
- $x-1=-\frac{y-3}{5}=-\frac{z-5}{2}$
- $x+1=\frac{y+3}{5}=\frac{z-5}{5}$

Question No: 3 (Marks: 1) - Please choose one

The level curves of f(x, y) = y Cscx are parabolas.

- True.
- False.

Question No: 4 (Marks: 1) - Please choose one

The equation $z=r$ is written in

- Rectangular coordinates
- Cylindrical coordinates

- Spherical coordinates
- None of the above

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Final term mcq calculus (Mth301) Solved By Rabia Rauf

FINALTERM EXAMINATION

MTH301- Calculus II

Question No: 1 (Marks: 1) - Please choose one

Q 1. Intersection of two straight lines is --------------

▶Surface

▶Curve

a. **▶Plane pg 13**

• ▶Point

Question No: 2 (Marks: 1) - Please choose one

Plane is a --------------- surface.

▶One-dimensional

▶Two-dimensional

▶Three-dimensional

▶Dimensionless

Question No: 3 (Marks: 1) - Please choose one

Let w = f(x, y, z) and x = g(r, s), y = h(r, s), z = t(r, s) then by chain rule

$$\frac{\partial w}{\partial r} =$$

- ► $\frac{\partial w}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial r} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial r}$ pg 50
- ► $\frac{\partial w}{\partial r}\frac{\partial x}{\partial r} + \frac{\partial w}{\partial r}\frac{\partial y}{\partial r} + \frac{\partial w}{\partial r}\frac{\partial z}{\partial r}$
- ► $\frac{\partial w}{\partial x}\frac{\partial x}{\partial r}\frac{\partial x}{\partial s} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial r}\frac{\partial y}{\partial s} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial r}\frac{\partial z}{\partial s}$
- ► $\frac{\partial w}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial w}{\partial r}\frac{\partial r}{\partial y} + \frac{\partial w}{\partial r}\frac{\partial r}{\partial z}$

Question No: 4 (Marks: 1) - Please choose one

What are the parametric equations that correspond to the following vector equation?

$$\vec{r}(t) = \sin^2 t\,\hat{i} + (1-\cos 2t)\,\hat{j}$$

- ► $x = \sin^2 t \quad , \quad y = 1-\cos 2t \quad , \quad z = 0$ pg 140
- ► $y = \sin^2 t \quad , \quad x = 1-\cos 2t \quad , \quad z = 0$
- ► $x = \sin^2 t \quad , \quad y = 1-\cos 2t \quad , \quad z = 1$
- ► $x = \sin^2 t \quad , \quad y = \cos 2t \quad , \quad z = 1$

Question No: 5 (Marks: 1) - Please choose one

What are the parametric equations that correspond to the following vector equation?

$$r(t) = (2t-1)\hat{i} - 3\sqrt{t}\,\hat{j} + \sin 3t\,\hat{k}$$

- ► $z = 2t-1 \quad , \quad x = -3\sqrt{t} \quad , \quad y = \sin 3t$
- ► $y = 2t-1 \quad , \quad x = -3\sqrt{t} \quad , \quad z = \sin 3t$
- ► $x = 2t-1 \quad , \quad z = -3\sqrt{t} \quad , \quad y = \sin 3t$
- ► $x = 2t-1 \quad , \quad y = -3\sqrt{t} \quad , \quad z = \sin 3t$ pg 140

Question No: 6 (Marks: 1) - Please choose one

What is the derivative of following vector-valued function?

$$\vec{r}(t) = (\cos 5t,\ \tan t\ ,\ 6\sin t)$$

- ► $\vec{r}'(t) = \left(\dfrac{\sin 5t}{5},\ \sec t\ ,\ 6\cos t\right)$
- ► $\vec{r}'(t) = (\dfrac{-\sin 5t}{5},\ \sec t\ ,\ 6\cos t)$
- ► $\vec{r}'(t) = (-5\sin 5t,\ \sec^2 t\ ,\ 6\cos t)$ According to formula at pg 145
- ► $\vec{r}'(t) = (\sin 5t,\ \sec^2 t\ ,\ -6\cos t)$

Question No: 7 (Marks: 1) - Please choose one

What is the derivative of following vector-valued function?

$$\vec{r}(t) = \left(t^4, \sqrt{t+1}, \frac{3}{t^2} \right)$$

- ► $\vec{r}'(t) = \left(4t^3, \frac{1}{\sqrt{t+1}}, \frac{-6}{t^3} \right)$
- ► $\vec{r}'(t) = \left(4t^3, \frac{1}{2\sqrt{t+1}}, \frac{6}{t^3} \right)$
- ► $\vec{r}'(t) = \left(4t^4, \frac{1}{2\sqrt{t+1}}, \frac{-6}{t^3} \right)$
- ► $\vec{r}'(t) = \left(4t^3, \frac{1}{2\sqrt{t+1}}, \frac{-6}{t^3} \right)$ Correct answer according to derivative property

Question No: 8 (Marks: 1) - Please choose one

The following differential is exact

$$dz = (x^2 y + y)\, dx - x\, dy$$

- ►True
- ►**False** **formula at pg 155**

Question No: 9 (Marks: 1) - Please choose one

Which one of the following is correct Wallis Sine formula when n is even and $n \geq 2$?

► $$\int_0^{\frac{\pi}{2}} \sin^n x\, dx = \frac{\pi}{2}\frac{(n-1)}{n}\frac{(n-3)}{(n-2)}\frac{(n-5)}{(n-4)}--------\frac{5}{6}\frac{3}{4}\frac{1}{2}$$ pg 182

► $$\int_0^{\frac{\pi}{2}} \sin^n x\, dx = \frac{(n-1)}{n}\frac{(n-3)}{(n-2)}\frac{(n-5)}{(n-4)}--------\frac{6}{7}\frac{4}{5}\frac{2}{3}$$

► $$\int_0^{\frac{\pi}{2}} \sin^n x\, dx = \frac{\pi}{2}\frac{(n)}{(n-1)}\frac{(n-2)}{(n-3)}\frac{(n-4)}{(n-5)}--------\frac{6}{5}\frac{4}{3}\frac{2}{1}$$

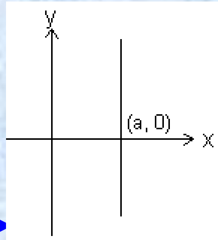
► $$\int_0^{\frac{\pi}{2}} \sin^n x\, dx = \frac{(n)}{(n-1)}\frac{(n-2)}{(n-3)}\frac{(n-4)}{(n-5)}--------\frac{6}{5}\frac{4}{3}\frac{2}{1}$$

Question No: 10 (Marks: 1) - Please choose one

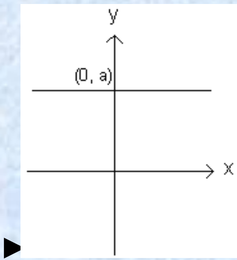
Match the following equation in polar co-ordinates with its graph.

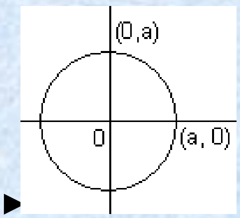
$r\ \cos\theta = a$

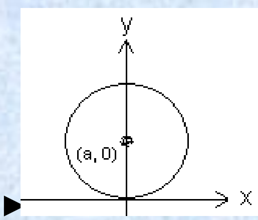
$where\ a\ is\ an\ arbitrary\ cons\tan t$

► 

pg 126

► 

► 

► 

Question No: 11 (Marks: 1) - Please choose one

If the equation of a curve, in polar co-ordinates, remains unchanged after replacing (r, θ) by $(r, \pi - \theta)$ then the curve is said to be symmetric about which of the following?

►Initial line

►y-axis pg 128

►Pole

Question No: 12 (Marks: 1) - Please choose one

If the equation of a curve, in polar co-ordinates, remains unchanged after replacing (r, θ) by $(-r, \theta)$ then the curve is said to be symmetric about which of the following?

▶Initial line

▶y-axis

▶Pole pg 128

Question No: 13 (Marks: 1) - Please choose one

What is the amplitude of a periodic function defined by $f(x) = \sin\frac{x}{3}$?

▶0

▶1 pg 197

▶ $\frac{1}{3}$

▶Does not exist

Question No: 14 (Marks: 1) - Please choose one

What is the period of a periodic function defined by $f(x) = 4\cos 3x$?

► $\frac{\pi}{4}$

► $\frac{\pi}{3}$

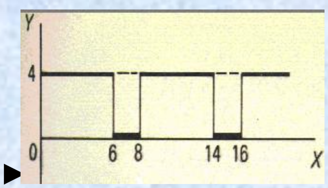
► $\frac{2\pi}{3}$ pg 197

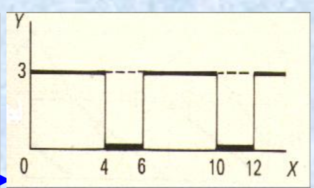
► π

Question No: 15 (Marks: 1) - Please choose one

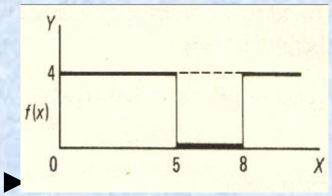
Match the following periodic function with its graph.

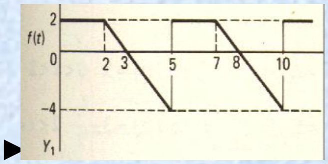
$$f(x) = \begin{cases} 3 & 0 < x < 4 \\ 0 & 4 < x < 6 \end{cases}$$

► 

► 

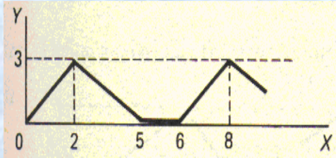
pg 197

▶ 

▶ 

Question No: 16 (Marks: 1) - Please choose one

What is the period of periodic function whose graph is as below?



▶2

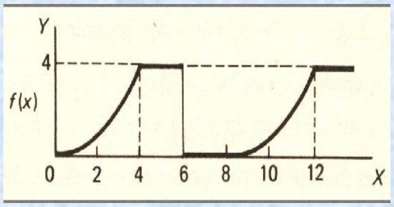
▶5

▶**6 pg 197**

▶8

Question No: 17 (Marks: 1) - Please choose one

What is the period of periodic function whose graph is as below?



►0

►4

►6

►8 pg 198

Question No: 18 (Marks: 1) - Please choose one

Let L denotes the Laplace Transform.

If $L\{F(t)\} = f(s)$ where s is a constant.and $\lim_{t \to 0}\left(\frac{F(t)}{t}\right)$ exists then which of the following equation holds?

► $L\left(\frac{F(t)}{t}\right) = f(s+a)$

$$L\left(\frac{F(t)}{t}\right)=f(s-a)$$

▶

$$L\left(\frac{F(t)}{t}\right)=\int_s^\infty f(s)\ ds$$

▶ pg 227

$$L\left(\frac{F(t)}{t}\right)=-\frac{d}{ds}\{f(s)\}$$

▶

Question No: 19 (Marks: 1) - Please choose one

Which of the following is Laplace inverse transform of the function $f(s)$ defined by

$$f(s)=\frac{3}{s-2}-\frac{2}{s}$$

?

▶ $3te^{2t}-2$

▶ $3e^{2t}-2t$

▶ $3e^{2t}-2$ correct answer

▶ None of these.

Question No: 20 **(Marks: 1) - Please choose one**

Let (x_1, y_1, z_1) and (x_2, y_2, z_2) be any two points in three dimensional space. What does the formula $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$ calculates?

► Distance between these two points pg 11

► Midpoint of the line joining these two points

► Ratio between these two points

Question No: 21 **(Marks: 1) - Please choose one**

Let the functions $P(x, y)$ and $Q(x, y)$ are finite and continuous inside and at the boundary of a closed curve C in the xy-plane. If $(P\,dx + Q\,dy)$ is an exact differential then

$$\oint_C (P\,dx + Q\,dy) =$$

► Zero pg 170

► One

► Infinite

Question No: 22 (Marks: 1) - Please choose one

What is Laplace transform of the function $F(t)$ if $F(t)=t$?

- ► $L\{t\}=\frac{1}{s}$ pg 222 Laplace transform of constant =a/s
- ► $L\{t\}=\frac{1}{s^2}$
- ► $L\{t\}=e^{-s}$
- ► $L\{t\}=s$

Question No: 23 (Marks: 1) - Please choose one

What is the value of $L\{e^{5t}\}$ if L denotes laplace transform?

- ► $L\{e^{5t}\}=\frac{1}{s-5}$ pg 222
- ► $L\{e^{5t}\}=\frac{s}{s^2+25}$
- ► $L\{e^{5t}\}=\frac{5}{s^2+25}$
- ► $L\{e^{5t}\}=\frac{5!}{s^6}$

Question No: 24 (Marks: 1) - Please choose one

Evaluate the line integral $\int_C (3x+2y)\,dx+(2x-y)\,dy$ where C is the line segment from (0, 0) to (0, 2).

►1

►0

►2 keeping x=0,dx=0 and limits of y 0 to 2

►-2

Question No: 25 (Marks: 1) - Please choose one

Evaluate the line integral $\int_C (2x+y)\,dx+(x^2-y)\,dy$ where C is the line segment from (0, 0) to (2, 0).

►0

►-4

►4 putting y=0,dy=0 and x from 0 to 2

►Do not exist

Question No: 26 (Marks: 1) - Please choose one

Which of the following are direction ratios for the line joining the points $(1, 3, 5)$ and $(2, -1, 4)$?

► 3, 2 and 9

► 1, -4 and -1 pg 13

► 2, -3 and 20

► 0.5, -3 and 5/4

Question No: 27 (Marks: 1) - Please choose one

$$If\ R = \{(x, y)/0 \le x \le 2\, and\, 1 \le y \le 4\}, then$$
$$\iint_R (6x^2 + 4xy^3)dA =$$

► $\int_1^4 \int_0^2 (6x^2 + 4xy^3)dydx$

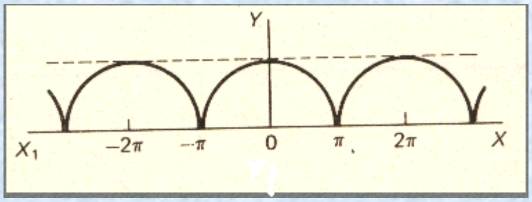
► $\int_0^2 \int_1^4 (6x^2 + 4xy^3)dxdy$

► $\int_1^4 \int_0^2 (6x^2 + 4xy^3)dxdy$ Correct answer

► $\int_2^4 \int_0^1 (6x^2 + 4xy^3)dxdy$

Question No: 28 (Marks: 1) - Please choose one

Which of the following is true for a periodic function whose graph is as below?

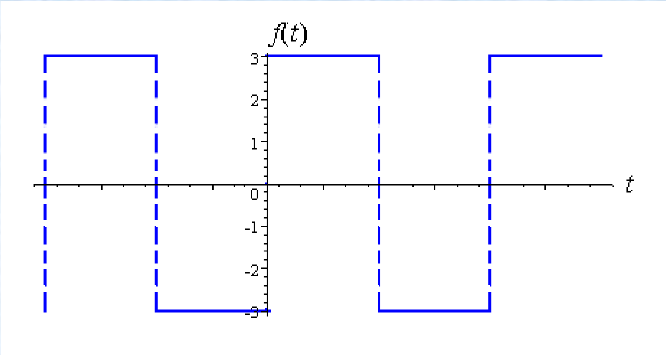


▶Even function pg 208

▶Odd function

▶Neither even nor odd function

Question No: 29 (Marks: 1) - Please choose one



Which of the following is true for a function whose graph is given above

▶An odd function pg 207

▶An even function

▶Neither even nor odd

Question No: 30 (Marks: 1) - Please choose one

At each point of domain, the function ----------------

► Is defined

► Is continuous

► Is infinite

► Has a limit

Solution:

In domain function is always defined but may be discontinuous

Question No: 1 (Marks: 1) - Please choose one

π is an example of -----------

► Irrational numbers pg 3

► Rational numbers

► Integers

Question No: 2 (Marks: 1) - Please choose one

Straight line is a special kind of ----------------

► Surface

► Curve

► Plane pg 13

► Parabola

An ordered triple corresponds to ----------- in three dimensional space.

► A unique point

► A point in each octant

► Three points

► Infinite number of points

Question No: 4 (Marks: 1) - Please choose one

The angles which a line makes with positive x ,y and z-axis are known as ------------

► Direction cosines

► Direction ratios

► Direction angles pg 11

Question No: 5 (Marks: 1) - Please choose one

Is the function $f(x, y)$ continuous at origin? If not, why?

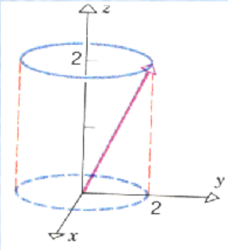
$f(x, y) = 4xy + \sin 3x^2 y$

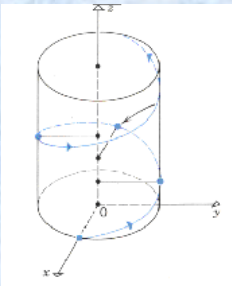
- ► $f(x, y)$ is continuous at origin
- ► $f(0, 0)$ is not defined
- ► $f(0, 0)$ is defined but $\lim_{(x,y)\to(0,0)} f(x, y)$ does not exist
- ► $f(0, 0)$ is defined and $\lim_{(x,y)\to(0,0)} f(x, y)$ exists but these two numbers are not equal.

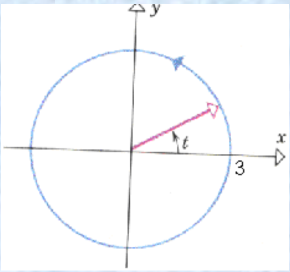
Question No: 6 (Marks: 1) - Please choose one

Match the following vector-valued function with its graph.

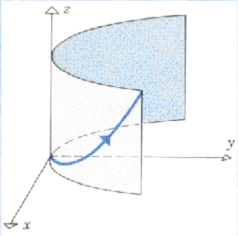
$r(t) = 3\cos t\,\hat{i} + 3\sin t\,\hat{j}$ and $0 \le t \le 2\pi$

► 

▶ 

▶ 

pg 141

▶ 

Question No: 9 (Marks: 1) - Please choose one

Is the following vector-valued function $\vec{r}(t)$ continuous at $t=0$? If not, why?

$\vec{r}(t) = (4\cos t, \sqrt{t}\ , 4\sin t)$

▶ $\vec{r}(0)$ is not defined

► $\vec{r}(0)$ is defined but $\lim_{t \to 0} \vec{r}(t)$ does not exist

► $\vec{r}(0)$ is defined and $\lim_{t \to 0} \vec{r}(t)$ exists but these two numbers are not equal.

► $\vec{r}(t)$ is continuous at $t=0$

Question No: 13 (Marks: 1) - Please choose one

Which one of the following is correct Wallis Sine formula when n is odd and $n \geq 3$?

► $\int_0^{\frac{\pi}{2}} \sin^n x \, dx = \frac{\pi}{2} \frac{(n-1)}{n} \frac{(n-3)}{(n-2)} \frac{(n-5)}{(n-4)} - - - - - - - - \frac{5}{6} \frac{3}{4} \frac{1}{2}$

► $\int_0^{\frac{\pi}{2}} \sin^n x \, dx = \frac{\pi}{2} \frac{(n)}{(n-1)} \frac{(n-2)}{(n-3)} \frac{(n-4)}{(n-5)} - - - - - - - - \frac{6}{5} \frac{4}{3} \frac{2}{1}$

► $\int_0^{\frac{\pi}{2}} \sin^n x \, dx = \frac{(n-1)}{n} \frac{(n-3)}{(n-2)} \frac{(n-5)}{(n-4)} - - - - - - - - \frac{6}{7} \frac{4}{5} \frac{2}{3}$ pg 182

$$\int_0^{\frac{\pi}{2}} \sin^n x \, dx = \frac{(n)}{(n-1)} \frac{(n-2)}{(n-3)} \frac{(n-4)}{(n-5)} ------- \frac{6}{5} \frac{4}{3} \frac{2}{1}$$

►

Question No: 14 (Marks: 1) - Please choose one

Which of the following is correct?

$$\int_0^{\frac{\pi}{2}} \sin^4 x \, dx = \frac{3}{16}$$

►

$$\int_0^{\frac{\pi}{2}} \sin^4 x \, dx = \frac{3\pi}{16}$$

pg 181

►

$$\int_0^{\frac{\pi}{2}} \sin^4 x \, dx = \frac{3}{8}$$

►

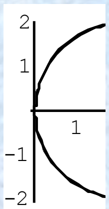
$$\int_0^{\frac{\pi}{2}} \sin^4 x \, dx = \frac{2\pi}{3}$$

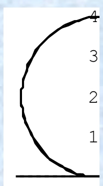
►

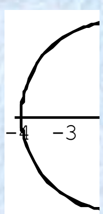
Question No: 15   (Marks: 1) - Please choose one

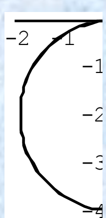
Match the following equation in polar co-ordinates with its graph.

$r = 4\sin\theta$

► 

►  not sure

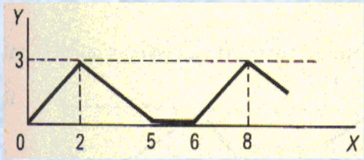
► 

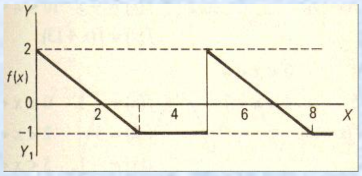
► 

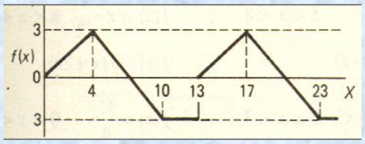
Question No: 18   (Marks: 1) - Please choose one

Match the following periodic function with its graph.

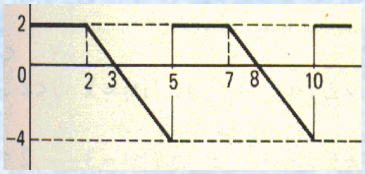
$$f(x)=\begin{cases}\frac{3}{4}x & 0<x<4\\ 7-x & 4<x<10\\ -3 & 10<x<13\end{cases}$$

► 

► 

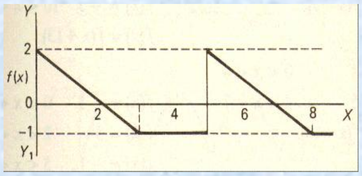
► 

pg 198

► 

Question No: 19   (Marks: 1) - Please choose one

What is the period of periodic function whose graph is as below?



► 2

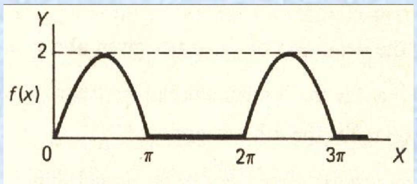
► 3

► 4

► 5   pg 198

Question No: 20   (Marks: 1) - Please choose one

What is the period of periodic function whose graph is as below?



► 0

► 2

► π

► 2π pg 198

Question No: 21 (Marks: 1) - Please choose one

$$\left(-2, \frac{-3\pi}{2}\right)$$

Polar co-ordinates of a point are . Which of the following is another possible polar co-ordinates representation of this point?

$\left(2, \frac{-\pi}{4}\right)$

▶

$\left(2, \frac{-\pi}{2}\right)$

▶

$\left(2, \frac{-\pi}{3}\right)$

▶

$\left(2, \frac{3\pi}{4}\right)$ not sure

▶

Question No: 22 (Marks: 1) - Please choose one

The function $f(x) = x^3 e^x$ is -------------

▶ Even function

▶ Odd function

▶ Neither even nor odd pg 209

Question No: 23 (Marks: 1) - Please choose one

The graph of an even function is symmetrical about ---------------

- ► x-axis
- ► y-axis pg 207
- ► origin

Question No: 24 (Marks: 1) - Please choose one

At which point the vertex of parabola, represented by the equation $y = x^2 - 4x + 3$, occurs?

- ► $(0, 3)$
- ► $(2, -1)$ pg 9
- ► $(-2, 15)$
- ► $(1, 0)$

Question No: 25 (Marks: 1) - Please choose one

The equation $y = x^2 - 4x + 2$ represents a parabola. Find a point at which the vertex of given parabola occurs?

- ► $(2, -2)$ pg 9
- ► $(-4, 34)$
- ► $(0, 0)$
- ► $(-2, 14)$

Question No: 26 (Marks: 1) - Please choose one

Is the function $f(x, y)$ continuous at origin? If not, why?

$$f(x, y) = \frac{xy}{x^2 + y^2}$$

- ► $f(x, y)$ is continuous at origin
- ► $\lim_{(x,y)\to(0,0)} f(x, y)$ limt does not exist pg 27
- ► $f(0, 0)$ is defined and $\lim_{(x,y)\to(0,0)} f(x, y)$ exists but these two numbers are not equal.

Question No: 27 (Marks: 1) - Please choose one

Sign of line integral is reversed when -----------

- ► path of integration is divided into parts.
- ► path of integration is parallel to y-axis.
- ► direction of path of integration is reversed. Pg 175
- ► path of integration is parallel to x-axis.

Question No: 28 (Marks: 1) - Please choose one

What is Laplace transform of a function F(t)?

(s is a constant)

► $\int_0^s e^{-st} F(t)\, dt$

► $\int_0^\infty e^{st} F(t)\, dt$

► $\int_{-\infty}^\infty e^{-st} F(t)\, dt$

► $\int_0^\infty e^{-st} F(t)\, dt$ pg 222

Question No: 29 (Marks: 1) - Please choose one

What is the value of $L\{e^{5t}\}$ if L denotes laplace transform?

► $L\{e^{5t}\} = \frac{1}{s-5}$ according to pg 223

► $L\{e^{5t}\} = \frac{s}{s^2+25}$

► $L\{e^{5t}\} = \frac{5}{s^2+25}$

► $L\{e^{5t}\} = \frac{5!}{s^6}$

Question No: 30 (Marks: 1) - Please choose one

What is the Laplace Inverse Transform of $\frac{1}{s+1}$

► $L^{-1}\left\{\frac{1}{s+1}\right\} = t+1$

► $L^{-1}\left\{\frac{1}{s+1}\right\} = e^{-t} + e^{t}$

► $L^{-1}\left\{\frac{1}{s+1}\right\} = e^{t}$

► $L^{-1}\left\{\frac{1}{s+1}\right\} = e^{-t}$ **pg 231 formula**

Question No: 31 (Marks: 1) - Please choose one

What is Laplace Inverse Transform of $\frac{5}{s^2+25}$

► $L^{-1}\left\{\frac{5}{s^2+25}\right\} = \sin 5t$ **pg 231**

► $L^{-1}\left\{\frac{5}{s^2+25}\right\} = \cos 5t$

► $L^{-1}\left\{\frac{5}{s^2+25}\right\} = \sin 25t$

► $L^{-1}\left\{\frac{5}{s^2+25}\right\} = \cos 25t$

Question No: 32 (Marks: 1) - Please choose one

What is $L\{-6\}$ if L denotes Laplace Transform?

- ▶ $L\{-6\} = \frac{1}{s+6}$
- ▶ $L\{-6\} = \frac{-6}{s}$ pg 223
- ▶ $L\{-6\} = \frac{s}{s^2+36}$
- ▶ $L\{-6\} = \frac{-6}{s^2+36}$

Question No: 37 (Marks: 1) - Please choose one

To evaluate the line integral, the integrand is expressed in terms of x, y, z with

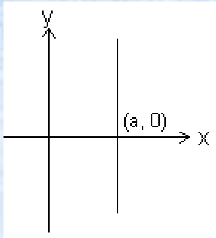
- ▶ $dr = dx\hat{i} + dy\,\hat{j}$
- ▶ $dr = dx\hat{i} + dy\,\hat{j} + dy\,\hat{k}$ pg 185
- ▶ $dr = dx + dy + dz$
- ▶ $dr = dx + dy$

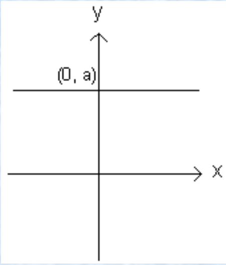
Question No: 38 (Marks: 1) - Please choose one

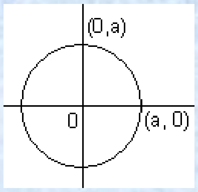
Match the following equation in polar co-ordinates with its graph.

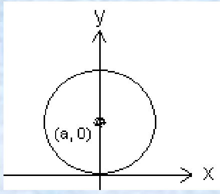
$r = a$

where a is an arbitrary constant.

► 

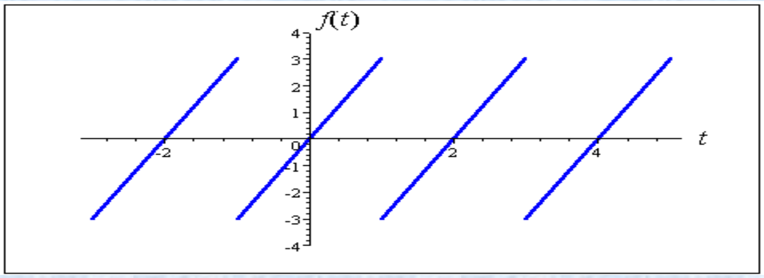
► 

► 

► 

pg 127

Question No: 40 (Marks: 1) - Please choose one



The graph of “saw tooth wave” given above is --------------

► An odd function because it is passing through the region

► An even function

► Neither even nor odd

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Paper Pattern

MCQS 40 each 1 mark
Short 4 each 2 marks
Short 4 each 3 marks
long 4 each 5 marks

Question No : 1 of 52 Marks: 1 (Budgeted Time 1 Min)

What is the derivative of following vector-valued function?

$\vec{r}(t) = \left(t^4, \sqrt{t+1}, \frac{3}{t^2}\right)$

Answer (Please select your correct option)

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- $\vec{r}'(t) = \left(4t^3, \frac{1}{\sqrt{t+1}}, \frac{-6}{t^3}\right)$
- $\vec{r}'(t) = \left(4t^3, \frac{1}{2\sqrt{t+1}}, \frac{6}{t^3}\right)$
- $\vec{r}'(t) = \left(4t^4, \frac{1}{2\sqrt{t+1}}, \frac{-6}{t^3}\right)$
- $\vec{r}'(t) = \left(4t^3, \frac{1}{2\sqrt{t+1}}, \frac{-6}{t^3}\right)$ correct

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Question No : 2 of 52 Marks: 1 (Budgeted Time 1 Min)

What is the derivative of following vector-valued function?

$\vec{r}(t) = \left(e^{t^2}, t^2, \sec 2t\right)$

Answer (Please select your correct option)

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- $\vec{r}'(t) = \left(2te^{t^2}, 2t, 2\sec 2t \tan 2t\right)$
- $\vec{r}'(t) = \left(te^{t^2}, 2t, \sec 2t \tan 2t\right)$ correct
- $\vec{r}'(t) = \left(2te^{t^2}, 2t, \tan 2t\right)$
- $\vec{r}'(t) = \left(t^2e^{t^2}, 2t, \sec 2t \tan 2t\right)$

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Question No : 3 of 52 Marks: 1 (Budgeted Time 1 Min)

Evaluate the integral $\int \left[(3t-1)\hat{i}+\sqrt{t}\,\hat{j}\right] dt$

Answer (Please select your correct option) WWW.VirtualAcademyLive.com

- $\int \left[(3t-1)\hat{i}+\sqrt{t}\,\hat{j}\right] dt = \left(\frac{3}{2}t^2-t\right)\hat{i}+\left(\frac{3}{2}t^{\frac{3}{2}}\right)\hat{j}+C$
- $\int \left[(3t-1)\hat{i}+\sqrt{t}\,\hat{j}\right] dt = \left(\frac{3}{2}t^2-t\right)\hat{i}+\left(\frac{2}{3}t^{\frac{3}{2}}\right)\hat{j}+C$ correct
- $\int \left[(3t-1)\hat{i}+\sqrt{t}\,\hat{j}\right] dt = \left(\frac{3}{2}t-t\right)\hat{i}+\left(\frac{2}{3}t^{\frac{3}{2}}\right)\hat{j}+C$
- $\int \left[(3t-1)\hat{i}+\sqrt{t}\,\hat{j}\right] dt = \left(\frac{3}{2}t^2-t\right)\hat{i}+\left(\frac{1}{2}t^{\frac{3}{2}}\right)\hat{j}+C$

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Question No : 4 of 52 Marks: 1 (Budgeted Time 1 Min)

The following differential is exact
$dz = (6xy+2y^2-5)\,dx+(3x^2+4xy-6)\,dy$

Answer (Please select your correct option) WWW.VirtualAcademyLive.com

- True correct
- False

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Question No : 5 of 52 Marks: 1 (Budgeted Time 1 Min)

What is the amplitude of a periodic function defined by $f(x) = 4\cos 3x$?

Answer (Please select your correct option) WWW.VirtualAcademyLive.com

- 1
- 3
- 4 correct
- 12

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Question No : 6 of 52 Marks: 1 (Budgeted Time 1 Min)

What is the period of a periodic function defined by $f(x) = 4\sin 2x$?

Answer (Please select your correct option)

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- 2^0
- 4^0
- 8^0
- 180^0 corrrect
- 360^0

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Question No : 7 of 52 Marks: 1 (Budgeted Time 1 Min)

What is the period of a periodic function defined by $f(x) = \sin\frac{x}{2}$?

Answer (Please select your correct option)

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- $\frac{\pi}{2}$
- π
- $\frac{3\pi}{2}$
- 4π

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Question No : 8 of 52 Marks: 1 (Budgeted Time 1 Min)

Match the following periodic function with its graph.

$$f(x) = \begin{cases} 4 & 0 < x < 5 \\ 0 & 5 < x < 8 \end{cases}$$

Answer (Please select your correct option)

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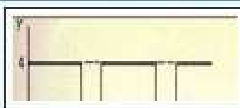
- 
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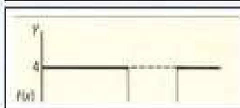
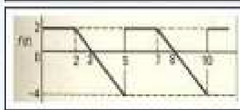
-  correct
- 

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Question No : 9 of 52 Marks: 1 (Budgeted Time 1 Min)

Match the following periodic function with its graph.

$$f(x) = \begin{cases} 4 & 0 < x < 6 \\ 0 & 6 < x < 8 \end{cases}$$

Answer (Please select your correct option)

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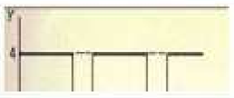
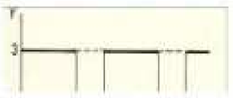
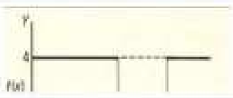
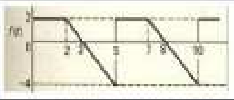


correct







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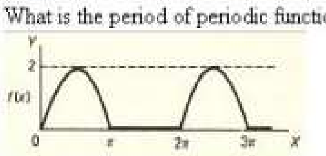
Question No : 10 of 52 Marks: 1 (Budgeted Time 1 Min)

What is the period of periodic function whose graph is as below?



Answer (Please select your correct option)

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- 0
- 2
- π
- 2π correct

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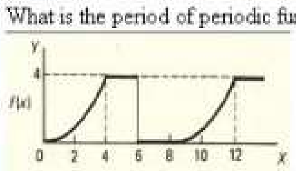
Question No : 11 of 52 Marks: 1 (Budgeted Time 1 Min)

What is the period of periodic function whose graph is as below?



Answer (Please select your correct option)

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- 0
- 4
- 6
- 8 correct

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Question No : 12 of 52 Marks: 1 (Budgeted Time 1 Min)

Let L denotes the Laplace Transform.

If $L(F(t)) = f(s)$ where s is a constant and $\lim_{t\to 0}\left(\frac{F(t)}{t}\right)$ exists then which of the following equation holds?

Answer (Please select your correct option) WWW.VirtualAcademyLive.com

- $L\left(\frac{F(t)}{t}\right) = f(s+a)$
- $L\left(\frac{F(t)}{t}\right) = f(s-a)$
- $L\left(\frac{F(t)}{t}\right) = \int_s^\infty f(s)\, ds$ correct
- $L\left(\frac{F(t)}{t}\right) = -\frac{d}{ds}(f(s))$

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Question No : 13 of 52 Marks: 1 (Budgeted Time 1 Min)

The function $f(x) = x^2 \cos 2x$ is -------------

Answer (Please select your correct option) WWW.VirtualAcademyLive.com

- Neither even nor odd
- Odd function
- Even function correct

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Question No : 14 of 52 Marks: 1 (Budgeted Time 1 Min)

The graph of an even function is symmetrical about ---------------

Answer (Please select your correct option) WWW.VirtualAcademyLive.com

- x-axis
- y-axis correct
- origin

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Question No : 15 of 52 Marks: 1 (Budgeted Time 1 Min)

The graph of an odd function is symmetrical about ----------------

Answer (Please select your correct option)

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- x-axis
- y-axis
- origin

correct

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Question No : 16 of 52 Marks: 1 (Budgeted Time 1 Min)

Sign of line integral is reversed when -----------

Answer (Please select your correct option)

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- path of integration is divided into parts.
- path of integration is parallel to y-axis.
- direction of path of integration is reversed.

correct

- path of integration is parallel to x-axis.

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Question No : 17 of 52 Marks: 1 (Budgeted Time 1 Min)

What is laplace transform of the function $F(t)$ if $F(t)=\cos 2t$?

Answer (Please select your correct option)

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- $L(\cos 2t)=\frac{2}{s^2+4}$

correct

- $L(\cos 2t)=\frac{1}{s-2}$
- $L(\cos 2t)=\frac{s}{s^2+4}$
- $L(\cos 2t)=\frac{2!}{s^3}$

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Question No : 18 of 52 Marks: 1 (Budgeted Time 1 Min)

What is Laplace Inverse Transform of $\frac{5}{s^2+25}$

Answer (Please select your correct option)

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- $L^{-1}\left\{\frac{5}{s^2+25}\right\} = \sin 5t$ correct
- $L^{-1}\left\{\frac{5}{s^2+25}\right\} = \cos 5t$
- $L^{-1}\left\{\frac{5}{s^2+25}\right\} = \sin 25t$
- $L^{-1}\left\{\frac{5}{s^2+25}\right\} = \cos 25t$

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Question No : 19 of 52 Marks: 1 (Budgeted Time 1 Min)

Evaluate the line integral $\int_C (3x+2y)\,dx + (2x-y)\,dy$ where C is the line segment from (0, 0) to (0, 2).

Answer (Please select your correct option)

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- 1
- 0
- 2 correct
- -2

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Question No : 20 of 52 Marks: 1 (Budgeted Time 1 Min)

Evaluate the line integral $\int_C (xy)\,dx + (1+y^2)\,dy$ where C is the line segment from (1, 0) to (3, 0).

Answer (Please select your correct option)

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- -4
- 0 correct
- 4
- Do not exist

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Question No : 21 of 52 Marks: 1 (Budgeted Time 1 Min)

If p is the period of a function then that function is said to be periodic if $f(x+p)=f(x)$, ______________

Answer (Please select your correct option)

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- For all values of x in the domain of f

correct

- For positive values of x in the domain of f
- For negative values of x in the domain of f

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Question No : 22 of 52 Marks: 1 (Budgeted Time 1 Min)

A vector field is a vector each of whose components is a scalar field

Answer (Please select your correct option)

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- True
- False

correct

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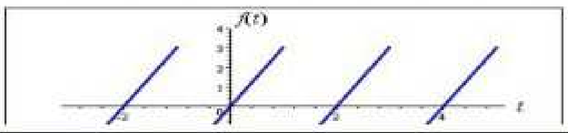
Question No : 23 of 52 Marks: 1 (Budgeted Time 1 Min)



Answer (Please select your correct option)

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- An odd function
- An even function
- Neither even nor odd

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Question No : 24 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following set is the union of sets of rational and irrational numbers?

Answer (Please select your correct option)

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- Set of rational numbers
- Set of integers
- Set of real numbers

correct

- Empty set.

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Question No : 25 of 52 Marks: 1 (Budgeted Time 1 Min)

$\sqrt{(x_2-x_1)^2+(y_2-y_1)^2+(z_2-z_1)^2}$ shows .... (x_1, y_1, z_1) and (x_2, y_2, z_2)

Answer (Please select your correct option)

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- Distance between the points

correct

- Midpoint of the line joining the points
- Ratio between the points

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Question No : 26 of 52 Marks: 1 (Budgeted Time 1 Min)

Domain of the function $f(x, y)=\sqrt{y-x^2}$ satisfies the condition ....

Answer (Please select your correct option)

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- $y < x^2$
- $y \geq x^2$

correct

- $y \neq x^2$
- Entire space

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Question No : 27 of 52 Marks: 1 (Budgeted Time 1 Min)

If rectangular co-ordinates of a point are $(1, \sqrt{3}, -2)$, then value of "r" in cylindrical co-ordinates is ....

Answer (Please select your correct option)

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- $\sqrt{2}$
- 2
- $2\sqrt{2}$
- $-2\sqrt{2}$

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Question No : 28 of 52 Marks: 1 (Budgeted Time 1 Min)

Suppose $f(x, y) = x^3 e^{xy}$. Which of the following options is correct?

Answer (Please select your correct option)

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- $\frac{\partial f}{\partial y} = 3x^3 e^{xy}$
- $\frac{\partial f}{\partial y} = x^3 e^{xy}$
- $\frac{\partial f}{\partial y} = x^4 e^{xy}$ correct
- $\frac{\partial f}{\partial y} = x^3 y e^{xy}$

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Question No : 29 of 52 Marks: 1 (Budgeted Time 1 Min)

The function decreases most rapidly in the direction of ....

Answer (Please select your correct option)

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- $-\nabla f$ correct
- $-|\nabla f|$
- $\nabla f \times \hat{u}$
- $|\nabla f|$

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Question No : 30 of 52 Marks: 1 (Budgeted Time 1 Min)

Two surfaces are said to be orthogonal at the point of their intersection if their normals at that point are ---------

Answer (Please select your correct option)

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- Parallel
- Perpendicular

corrrect

- In opposite direction
- Overlapping

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Question No : 31 of 52 Marks: 1 (Budgeted Time 1 Min)

For a function $f(x,y)$ to have both absolute maximum and minimum, it must be Continuous on . . . . set R.

Answer (Please select your correct option)

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- a closed and bounded
- an open and bounded
- a closed and unbounded
- an open and unbounded

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Question No : 32 of 52 Marks: 1 (Budgeted Time 1 Min)

If $R = ((x,y): 2 \le x \le 4 \, and \, 0 \le y \le 1)$, then $\iint_R (4xe^{2y})dA =$ . . . .

Answer (Please select your correct option)

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- $\int_0^1 \int_2^4 (4xe^{2y})dydx$
- $\int_0^1 \int_2^4 (4xe^{2y})dxdy$
- $\int_1^4 \int_0^2 (4xe^{2y})dxdy$
- $\int_1^4 \int_0^2 (4xe^{2y})dydx$

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Question No : 33 of 52 Marks: 1 (Budgeted Time 1 Min)

Let R be a closed region in two dimensional space then the double integral over R calculates. . . . .

Answer (Please select your correct option) WWW.VirtualAcademyLive.com

- Area of R
- Radius of inscribed circle in R.
- Distance between two endpoints of R.
- None of these

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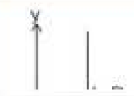
Question No : 34 of 52 Marks: 1 (Budgeted Time 1 Min)

Match the following equation in polar co-ordinates with its graph.

$r\cos\theta = a$

where a is an arbitrary constant

Answer (Please select your correct option) WWW.VirtualAcademyLive.com

- 
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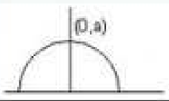
- 

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Question No : 35 of 52 Marks: 1 (Budgeted Time 1 Min)

Polar co-ordinates of a point are $\left(-2, \frac{-3\pi}{2}\right)$. Which of the following is another possible polar co-ordinates representation of this point?

Answer (Please select your correct option) WWW.VirtualAcademyLive.com

- $\left(2, \frac{-\pi}{4}\right)$
- $\left(2, \frac{-\pi}{2}\right)$
- $\left(2, \frac{-\pi}{3}\right)$
- $\left(2, \frac{3\pi}{4}\right)$

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Question No : 36 of 52 Marks: 1 (Budgeted Time 1 Min)

If the equation of a curve, in polar co-ordinates, remains unchanged after replacing (r, θ) by $(r, -\theta)$ then the curve is said to be symmetric about ....

Answer (Please select your correct option) WWW.VirtualAcademyLive.com

- Initial line
- y-axis correct
- Pole
- origin

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Question No : 37 of 52 Marks: 1 (Budgeted Time 1 Min)

The graph of curve $r^2 = 4\cos 2\theta$ is symmetric about ....

Answer (Please select your correct option) WWW.VirtualAcademyLive.com

- Initial line correct
- y-axis
- both initial line and y-axis
- none of these

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Question No : 38 of 52 Marks: 1 (Budgeted Time 1 Min)

If $p(r,\theta)$ is a point in polar coordinate system, then r is the distance of p from ....

Answer (Please select your correct option) WWW.VirtualAcademyLive.com

- Pole
- Imaginary axis
- None of these
- Polar axis

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Question No : 39 of 52 Marks: 1 (Budgeted Time 1 Min)

The point $p(0,\theta)$ in polar coordinate system lies on . . . .

Answer (Please select your correct option)

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- Polar axis
- y-axis
- Pole
- None of these

Made by: Waqar Siddhu

Question No : 40 of 52 Marks: 1 (Budgeted Time 1 Min)

Given the integral $\iint_R f(x,y)dxdy$, after converting to polar coordinates, it will become . . . . where $a \le \theta \le b$ and $c \le r \le d$.

Answer (Please select your correct option)

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- $\int_a^b \int_c^d f(r,\theta) r dr d\theta$
- $\int_a^b \int_c^d f(r,\theta) dr d\theta$
- $\int_a^b \int_c^d f(r,\theta) r d\theta dr$
- $\int_a^b \int_c^d f(r,\theta) d\theta dr$

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Question No : 41 of 52 Marks: 2 (Budgeted Time 4 Min)

Determine whether the following differential is exact or not.

$dz = 4x^3y^3\ dx + 3x^4y^2\ dy$

Answer (Please click here to Add Answer)

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Question No : 42 of 52 — Marks: 2 (Budgeted Time 4 Min)

Evaluate

$$\int_{-\pi}^{\pi} \cos nx \, dx$$

where n is an integer other than zero.

Answer (Please click here to Add Answer)

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Question No : 43 of 52 — Marks: 2 (Budgeted Time 4 Min)

Prove whether the following function is even, odd or neither.

$f(x) = x^2 - 4\sin x$

Answer (Please click here to Add Answer)

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Question No : 44 of 52 — Marks: 2 (Budgeted Time 4 Min)

Given $\vec{a} \times \vec{b} = 3xi + 2yj + xk$ and $\vec{c} = 7xi + 4yk$. Find scalar triple product of these vectors.

Answer (Please click here to Add Answer)

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Question No : 45 of 52 Marks: 3 (Budgeted Time 6 Min)

What is the arc-length of the curve $\vec{r}(t) = (4+3t)\hat{i} + (2-2t)\hat{j} + (5+t)\hat{k}$ when $3 \le t \le 4$?

Answer (Please click here to Add Answer)

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Question No : 46 of 52 Marks: 3 (Budgeted Time 6 Min)

Find $div\ \vec{F}$, if $\vec{F} = (3x+y)\hat{i} + xy^2z\hat{j} + (xz^2)\hat{k}$

Answer (Please click here to Add Answer)

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Question No : 47 of 52 Marks: 3 (Budgeted Time 6 Min)

Determine the Fourier co-efficient a_0 of the periodic function defined below:

$f(x) = 2x+1 \qquad 0 < x < 2$

Answer (Please click here to Add Answer)

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Question No : 48 of 52 Marks: 3 (Budgeted Time 6 Min)

A line, in three dimensional space, passes through the point $(3, -4, 2)$ and parallel to the vector $\vec{n} = 4\hat{i} + 3\hat{j} + 6\hat{k}$. Write down the equation of this line in parametric and symmetric form.

Answer (Please click here to Add Answer)

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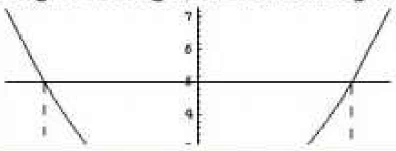
Question No : 49 of 52 Marks: 5 (Budgeted Time 10 Min)

Using definite integral, find area of the region bounded by the curves of $y = x^2 + 1$ *and* $y = 5$



Answer (Please click here to Add Answer)

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Question No : 50 of 52 Marks: 5 (Budgeted Time 10 Min)

Show that Laplace transform of the function

$f(t) = 1$ is

$\frac{1}{s}$ where s is a constant for the integration and s > 0.

Answer (Please click here to Add Answer)

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Question No : 51 of 52 Marks: 5 (Budgeted Time 10 Min)

Determine whether the following vector field $\vec{F}$ is conservative or not.

$\vec{F}(x, y, z) = (4x - z)\hat{i} + (3y + z)\hat{j} + (y - x)\hat{k}$

Answer (Please click here to Add Answer)

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Question No : 52 of 52 Marks: 5 (Budgeted Time 10 Min)

Consider the point $(-5, 5, 6)$ in rectangular coordinate system. Convert it into Spherical coordinates.

Answer (Please click here to Add Answer)

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Paper Pattern

MCQS 40 each 1 mark
Short 4 each 2 marks
Short 4 each 3 marks
long 4 each 5 marks

Question No : 1 of 52 Marks: 1 (Budgeted Time 1 Min)

Which one of the following is correct Wallis Sine formula when n is odd and $n \geq 3$?

Answer (Please select your correct option) WWW.VirtualAcademyLive.com

- $\int_0^{\frac{\pi}{2}} \sin^n x\, dx = \frac{\pi}{2}\frac{(n-1)}{n}\frac{(n-3)}{(n-2)}\frac{(n-5)}{(n-4)}-------\frac{5}{6}\frac{3}{4}\frac{1}{2}$
- $\int_0^{\frac{\pi}{2}} \sin^n x\, dx = \frac{\pi}{2}\frac{(n)}{(n-1)}\frac{(n-2)}{(n-3)}\frac{(n-4)}{(n-5)}-------\frac{6}{5}\frac{4}{3}\frac{2}{1}$
- $\int_0^{\frac{\pi}{2}} \sin^n x\, dx = \frac{(n-1)}{n}\frac{(n-3)}{(n-2)}\frac{(n-5)}{(n-4)}-------\frac{6}{7}\frac{4}{5}\frac{2}{3}$ correct
- $\int_0^{\frac{\pi}{2}} \sin^n x\, dx = \frac{(n)}{(n-1)}\frac{(n-2)}{(n-3)}\frac{(n-4)}{(n-5)}-------\frac{6}{5}\frac{4}{3}\frac{2}{1}$ Made by: Waqar Siddhu

Question No : 2 of 52 Marks: 1 (Budgeted Time 1 Min)

What is the amplitude of a periodic function defined by $f(x) = 4\sin 2x$?

Answer (Please select your correct option) WWW.VirtualAcademyLive.com

- 2
- 4 correct
- 8
- 16 Made by: Waqar Siddhu

Question No : 3 of 52 Marks: 1 (Budgeted Time 1 Min)

What is the period of a periodic function defined by $f(x) = \sin \frac{x}{2}$?

Answer (Please select your correct option) WWW.VirtualAcademyLive.com

- $\frac{\pi}{2}$
- π
- $\frac{3\pi}{2}$
- 4π correct

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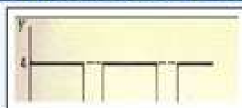
Question No : 4 of 52 Marks: 1 (Budgeted Time 1 Min)

Match the following periodic function with its graph.

$$f(x) = \begin{cases} 3 & 0 < x < 4 \\ 0 & 4 < x < 6 \end{cases}$$

Answer (Please select your correct option) WWW.VirtualAcademyLive.com

- 
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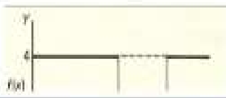
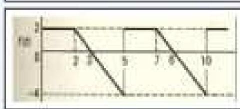
- 
- 

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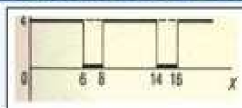
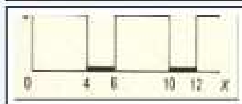
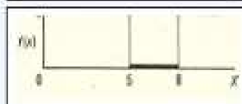
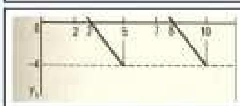
Question No : 4 of 52 Marks: 1 (Budgeted Time 1 Min)

Match the following periodic function with its graph.

$$f(x) = \begin{cases} 3 & 0 < x < 4 \\ 0 & 4 < x < 6 \end{cases}$$

Answer (Please select your correct option) WWW.VirtualAcademyLive.com

- 
-  correct
- 
- 

Made by: Waqar Siddhu

Question No : 5 of 52 Marks: 1 (Budgeted Time 1 Min)

Match the following periodic function with its graph.

$$f(x) = \begin{cases} 4 & 0 < x < 6 \\ 0 & 6 < x < 8 \end{cases}$$

Answer (Please select your correct option)

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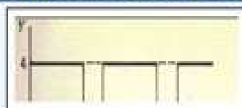
- 
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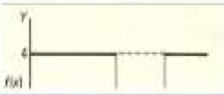
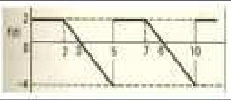
- 
- 

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Question No : 5 of 52 Marks: 1 (Budgeted Time 1 Min)

Match the following periodic function with its graph.

$$f(x) = \begin{cases} 4 & 0 < x < 6 \\ 0 & 6 < x < 8 \end{cases}$$

Answer (Please select your correct option)

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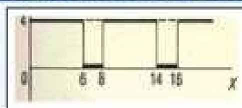
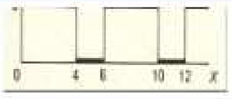
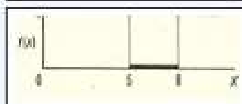
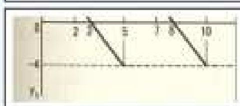
-  correct
- 
- 
- 

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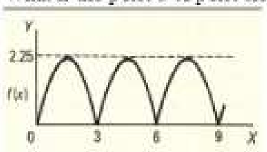
Question No : 6 of 52 Marks: 1 (Budgeted Time 1 Min)

What is the period of periodic function whose graph is as below?



Answer (Please select your correct option)

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- 0
- 2.25
- 3 correct
- 6

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Question No : 7 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following condition must be satisfied for a vector field $\vec{F}$ to be a conservative vector field?

Answer (Please select your correct option)

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- Line integral of $\vec{F}$ along a curve, depends only on the endpoints of that curve, not on the particular route taken. correct
- Divergence of $\vec{F}$ should be zero
- Gradient of $\vec{F}$ should be zero.
- $\vec{F} = 0$

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Question No : 8 of 52 Marks: 1 (Budgeted Time 1 Min)

Let L denotes the Laplace Transform.
According to First Shift Theorem, if $L(F(t)) = f(s)$ then which of the following equation holds?
s and a are constants.

Answer (Please select your correct option)

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- $L(e^{-at}F(t)) = f(s-a)$
- $L(e^{-at}F(t)) = f(s+a)$
- $L(e^{-at}F(t)) = f(s)$
- $L(e^{-at}F(t)) = f(a)$

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Question No : 9 of 52 Marks: 1 (Budgeted Time 1 Min)

The function $f(x) = x^2 \cos 2x$ is -------------

Answer (Please select your correct option)

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- Even function correct
- Odd function
- Neither even nor odd

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Question No : 10 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following is Laplace inverse transform of the function $f(s)$ defined by $f(s) = \frac{3}{s-2} - \frac{2}{s}$?

Answer (Please select your correct option)

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- $3te^{2t} - 2$
- $3e^{2t} - 2t$
- $3e^{2t} - 2$ correct
- None of these.

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Question No : 11 of 52 Marks: 1 (Budgeted Time 1 Min)

What is Laplace transform of a function F(t)?

(s is a constant)

Answer (Please select your correct option)

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- $\int_0^t e^{-st} F(t)\, dt$
- $\int_0^\infty e^{st} F(t)\, dt$
- $\int_{-\infty}^\infty e^{-st} F(t)\, dt$
- $\int_0^\infty e^{-st} F(t)\, dt$ correct

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Question No : 12 of 52 Marks: 1 (Budgeted Time 1 Min)

What is laplace transform of the function $F(t)$ if $F(t) = \sin 3t$?

Answer (Please select your correct option)

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- $L(\sin 3t) = \frac{3}{s^2+9}$ correct
- $L(\sin 3t) = \frac{s}{s^2+9}$
- $L(\sin 3t) = \frac{1}{s-3}$
- $L(\sin 3t) = \frac{3!}{s^4}$

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Question No : 13 of 52 Marks: 1 (Budgeted Time 1 Min)

If L denotes laplace transform then
$L(te^{5t}) =$

Answer (Please select your correct option) WWW.VirtualAcademyLive.com

- $L(te^{5t}) = \dfrac{1}{s^2-5}$
- $L(te^{5t}) = \dfrac{1}{s^2+5}$
- $L(te^{5t}) = \dfrac{1}{(s+5)^2}$
- $L(te^{5t}) = \dfrac{1}{(s-5)^2}$ correct

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Question No : 14 of 52 Marks: 1 (Budgeted Time 1 Min)

What is Laplace Inverse Transform of $\dfrac{s}{s^2+25}$

Answer (Please select your correct option) WWW.VirtualAcademyLive.com

- $L^{-1}\left\{\dfrac{s}{s^2+25}\right\} = \sin 5t$
- $L^{-1}\left\{\dfrac{s}{s^2+25}\right\} = \cos 5t$ correct
- $L^{-1}\left\{\dfrac{s}{s^2+25}\right\} = \sin 25t$
- $L^{-1}\left\{\dfrac{s}{s^2+25}\right\} = \cos 25t$

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Question No : 15 of 52 Marks: 1 (Budgeted Time 1 Min)

What is $L(-6)$ if L denotes Laplace Transform?

Answer (Please select your correct option) WWW.VirtualAcademyLive.com

- $L(-6) = \dfrac{1}{s+6}$
- $L(-6) = \dfrac{-6}{s}$ correct
- $L(-6) = \dfrac{s}{s^2+36}$
- $L(-6) = \dfrac{-6}{s^2+36}$

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Question No : 16 of 52 Marks: 1 (Budgeted Time 1 Min)

Curl of vector function is always a --------------

Answer (Please select your correct option)

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- Scalar
- Vector correct

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Question No : 17 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following is geometrical representation of set of real numbers?

Answer (Please select your correct option)

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- Co-ordinate line correct
- xy-plane
- Sphere
- Circular cylinder

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Question No : 18 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following is the interval notation of real line?

Answer (Please select your correct option)

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- $(-\infty, +\infty)$ correct
- $(-\infty, 0)$
- $(0, +\infty)$

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Question No : 19 of 52 Marks: 1 (Budgeted Time 1 Min)

An ordered triple corresponds to ----------- in a three dimensional space.

Answer (Please select your correct option)

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- A unique point correct
- A point in each octant
- Three points
- Infinite number of points

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Question No : 20 of 52 Marks: 1 (Budgeted Time 1 Min)

What is the distance between points (3, 2, 0) and (1, 0, -1)?

Answer (Please select your correct option)

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- 3 correct
- $\sqrt{6}$
- $\sqrt{3}$
- $\sqrt{10}$

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Question No : 21 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following are direction ratios for the line joining the points $(1, 3, 5)$ and $(2, -1, 4)$?

Answer (Please select your correct option)

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- 3, 2, 9
- 1, -4, -1
- 2, -3, 20
- 0.5, -3, 5/4

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Question No : 22 of 52 Marks: 1 (Budgeted Time 1 Min)

In three dimensional space, the equation $y = x^2$ always represents . . . .

Answer (Please select your correct option)

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- Parabola
- Straight line
- Half cylinder
- Cone

correct

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Question No : 23 of 52 Marks: 1 (Budgeted Time 1 Min)

For a function $f(x,y,z)$, the equation $\frac{\partial^2 f}{\partial^2 x}+\frac{\partial^2 f}{\partial^2 y}+\frac{\partial^2 f}{\partial^2 z}=0$ is known as -------------

Answer (Please select your correct option)

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- Gauss Equation
- Euler's equation
- Laplace's Equation correct
- Stoke's Equation

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Question No : 24 of 52 Marks: 1 (Budgeted Time 1 Min)

Every differentiable function is always . . . .

Answer (Please select your correct option)

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- Piece wise continuous
- Discontinuous
- Continuous correct

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Question No : 25 of 52 Marks: 1 (Budgeted Time 1 Min)

Gradient of a scalar function always results in a . . . . function.

Answer (Please select your correct option) WWW.VirtualAcademyLive.com

- Scalar
- Continuous
- Vector

correct

- Constant

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Question No : 26 of 52 Marks: 1 (Budgeted Time 1 Min)

The two normal lines for the surfaces $f(x,y,z)=0$ and $g(x,y,z)=0$, are orthogonal if and only if . . . . where f_x, f_y, f_z and g_x, g_y, g_z are direction ratios for the two normal lines respectively.

Answer (Please select your correct option) WWW.VirtualAcademyLive.com

- $f_x g_y + f_y g_z + f_z g_x = 0$
- $f_x g_x + f_y g_y + f_z g_z \geq 0$
- $f_x g_x + f_y g_y + f_z g_z = 0$

correct

- $f_x + f_y + f_z + g_x + g_y + g_z = 0$

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Question No : 27 of 52 Marks: 1 (Budgeted Time 1 Min)

Match the following equation in polar co-ordinates with its graph.

$r \cos\theta = a$

$where\ a\ is\ an\ arbitrary\ constant$

Answer (Please select your correct option) WWW.VirtualAcademyLive.com

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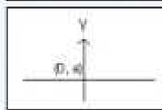
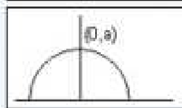
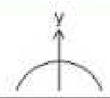
- 

- 

- 

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Question No : 28 of 52 Marks: 1 (Budgeted Time 1 Min)

Match the following equation in polar co-ordinates with its graph.
$r \sin\theta = a$
where a is an arbitrary constant

Answer (Please select your correct option)

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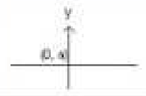
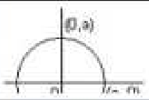
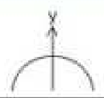




correct



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Question No : 29 of 52 Marks: 1 (Budgeted Time 1 Min)

Polar co-ordinates of a point are $\left(-2, \frac{-3\pi}{2}\right)$. Which of the following is another possible polar co-ordinates representation of this point?

Answer (Please select your correct option)

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- $\left(2, \frac{-\pi}{4}\right)$
- $\left(2, \frac{-\pi}{2}\right)$ correct
- $\left(2, \frac{-\pi}{3}\right)$
- $\left(2, \frac{3\pi}{4}\right)$

Made by: Waqar Siddhu

Question No : 28 of 52 Marks: 1 (Budgeted Time 1 Min)

Match the following equation in polar co-ordinates with its graph.
$r \sin\theta = a$
where a is an arbitrary constant

Answer (Please select your correct option)

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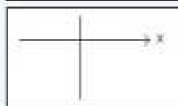

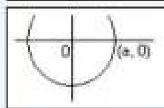

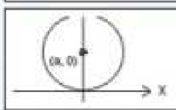

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Question No : 27 of 52 Marks: 1 (Budgeted Time 1 Min)

Match the following equation in polar co-ordinates with its graph.
$r\ \cos\theta = a$
where a is an arbitrary constant

Answer (Please select your correct option)

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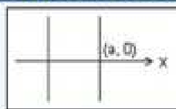
- 

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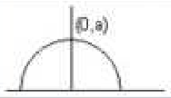
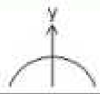
- 

- 

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Question No : 27 of 52 Marks: 1 (Budgeted Time 1 Min)

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$r\ \cos\theta = a$
where a is an arbitrary constant

Answer (Please select your correct option)

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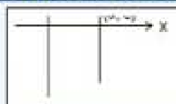
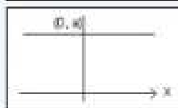
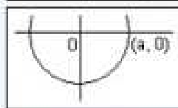
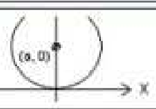
- 

- 

- 

- 

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Question No : 29 of 52 Marks: 1 (Budgeted Time 1 Min)

Polar co-ordinates of a point are $\left(-2, \frac{-3\pi}{2}\right)$. Which of the following is another possible polar co-ordinates representation of this point?

Answer (Please select your correct option)

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- $\left(2, \frac{-\pi}{4}\right)$
- $\left(2, \frac{-\pi}{2}\right)$
- $\left(2, \frac{-\pi}{3}\right)$
- $\left(2, \frac{3\pi}{4}\right)$

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Question No : 30 of 52 Marks: 1 (Budgeted Time 1 Min)

If the equation of a curve, in polar co-ordinates, remains unchanged after replacing (r, θ) by $(r, -\theta)$ then the curve is said to be symmetric about ....

Answer (Please select your correct option)

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- Initial line correct
- y-axis
- Pole
- origin

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Question No : 31 of 52 Marks: 1 (Budgeted Time 1 Min)

Given the integral $\iint_R f(x,y)dxdy$, after converting to polar coordinates, it will become .... where $a \le \theta \le b$ and $c \le r \le d$.

Answer (Please select your correct option)

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- $\int_a^b \int_c^d f(r,\theta)rdrd\theta$
- $\int_a^b \int_c^d f(r,\theta)drd\theta$ correct
- $\int_a^b \int_c^d f(r,\theta)rd\theta dr$
- $\int_a^b \int_c^d f(r,\theta)d\theta dr$

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Question No : 32 of 52 Marks: 1 (Budgeted Time 1 Min)

The parametric equations that correspond to the vector equation $\vec{r}(t) = \sin^2 t\hat{i} + (1-\cos 2t)\hat{j}$ are ....

Answer (Please select your correct option)

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- $x = \sin^2 t$, $y = 1-\cos 2t$, $z = 0$ correct
- $y = \sin^2 t$, $x = 1-\cos 2t$, $z = 0$
- $x = \sin^2 t$, $y = 1-\cos 2t$, $z = 1$
- $x = \sin^2 t$, $y = \cos 2t$, $z = 1$

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Question No : 33 of 52 Marks: 1 (Budgeted Time 1 Min)

The graph of the equation $r = 3\hat{i} - 2\hat{j} - \hat{k}$ is the point ....

Answer (Please select your correct option)
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- $(1, -2, 3)$
- $(3, -2, -1)$ correct
- $(3, 2, 1)$
- $(-3, -2, -1)$

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Question No : 34 of 52 Marks: 1 (Budgeted Time 1 Min)

If $\vec{R}(t)$ is the anti-derivative of a given vector valued function $\vec{r}(t)$ i.e., $\vec{R}'(t) = \vec{r}(t)$ then $\int_a^b \vec{r}(t)dt =$ ....

Answer (Please select your correct option)
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- $\vec{R}(a) + \vec{R}(b)$
- $\vec{R}(b) - \vec{R}(a) + c$ where c is a non-zero constant. correct
- $\vec{R}(a).\vec{R}(b)$
- $\vec{R}(b) - \vec{R}(a)$

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Question No : 35 of 52 Marks: 1 (Budgeted Time 1 Min)

For the given vector valued functions $\vec{r}_1(t)$ and $\vec{r}_2(t)$, $\frac{d}{dt}[\vec{r}_1(t) \times \vec{r}_2(t)] =$ .... where "×" and "•" represent the cross and dot product respectively.

Answer (Please select your correct option)
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- $\vec{r}_1 \cdot \frac{d\vec{r}_2}{dt} + \frac{d\vec{r}_1}{dt} \cdot \vec{r}_2$
- $\vec{r}_1 \times \frac{d\vec{r}_2}{dt} + \frac{d\vec{r}_1}{dt} \times \vec{r}_2$
- $\frac{d\vec{r}_2}{dt} \times \vec{r}_1 + \vec{r}_2 \times \frac{d\vec{r}_1}{dt}$
- $\frac{d\vec{r}_2}{dt} \times \vec{r}_1 + \frac{d\vec{r}_1}{dt} \times \vec{r}_2$

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Question No : 36 of 52 Marks: 1 (Budgeted Time 1 Min)

A vector valued function $\vec{r}(t) = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}$ is continuous if ....

Answer (Please select your correct option)

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- Atleast one of its components is continuous.
- All of its components are necessarily differentiable.
- All of its components are continuous.
- Limit exists for all of its components.

correct

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Question No : 37 of 52 Marks: 1 (Budgeted Time 1 Min)

Given a vector valued function $\vec{r}(t) = \frac{1}{(t-3)}\hat{i} + e^t\hat{j}$ and its anti-derivative $\vec{R}(t) = \ln(t-3)\hat{i} + e^t\hat{j}$, then $\int \vec{r}(t)dt =$ ....

Answer (Please select your correct option)

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- $\ln(t-3)\hat{i} + e^t\hat{j} + c$ correct
- $(t-3)\hat{i} + \frac{e^t}{2}\hat{j} + c$
- $(t-3)^{-1}\hat{i} + \frac{e^t}{2}\hat{j} + c$
- $\frac{1}{(t-3)}\hat{i} + e^t\hat{j}$

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Question No : 38 of 52 Marks: 1 (Budgeted Time 1 Min)

For any two vector valued functions $\vec{r}_1(t)$ and $\vec{r}_2(t)$, $\frac{d}{dt}[\vec{r}_1(t).\vec{r}_2(t)] =$ .... where "×" and "•" represent the cross and dot product respectively.

Answer (Please select your correct option)

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- $\vec{r}_1 \cdot \frac{d\vec{r}_2}{dt} - \frac{d\vec{r}_1}{dt} \cdot \vec{r}_2$
- $\vec{r}_1 \cdot \frac{d\vec{r}_2}{dt} + \frac{d\vec{r}_1}{dt} \cdot \vec{r}_2$
- $\vec{r}_1 \times \frac{d\vec{r}_2}{dt} + \frac{d\vec{r}_1}{dt} \times \vec{r}_2$
- $\vec{r}_1 \cdot \frac{d\vec{r}_2}{dt} - \frac{d\vec{r}_1}{dt} \times \vec{r}_2$ correct

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Question No : 39 of 52 Marks: 1 (Budgeted Time 1 Min)

A single curve can be represented by . . . . vector valued function(s).

Answer (Please select your correct option)

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- Two
- Infinitely many
- Single correct
- Three

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Question No : 40 of 52 Marks: 1 (Budgeted Time 1 Min)

A function is said to be smooth if it's derivative is . . . . on any value of its domain.

Answer (Please select your correct option)

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- continuous and non zero
- piecewise continuous
- defined and non zero
- differentiable correct

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Question No : 41 of 52 Marks: 2 (Budgeted Time 4 Min)

Use Wallis cosine formula to evaluate $\int_0^{\frac{\pi}{2}} \cos^6 x \, dx$

Answer (Please click here to Add Answer)

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Question No : 42 of 52 Marks: 2 (Budgeted Time 4 Min)

Prove whether the following function is even, odd or neither.

$f(x) = x^3 + x^2$

Answer (Please click here to Add Answer)

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Question No : 43 of 52 Marks: 2 (Budgeted Time 4 Min)

Let $f(x, y) = \tan^{-1}\frac{y}{x} - y^2 \tan^{-1}\frac{x}{y}$. Is the function defined at (1, 1)? If yes, what is its value and if no, give the reason.

Answer (Please click here to Add Answer)

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Question No : 44 of 52 Marks: 2 (Budgeted Time 4 Min)

Evaluate the following limit.

$$\lim_{t \to \frac{\pi}{4}} \left[(\cos t)\hat{i} + (\sin t)\hat{j} \right]$$

Answer (Please click here to Add Answer)

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Question No : 45 of 52 Marks: 3 (Budgeted Time 6 Min)

Determine the fourier co-efficient a_0, of periodic function defined by

$f(x) = x \qquad 0 < x < 1$

Answer (Please click here to Add Answer)

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Question No : 46 of 52 Marks: 3 (Budgeted Time 6 Min)

Use Wallis sine formula to evaluate $\int_0^{\frac{\pi}{2}} \left(\sin^3 x + \sin^4 x\right) dx$

Answer (Please click here to Add Answer)

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Question No : 47 of 52 Marks: 3 (Budgeted Time 6 Min)

If the order of integration for the integral $\int_0^2 \int_{y^2}^4 y \cos x^2 \, dx dy$ is changed. Find the change in the limits of new integral.

Answer (Please click here to Add Answer)

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Question No : 48 of 52 Marks: 3 (Budgeted Time 6 Min)

What is the arc-length of the curve $\vec{r}(t) = 3\cos t\,\hat{i} + 3\sin t\,\hat{j}$ when $0 \le t \le 2\pi$?

Answer (Please click here to Add Answer)

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Question No : 49 of 52 Marks: 5 (Budgeted Time 10 Min)

Consider a periodic function defined by

$f(x) = 3x \qquad -\pi \le x \le \pi$

(i) Find whether the given function is even or odd?

(ii) Determine Fourier Co-efficients a_0, a_n and b_n

Answer (Please click here to Add Answer)

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Question No : 50 of 52 Marks: 5 (Budgeted Time 10 Min)

Determine whether the following vector field $\vec{F}$ is conservative or not.

$\vec{F}(x, y, z) = (3x + y)\hat{i} + xy^2z\,\hat{j} + xz^2\,\hat{k}$

Answer (Please click here to Add Answer)

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Question No : 51 of 52 Marks: 5 (Budgeted Time 10 Min)

Find Equation of a Tangent plane to the surface $f(x, y, z) = x^2 + 3y + z^3 - 9$ at the point $(2, -1, 2)$

Answer (Please click here to Add Answer)

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Question No : 52 of 52 Marks: 5 (Budgeted Time 10 Min)

Let $\vec{r}(t) = t^2\hat{i} + t\hat{j} + (t^2 - 5)\hat{k}$. Find t, such that $\vec{r}(t)$ and $\vec{r'}(t)$ are perpendicular to each other.

Answer (Please click here to Add Answer)

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MTH301
Solved Final Term Paper 5

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2017

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بِسْمِ ٱللَّهِ ٱلرَّحْمَٰنِ ٱلرَّحِيمِ

In the Name of Allāh, the Most Gracious, the Most Merciful

Paper Pattern

MCQS 40 each 1 mark
Short 4 each 2 marks
Short 4 each 3 marks
long 4 each 5 marks

Question No : 1 of 52 Marks: 1 (Budgeted Time 1 Min)

What is the derivative of following vector-valued function?

$\vec{r}(t) = \left(t^4, \sqrt{t+1}, \frac{3}{t^2}\right)$

Answer (Please select your correct option)

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- $\vec{r}'(t) = \left(4t^3, \frac{1}{\sqrt{t+1}}, \frac{-6}{t^3}\right)$
- $\vec{r}'(t) = \left(4t^3, \frac{1}{2\sqrt{t+1}}, \frac{6}{t^3}\right)$
- $\vec{r}'(t) = \left(4t^4, \frac{1}{2\sqrt{t+1}}, \frac{-6}{t^3}\right)$
- $\vec{r}'(t) = \left(4t^3, \frac{1}{2\sqrt{t+1}}, \frac{-6}{t^3}\right)$ correct

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Question No : 2 of 52 Marks: 1 (Budgeted Time 1 Min)

What is the derivative of following vector-valued function?

$\vec{r}(t) = \left(e^{t^2}, t^2, \sec 2t\right)$

Answer (Please select your correct option)

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- $\vec{r}'(t) = \left(2te^{t^2}, 2t, 2\sec 2t \tan 2t\right)$
- $\vec{r}'(t) = \left(te^{t^2}, 2t, \sec 2t \tan 2t\right)$ correct
- $\vec{r}'(t) = \left(2te^{t^2}, 2t, \tan 2t\right)$
- $\vec{r}'(t) = \left(t^2e^{t^2}, 2t, \sec 2t \tan 2t\right)$

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Question No : 3 of 52 Marks: 1 (Budgeted Time 1 Min)

Evaluate the integral $\int \left[(3t-1)\hat{i}+\sqrt{t}\,\hat{j}\right] dt$

Answer (Please select your correct option) WWW.VirtualAcademyLive.com

- $\int \left[(3t-1)\hat{i}+\sqrt{t}\,\hat{j}\right] dt = \left(\frac{3}{2}t^2-t\right)\hat{i}+\left(\frac{3}{2}t^{\frac{3}{2}}\right)\hat{j}+C$
- $\int \left[(3t-1)\hat{i}+\sqrt{t}\,\hat{j}\right] dt = \left(\frac{3}{2}t^2-t\right)\hat{i}+\left(\frac{2}{3}t^{\frac{3}{2}}\right)\hat{j}+C$ correct
- $\int \left[(3t-1)\hat{i}+\sqrt{t}\,\hat{j}\right] dt = \left(\frac{3}{2}t-t\right)\hat{i}+\left(\frac{2}{3}t^{\frac{3}{2}}\right)\hat{j}+C$
- $\int \left[(3t-1)\hat{i}+\sqrt{t}\,\hat{j}\right] dt = \left(\frac{3}{2}t^2-t\right)\hat{i}+\left(\frac{1}{2}t^{\frac{3}{2}}\right)\hat{j}+C$

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Question No : 4 of 52 Marks: 1 (Budgeted Time 1 Min)

The following differential is exact
$dz = (6xy+2y^2-5)\,dx+(3x^2+4xy-6)\,dy$

Answer (Please select your correct option) WWW.VirtualAcademyLive.com

- True correct
- False

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Question No : 5 of 52 Marks: 1 (Budgeted Time 1 Min)

What is the amplitude of a periodic function defined by $f(x)=4\cos 3x$?

Answer (Please select your correct option) WWW.VirtualAcademyLive.com

- 1
- 3
- 4 correct
- 12

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Question No : 6 of 52 Marks: 1 (Budgeted Time 1 Min)

What is the period of a periodic function defined by $f(x) = 4\sin 2x$?

Answer (Please select your correct option)

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- 2^0
- 4^0
- 8^0
- 180^0

corrrect

- 360^0

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Question No : 7 of 52 Marks: 1 (Budgeted Time 1 Min)

What is the period of a periodic function defined by $f(x) = \sin\frac{x}{2}$?

Answer (Please select your correct option)

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- $\frac{\pi}{2}$
- π
- $\frac{3\pi}{2}$
- 4π

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Question No : 8 of 52 Marks: 1 (Budgeted Time 1 Min)

Match the following periodic function with its graph.

$$f(x) = \begin{cases} 4 & 0 < x < 5 \\ 0 & 5 < x < 8 \end{cases}$$

Answer (Please select your correct option)

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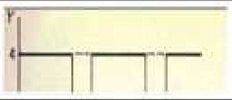
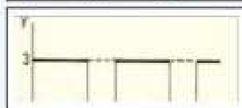
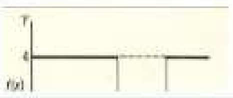
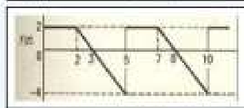
- 
- 
- 

correct

- 

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Question No : 9 of 52 Marks: 1 (Budgeted Time 1 Min)

Match the following periodic function with its graph.

$$f(x)=\begin{cases}4 & 0<x<6\\ 0 & 6<x<8\end{cases}$$

Answer (Please select your correct option)

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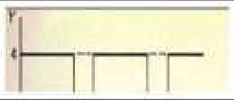
-  correct
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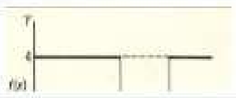
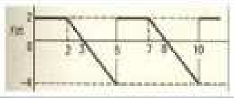
- 
- 

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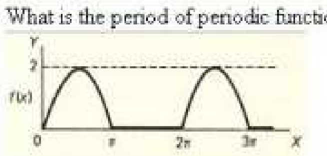
Question No : 10 of 52 Marks: 1 (Budgeted Time 1 Min)

What is the period of periodic function whose graph is as below?



Answer (Please select your correct option)

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- 0
- 2
- π
- 2π correct

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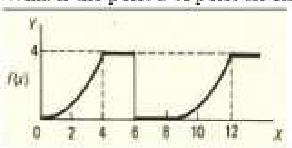
Question No : 11 of 52 Marks: 1 (Budgeted Time 1 Min)

What is the period of periodic function whose graph is as below?



Answer (Please select your correct option)

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- 0
- 4
- 6
- 8 correct

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Question No : 12 of 52 Marks: 1 (Budgeted Time 1 Min)

Let L denotes the Laplace Transform

If $L(F(t)) = f(s)$ where s is a constant and $\lim_{t\to 0}\left(\frac{F(t)}{t}\right)$ exists then which of the following equation holds?

Answer (Please select your correct option)

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- $L\left(\frac{F(t)}{t}\right) = f(s+a)$
- $L\left(\frac{F(t)}{t}\right) = f(s-a)$
- $L\left(\frac{F(t)}{t}\right) = \int_s^\infty f(s)\,ds$ correct
- $L\left(\frac{F(t)}{t}\right) = -\frac{d}{ds}(f(s))$

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Question No : 13 of 52 Marks: 1 (Budgeted Time 1 Min)

The function $f(x) = x^2 \cos 2x$ is -------------

Answer (Please select your correct option)

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- Neither even nor odd
- Odd function
- Even function correct

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Question No : 14 of 52 Marks: 1 (Budgeted Time 1 Min)

The graph of an even function is symmetrical about ---------------

Answer (Please select your correct option)

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- x-axis
- y-axis correct
- origin

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Question No : 15 of 52 Marks: 1 (Budgeted Time 1 Min)

The graph of an odd function is symmetrical about ----------------

Answer (Please select your correct option)

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- x-axis
- y-axis
- origin

correct

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Question No : 16 of 52 Marks: 1 (Budgeted Time 1 Min)

Sign of line integral is reversed when -----------

Answer (Please select your correct option)

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- path of integration is divided into parts.
- path of integration is parallel to y-axis.
- direction of path of integration is reversed.

correct

- path of integration is parallel to x-axis.

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Question No : 17 of 52 Marks: 1 (Budgeted Time 1 Min)

What is laplace transform of the function $F(t)$ if $F(t) = \cos 2t$?

Answer (Please select your correct option)

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- $L(\cos 2t) = \dfrac{2}{s^2+4}$

correct

- $L(\cos 2t) = \dfrac{1}{s-2}$
- $L(\cos 2t) = \dfrac{s}{s^2+4}$
- $L(\cos 2t) = \dfrac{2!}{s^3}$

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Question No : 18 of 52 Marks: 1 (Budgeted Time 1 Min)

What is Laplace Inverse Transform of $\frac{5}{s^2+25}$

Answer (Please select your correct option)

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- $L^{-1}\left\{\frac{5}{s^2+25}\right\} = \sin 5t$ correct
- $L^{-1}\left\{\frac{5}{s^2+25}\right\} = \cos 5t$
- $L^{-1}\left\{\frac{5}{s^2+25}\right\} = \sin 25t$
- $L^{-1}\left\{\frac{5}{s^2+25}\right\} = \cos 25t$

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Question No : 19 of 52 Marks: 1 (Budgeted Time 1 Min)

Evaluate the line integral $\int_C (3x+2y)\,dx + (2x-y)\,dy$ where C is the line segment from (0, 0) to (0, 2).

Answer (Please select your correct option)

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- 1
- 0
- 2 correct
- -2

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Question No : 20 of 52 Marks: 1 (Budgeted Time 1 Min)

Evaluate the line integral $\int_C (xy)\,dx + (1+y^2)\,dy$ where C is the line segment from (1, 0) to (3, 0).

Answer (Please select your correct option)

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- -4
- 0 correct
- 4
- Do not exist

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Question No : 21 of 52 Marks: 1 (Budgeted Time 1 Min)

If p is the period of a function then that function is said to be periodic if $f(x+p)=f(x)$, ______________

Answer (Please select your correct option)

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- For all values of x in the domain of f

correct

- For positive values of x in the domain of f
- For negative values of x in the domain of f

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Question No : 22 of 52 Marks: 1 (Budgeted Time 1 Min)

A vector field is a vector each of whose components is a scalar field

Answer (Please select your correct option)

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- True
- False

correct

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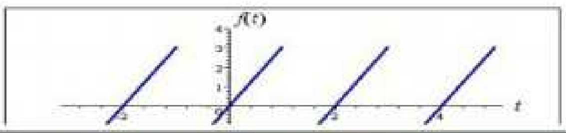
Question No : 23 of 52 Marks: 1 (Budgeted Time 1 Min)



Answer (Please select your correct option)

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- An odd function
- An even function
- Neither even nor odd

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Question No : 24 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following set is the union of sets of rational and irrational numbers?

Answer (Please select your correct option)

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- Set of rational numbers
- Set of integers
- Set of real numbers correct
- Empty set.

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Question No : 25 of 52 Marks: 1 (Budgeted Time 1 Min)

$\sqrt{(x_2-x_1)^2+(y_2-y_1)^2+(z_2-z_1)^2}$ shows .... (x_1, y_1, z_1) and (x_2, y_2, z_2).

Answer (Please select your correct option)

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- Distance between the points correct
- Midpoint of the line joining the points
- Ratio between the points

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Question No : 26 of 52 Marks: 1 (Budgeted Time 1 Min)

Domain of the function $f(x, y) = \sqrt{y-x^2}$ satisfies the condition ....

Answer (Please select your correct option)

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- $y < x^2$
- $y \geq x^2$ correct
- $y \neq x^2$
- Entire space

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Question No : 27 of 52 Marks: 1 (Budgeted Time 1 Min)

If rectangular co-ordinates of a point are $(1, \sqrt{3}, -2)$, then value of "r" in cylindrical co-ordinates is . . . .

Answer (Please select your correct option)

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- $\sqrt{2}$
- 2
- $2\sqrt{2}$
- $-2\sqrt{2}$

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Question No : 28 of 52 Marks: 1 (Budgeted Time 1 Min)

Suppose $f(x, y) = x^3 e^{xy}$. Which of the following options is correct?

Answer (Please select your correct option)

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- $\frac{\partial f}{\partial y} = 3x^3 e^{xy}$
- $\frac{\partial f}{\partial y} = x^3 e^{xy}$
- $\frac{\partial f}{\partial y} = x^4 e^{xy}$ correct
- $\frac{\partial f}{\partial y} = x^3 y e^{xy}$

Made by: Waqar Siddhu

Question No : 29 of 52 Marks: 1 (Budgeted Time 1 Min)

The function decreases most rapidly in the direction of . . . .

Answer (Please select your correct option)

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- $-\nabla f$ correct
- $-|\nabla f|$
- $\nabla f \times \hat{u}$
- $|\nabla f|$

Made by: Waqar Siddhu

Question No : 30 of 52 Marks: 1 (Budgeted Time 1 Min)

Two surfaces are said to be orthogonal at the point of their intersection if their normals at that point are ---------

Answer (Please select your correct option)

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- Parallel
- Perpendicular

  corrrect
- In opposite direction
- Overlapping

Made by: Waqar Siddhu

Question No : 31 of 52 Marks: 1 (Budgeted Time 1 Min)

For a function $f(x,y)$ to have both absolute maximum and minimum, it must be Continuous on . . . . set R.

Answer (Please select your correct option)

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- a closed and bounded
- an open and bounded
- a closed and unbounded
- an open and unbounded

Made by: Waqar Siddhu

Question No : 32 of 52 Marks: 1 (Budgeted Time 1 Min)

If $R = \{(x,y) : 2 \le x \le 4 \, and \, 0 \le y \le 1\}$, then $\iint_R (4xe^{2y})dA =$ . . . .

Answer (Please select your correct option)

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- $\int_0^1 \int_2^4 (4xe^{2y})dydx$
- $\int_0^1 \int_2^4 (4xe^{2y})dxdy$
- $\int_1^4 \int_0^2 (4xe^{2y})dxdy$
- $\int_1^4 \int_0^2 (4xe^{2y})dydx$

Made by: Waqar Siddhu

Question No : 33 of 52 Marks: 1 (Budgeted Time 1 Min)

Let R be a closed region in two dimensional space then the double integral over R calculates. . . . .

Answer (Please select your correct option)

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- Area of R
- Radius of inscribed circle in R
- Distance between two endpoints of R
- None of these

Made by: Waqar Siddhu

Question No : 34 of 52 Marks: 1 (Budgeted Time 1 Min)

Match the following equation in polar co-ordinates with its graph.

$r\ \cos\theta = a$

where a is an arbitrary constant

Answer (Please select your correct option)

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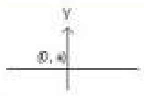
- 

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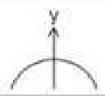
- 

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Question No : 35 of 52 Marks: 1 (Budgeted Time 1 Min)

Polar co-ordinates of a point are $\left(-2, \frac{-3\pi}{2}\right)$. Which of the following is another possible polar co-ordinates representation of this point?

Answer (Please select your correct option)

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- $\left(2, \frac{-\pi}{4}\right)$
- $\left(2, \frac{-\pi}{2}\right)$
- $\left(2, \frac{-\pi}{3}\right)$
- $\left(2, \frac{3\pi}{4}\right)$

Made by: Waqar Siddhu

Question No : 36 of 52 Marks: 1 (Budgeted Time 1 Min)

If the equation of a curve, in polar co-ordinates, remains unchanged after replacing (r, θ) by $(r, -\theta)$ then the curve is said to be symmetric about ....

Answer (Please select your correct option)

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- Initial line
- y-axis
- Pole
- origin

correct

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Question No : 37 of 52 Marks: 1 (Budgeted Time 1 Min)

The graph of curve $r^2 = 4 \cos 2\theta$ is symmetric about ....

Answer (Please select your correct option)

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- Initial line

correct

- y-axis
- both initial line and y-axis
- none of these

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Question No : 38 of 52 Marks: 1 (Budgeted Time 1 Min)

If $p(r,\theta)$ is a point in polar coordinate system, then r is the distance of p from ....

Answer (Please select your correct option)

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- Pole
- Imaginary axis
- None of these
- Polar axis

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Question No : 39 of 52 Marks: 1 (Budgeted Time 1 Min)

The point $p(0,\theta)$ in polar coordinate system lies on . . . .

Answer (Please select your correct option)

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- Polar axis
- y-axis
- Pole
- None of these

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Question No : 40 of 52 Marks: 1 (Budgeted Time 1 Min)

Given the integral $\iint_R f(x,y)dxdy$, after converting to polar coordinates, it will become . . . . where $a \le \theta \le b$ and $c \le r \le d$.

Answer (Please select your correct option)

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- $\int_a^b \int_c^d f(r,\theta) r dr d\theta$
- $\int_a^b \int_c^d f(r,\theta) dr d\theta$
- $\int_a^b \int_c^d f(r,\theta) r d\theta dr$
- $\int_a^b \int_c^d f(r,\theta) d\theta dr$

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Question No : 41 of 52 Marks: 2 (Budgeted Time 4 Min)

Determine whether the following differential is exact or not.

$dz = 4x^3y^3\,dx + 3x^4y^2\,dy$

Answer (Please click here to Add Answer)

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Question No : 42 of 52 Marks: 2 (Budgeted Time 4 Min)

Evaluate

$$\int_{-\pi}^{\pi} \cos nx \, dx$$

where n is an integer other than zero.

Answer (Please click here to Add Answer)

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Question No : 43 of 52 Marks: 2 (Budgeted Time 4 Min)

Prove whether the following function is even, odd or neither.

$f(x) = x^2 - 4\sin x$

Answer (Please click here to Add Answer)

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Question No : 44 of 52 Marks: 2 (Budgeted Time 4 Min)

Given $\vec{a} \times \vec{b} = 3xi + 2yj + xk$ and $\vec{c} = 7xi + 4yk$. Find scalar triple product of these vectors.

Answer (Please click here to Add Answer)

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Question No : 45 of 52 Marks: 3 (Budgeted Time 6 Min)

What is the arc-length of the curve $\vec{r}(t) = (4+3t)\,\hat{i} + (2-2t)\,\hat{j} + (5+t)\,\hat{k}$ when $3 \le t \le 4$?

Answer (Please click here to Add Answer)

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Question No : 46 of 52 Marks: 3 (Budgeted Time 6 Min)

Find $div\ \vec{F}$, if $\vec{F} = (3x+y)\hat{i} + xy^2z\,\hat{j} + (xz^2)\hat{k}$

Answer (Please click here to Add Answer)

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Question No : 47 of 52 Marks: 3 (Budgeted Time 6 Min)

Determine the Fourier co-efficient a_0 of the periodic function defined below:

$f(x) = 2x+1 \qquad 0 < x < 2$

Answer (Please click here to Add Answer)

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Question No : 48 of 52 Marks: 3 (Budgeted Time 6 Min)

A line, in three dimensional space, passes through the point $(3, -4, 2)$ and parallel to the vector $\vec{n} = 4\hat{i} + 3\hat{j} + 6\hat{k}$. Write down the equation of this line in parametric and symmetric form.

Answer (Please click here to Add Answer)

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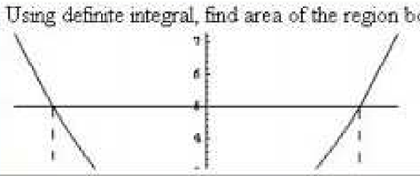
Question No : 49 of 52 Marks: 5 (Budgeted Time 10 Min)

Using definite integral, find area of the region bounded by the curves of $y = x^2 + 1$ *and* $y = 5$



Answer (Please click here to Add Answer)

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Question No : 50 of 52 Marks: 5 (Budgeted Time 10 Min)

Show that Laplace transform of the function

$f(t) = 1$ is

$\frac{1}{s}$ where s is a constant for the integration and s > 0.

Answer (Please click here to Add Answer)

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Question No : 51 of 52 Marks: 5 (Budgeted Time 10 Min)

Determine whether the following vector field $\vec{F}$ is conservative or not.

$\vec{F}(x, y, z) = (4x - z)\hat{i} + (3y + z)\hat{j} + (y - x)\hat{k}$

Answer (Please click here to Add Answer)

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Question No : 52 of 52 Marks: 5 (Budgeted Time 10 Min)

Consider the point $(-5, 5, 6)$ in rectangular coordinate system. Convert it into Spherical coordinates.

Answer (Please click here to Add Answer)

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FINALTERM EXAMINATION

Spring 2010

MTH301- Calculus II

Time: 90 min

Marks: 60

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Q No.	1	2	3	4	5	6	7	8	Total
Marks									
Q No.	9	10	11	12	13	14	15	16	
Marks									
Q No.	17	18	19	20	21	22	23	24	
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Q No.	25	26	27	28	29	30	31	32	
Marks									
Q No.	33	34	35	36	37	38	39		
Marks									

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Question No: 1 (Marks: 1) - Please choose one

Intersection of two straight lines is -------------

► Surface

► Curve

► Plane

► ***Point***

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Question No: 2 (Marks: 1) - Please choose one

Plane is a --------------- surface.

► One-dimensional

► ***Two-dimensional***

► Three-dimensional

► Dimensionless

Question No: 3 (Marks: 1) - Please choose one

Let w = f(x, y, z) and x = g(r, s), y = h(r, s), z = t(r, s) then by chain rule

$$\frac{\partial w}{\partial r} =$$

- $\frac{\partial w}{\partial x}\frac{\partial x}{\partial r}+\frac{\partial w}{\partial y}\frac{\partial y}{\partial r}+\frac{\partial w}{\partial z}\frac{\partial z}{\partial r}$
- $\frac{\partial w}{\partial r}\frac{\partial x}{\partial r}+\frac{\partial w}{\partial r}\frac{\partial y}{\partial r}+\frac{\partial w}{\partial r}\frac{\partial z}{\partial r}$
- $\frac{\partial w}{\partial x}\frac{\partial x}{\partial r}\frac{\partial x}{\partial s}+\frac{\partial w}{\partial y}\frac{\partial y}{\partial r}\frac{\partial y}{\partial s}+\frac{\partial w}{\partial z}\frac{\partial z}{\partial r}\frac{\partial z}{\partial s}$
- $\frac{\partial w}{\partial r}\frac{\partial r}{\partial x}+\frac{\partial w}{\partial r}\frac{\partial r}{\partial y}+\frac{\partial w}{\partial r}\frac{\partial r}{\partial z}$

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Question No: 4 (Marks: 1) - Please choose one

What are the parametric equations that correspond to the following vector equation?

$$\vec{r}(t) = \sin^2 t\,\hat{i} + (1-\cos 2t)\,\hat{j}$$

- $x = \sin^2 t \quad , \quad y = 1-\cos 2t \quad , \quad z = 0$
- $y = \sin^2 t \quad , \quad x = 1-\cos 2t \quad , \quad z = 0$

► $x = \sin^2 t$, $y = 1 - \cos 2t$, $z = 1$

► $x = \sin^2 t$, $y = \cos 2t$, $z = 1$

Question No: 5 (Marks: 1) - Please choose one

What are the parametric equations that correspond to the following vector equation?

$r(t) = (2t-1)\hat{i} - 3\sqrt{t}\,\hat{j} + \sin 3t\,\hat{k}$

► $z = 2t-1$, $x = -3\sqrt{t}$, $y = \sin 3t$

► $y = 2t-1$, $x = -3\sqrt{t}$, $z = \sin 3t$

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► $x = 2t-1$, $z = -3\sqrt{t}$, $y = \sin 3t$

► $x = 2t-1$, $y = -3\sqrt{t}$, $z = \sin 3t$

Question No: 6 (Marks: 1) - Please choose one

What is the derivative of following vector-valued function?

$\vec{r}(t) = (\cos 5t, \tan t\ , 6\sin t)$

► $\vec{r}'(t) = \left(\frac{\sin 5t}{5}, \sec t\ , 6\cos t \right)$

▶ $\vec{r}'(t) = (\frac{-\sin 5t}{5}, \sec t\ , 6\cos t)$

▶ $\vec{r}'(t) = (-5\sin 5t, \sec^2 t\ , 6\cos t)$

▶ $\vec{r}'(t) = (\sin 5t, \sec^2 t\ , -6\cos t)$

Question No: 7 (Marks: 1) - Please choose one

What is the derivative of following vector-valued function?

$\vec{r}(t) = \left(t^4, \sqrt{t+1}\ , \frac{3}{t^2} \right)$

▶ $\vec{r}'(t) = \left(4t^3, \frac{1}{\sqrt{t+1}}, \frac{-6}{t^3} \right)$

▶ $\vec{r}'(t) = \left(4t^3, \frac{1}{2\sqrt{t+1}}, \frac{6}{t^3} \right)$

▶ $\vec{r}'(t) = \left(4t^4, \frac{1}{2\sqrt{t+1}}, \frac{-6}{t^3} \right)$

▶ $\vec{r}'(t) = \left(4t^3, \frac{1}{2\sqrt{t+1}}, \frac{-6}{t^3} \right)$

Question No: 8 (Marks: 1) - Please choose one

The following differential is exact

$dz = (x^2 y + y)\ dx - x\ dy$

► True

► False

Question No: 9 (Marks: 1) - Please choose one

Which one of the following is correct Wallis Sine formula when n is even and $n \geq 2$?

► $\int_0^{\frac{\pi}{2}} \sin^n x\, dx = \frac{\pi}{2} \frac{(n-1)}{n} \frac{(n-3)}{(n-2)} \frac{(n-5)}{(n-4)} - - - - - - - - \frac{5}{6} \frac{3}{4} \frac{1}{2}$ page #182

► $\int_0^{\frac{\pi}{2}} \sin^n x\, dx = \frac{(n-1)}{n} \frac{(n-3)}{(n-2)} \frac{(n-5)}{(n-4)} - - - - - - - - \frac{6}{7} \frac{4}{5} \frac{2}{3}$

► $\int_0^{\frac{\pi}{2}} \sin^n x\, dx = \frac{\pi}{2} \frac{(n)}{(n-1)} \frac{(n-2)}{(n-3)} \frac{(n-4)}{(n-5)} - - - - - - - - \frac{6}{5} \frac{4}{3} \frac{2}{1}$

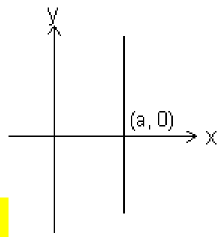
► $\int_0^{\frac{\pi}{2}} \sin^n x\, dx = \frac{(n)}{(n-1)} \frac{(n-2)}{(n-3)} \frac{(n-4)}{(n-5)} - - - - - - - - \frac{6}{5} \frac{4}{3} \frac{2}{1}$

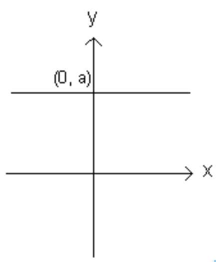
Question No: 10 (Marks: 1) - Please choose one

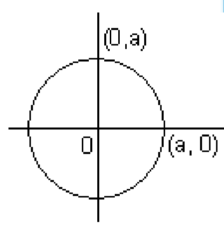
Match the following equation in polar co-ordinates with its graph.

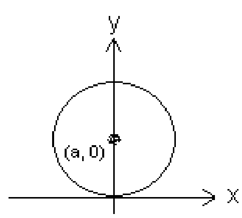
$r\ \cos\theta = a$

where a is an arbitrary constant

▶ 

▶ 

▶ 

▶ 

Question No: 11 (Marks: 1) - Please choose one

If the equation of a curve, in polar co-ordinates, remains unchanged after replacing (r, θ) by $(r, \pi - \theta)$ then the curve is said to be symmetric about which of the following?

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► Initial line

► Y-axis

► Pole

Question No: 12 (Marks: 1) - Please choose one

If the equation of a curve, in polar co-ordinates, remains unchanged after replacing (r, θ) by $(-r, \theta)$ then the curve is said to be symmetric about which of the following?

► Initial line

► y-axis

► *Pole*

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Question No: 13 (Marks: 1) - Please choose one

What is the amplitude of a periodic function defined by $f(x) = \sin\frac{x}{3}$?

► 0

► 1

▶ $\frac{1}{3}$

▶ Does not exist

Question No: 14 (Marks: 1) - Please choose one

What is the period of a periodic function defined by $f(x) = 4\cos 3x$ **?**

▶ $\frac{\pi}{4}$

▶ $\frac{\pi}{3}$

▶ $\frac{2\pi}{3}$

▶ π

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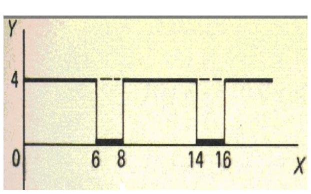
Question No: 15 (Marks: 1) - Please choose one

Match the following periodic function with its graph.

$$f(x) = \begin{cases} 3 & 0 < x < 4 \\ 0 & 4 < x < 6 \end{cases}$$

▶ 

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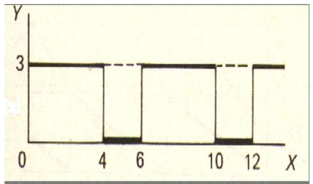
► 

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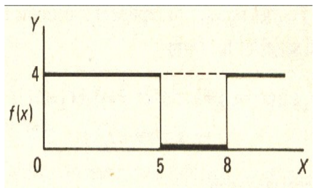
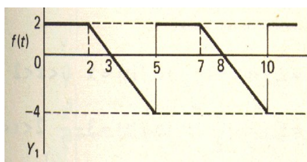
► 

► 

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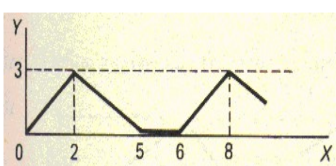
Question No: 16 (Marks: 1) - Please choose one

What is the period of periodic function whose graph is as below?



► 2

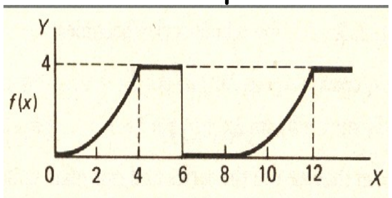
► 5

► 6

► 8

Question No: 17 (Marks: 1) - Please choose one

What is the period of periodic function whose graph is as below?



► 0

► 4

► 6

► **8**

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Question No: 18 (Marks: 1) - Please choose one

Let L **denotes the Laplace Transform.**

If $L\{F(t)\} = f(s)$ **where** s **is a constant.and** $\lim_{t\to 0}\left(\frac{F(t)}{t}\right)$ **exists then which of the following equation holds?**

► $L\left(\frac{F(t)}{t}\right) = f(s+a)$

► $L\left(\frac{F(t)}{t}\right) = f(s-a)$

► $L\left(\frac{F(t)}{t}\right)=\int_{s}^{\infty} f(s)\, ds$

► $L\left(\frac{F(t)}{t}\right)=-\frac{d}{ds}\{f(s)\}$

Question No: 19 (Marks: 1) - Please choose one

Which of the following is Laplace inverse transform of the function $f(s)$ defined by $f(s)=\frac{3}{s-2}-\frac{2}{s}$?

► $3te^{2t}-2$

► $3e^{2t}-2t$

► $3e^{2t}-2$

► None of these.

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Question No: 20 (Marks: 1) - Please choose one

Let (x_1, y_1, z_1) and (x_2, y_2, z_2) be any two points in three dimensional space. What does the formula $\sqrt{(x_2-x_1)^2+(y_2-y_1)^2+(z_2-z_1)^2}$ calculates?

► Distance between these two points

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► Midpoint of the line joining these two points

► Ratio between these two points

Question No: 21 (Marks: 1) - Please choose one

Let the functions $P(x, y)$ and $Q(x, y)$ are finite and continuous inside and at the boundary of a closed curve C in the xy-plane. If $(P\,dx + Q\,dy)$ is an exact differential then

$\oint_C (P\,dx + Q\,dy) =$

► Zero

► One

► Infinite

Question No: 22 (Marks: 1) - Please choose one

What is Laplace transform of the function $F(t)$ if $F(t) = t$?

► $L\{t\} = \frac{1}{s}$

► $L\{t\} = \frac{1}{s^2}$

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► $L\{t\} = e^{-s}$

► $L\{t\} = s$

Question No: 23 (Marks: 1) - Please choose one

What is the value of $L\{e^{5t}\}$ if L denotes laplace transform?

► $L\{e^{5t}\} = \frac{1}{s-5}$

► $L\{e^{5t}\} = \frac{s}{s^2+25}$

► $L\{e^{5t}\} = \frac{5}{s^2+25}$

► $L\{e^{5t}\} = \frac{5!}{s^6}$

Question No: 24 (Marks: 1) - Please choose one

Evaluate the line integral $\int_C (3x+2y)\,dx+(2x-y)\,dy$ **where** ***C*** **is the line segment from (0, 0) to (0, 2).**

► 1

► 0

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► **2**

► -2

Question No: 25 (Marks: 1) - Please choose one

Evaluate the line integral $\int_C (2x+y)\,dx+(x^2-y)\,dy$ where *C* is the line segment from (0, 0) to (2, 0).

► **0**

► -4

► 4

► Do not exist

Question No: 26 (Marks: 1) - Please choose one

Which of the following are direction ratios for the line joining the points $(1, 3, 5)$ and $(2, -1, 4)$?

► 3, 2 and 9

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► 1, -4 and -1

► 2, -3 and 20

► 0.5, -3 and 5/4

Question No: 27 (Marks: 1) - Please choose one

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$If\ R = \{(x, y) / 0 \le x \le 2\ and\ 1 \le y \le 4\}, then$

$\iint_R (6x^2 + 4xy^3)dA =$

► $\int_1^4 \int_0^2 (6x^2 + 4xy^3)dydx$

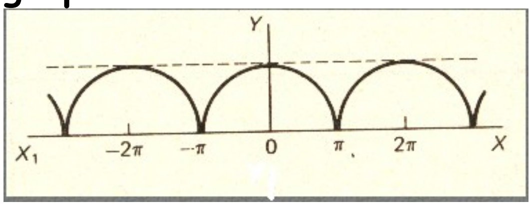
► $\int_0^2 \int_1^4 (6x^2 + 4xy^3)dxdy$

► $\int_1^4 \int_0^2 (6x^2 + 4xy^3)dxdy$

► $\int_2^4 \int_0^1 (6x^2 + 4xy^3)dxdy$

Question No: 28 (Marks: 1) - Please choose one

Which of the following is true for a periodic function whose graph is as below?

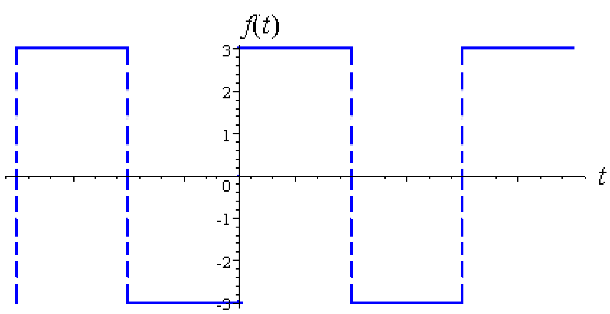


► Even function page209

► Odd function

► Neither even nor odd function

Question No: 29 (Marks: 1) - Please choose one



Which of the following is true for a function whose graph is given above.

► An odd function pg 212

► An even function

► Neither even nor odd

Question No: 30 (Marks: 1) - Please choose one

At each point of domain, the function ----------------

► **Is defined**

► Is continuous

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► Is infinite

► Has a limit

Question No: 31 (Marks: 2)

Determine whether the following differential is exact or not.

$dz = 4x^3y^3\ dx + 3x^4y^2\ dy$

Solution:

$$dz = 4x^3y^3 \ dx + 3x^4y^2 \ dy$$

$$\frac{\partial p}{\partial y} = 12x^3y^2$$

$$\frac{\partial Q}{\partial X} = 12x^3y^2$$

$$\frac{\partial p}{\partial y} = \frac{\partial Q}{\partial X}$$

yes

Question No: 32 (Marks: 2)

Evaluate

$$\int_{-\pi}^{\pi} \sin \ nx \ dx$$

where n is an integer other than zero.

Solution:

$$\int_{-\pi}^{\pi} \sin \ nx \ dx$$

$$= \left[\frac{-\cos nx}{n}\right]_{-\pi}^{\pi}$$

$$= \left[\frac{-\cos n\pi}{n} + \frac{\cos n\pi}{n}\right]$$

$$= \frac{1}{n}(-\cos n\pi + \cos n\pi)$$

$$= 0$$

Question No: 33 (Marks: 2)

Find Laplce transform of the function $F(t)$ if $F(t) = e^{3t}$

Solution:

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$$L(e^{3t}) = \int_0^{\infty} e^{3t} - e^{-st}$$

$$= \int_0^{\infty} e^{-(s-3)t}.dt$$

$$= \{\frac{e^{-(s-3)t}}{-(s-3)}\} \lim 0 - \infty$$

$$= \frac{-1}{s-3}(\frac{1}{e^{(s-3)t}})$$

$$= \frac{-1}{s-3}(0-1)$$

$$= \frac{1}{s-3} \text{......Ans}$$

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Question No: 34 (Marks: 3)

Determine the Fourier co-efficient a_0 of the periodic function defined below:

$f(x) = 2x+1 \qquad 0 < x < 2$

Solution:

$$a_\circ = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)dx$$

$$f(x) = (2x+1)$$

$$(0,2)$$

$$= \int_0^2 (2x+1)dx$$

$$= \left[x^2 + x\right]_0^2$$

$$= 6$$

Question No: 35 (Marks: 3)

Determine whether the following differential is exact or not.

$$dz = (3x^2e^{2y} - 2y^2e^{3x})\ dx + (2x^3e^{2y} - 2ye^{3x})\,dy$$

Solution:

dz = Pdx + Qdy

Therefore,

For dz to be an exact differential it must satisfy $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$

But this test fails becuase $\frac{\partial P}{\partial y} \neq \frac{\partial Q}{\partial x}$

Not Exact

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Question No: 36 (Marks: 3)

Use Wallis sine formula to evaluate $\int_0^{\frac{\pi}{2}} \left(\sin^3 x + \sin^5 x\right) dx$

Solution:

$$\int_0^{\frac{\pi}{2}} \sin^3 x dx$$

$$= \frac{n-1}{n}.$$

$$= \frac{3-1}{3}$$

$$= \frac{2}{3}$$

$$\int_0^{\frac{\pi}{2}} \sin^5 x dx$$

$$= \frac{n-1}{n}.\frac{n-3}{n-2}$$

$$= \frac{5-1}{5}.\frac{5-3}{5-2}$$

$$= \frac{4}{5}.\frac{2}{3}$$

$$\int_0^{\frac{\pi}{2}} \left(\sin^3 x + \sin^5 x\right) dx$$

$$= \frac{2}{3} + \frac{4}{5}.\frac{2}{3}$$

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Question No: 37 (Marks: 5)

Evaluate the following line integral which is independent of path.

$$\int_{(0,0)}^{(3,2)} (2xe^y)\, dx + (x^2 e^y)\, dy$$

Solution:

$$p = \frac{\partial z}{\partial x} = 2e^{y} \qquad \int 2e^{y}\,dx$$

$$Q = \frac{\partial z}{\partial y} = x^{2}e^{y} \qquad \int x^{2}e^{y}\,dy$$

$$z = \int_{(0,0)}^{(3,2)} 2xe^{y} + x^{2}ye^{y}$$

$$z = 6e^{2} + 18e^{2}$$

$$z = 24e^{2}$$

Question No: 38 (Marks: 5)

Determine the Fourier coefficients b_n for a periodic function $f(t)$ of period 2 defined by

$$f(t) = \begin{cases} 4\,(1 + t) & -1 < t < 0 \\ 0 & 0 < t < 1 \end{cases}$$

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Solution:

$$bn = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx dx$$

$$= \frac{1}{\pi}\int_{-1}^{1} 4(1+t)\sin nx dx$$

$$= \frac{1}{\pi}\left[\frac{-4(1+t)\cos nx}{n}\right]_{-1}^{1}$$

$$= \frac{-4(1+t)}{\pi n}\left[\cos n(1) - \cos n(-1)\right]$$

$$= \frac{-4(1+t)}{\pi n}(\cos n + \cos n)$$

Question No: 39 (Marks: 5)

Determine whether the following vector field $\vec{F}$ is conservative or not.

$$\vec{F}(x,\ y,\ z) = (4x - z)\hat{i} + (3y + z)\hat{j} + (y - x)\hat{k}$$

……………………………………

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ASSALAM O ALAIKUM

All Dearz fellows

ALL IN ONE MTH301

Final term PAPERS & MCQz

Created BY Farhan & Ali

BS (cs) 2nd sem

Hackers Group

From Mandi Bahauddin

Remember us in your prayers

Mindhacker124@gmail.com

Hearthacker124@gmail.com

FINALTERM EXAMINATION

Spring 2010

MTH301- Calculus II (Session - 2)

Time: 90 min

Marks: 60

Student Info	
StudentID:	$$
Center:	OPKST
ExamDate:	19 Aug 2010

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Q No.	1	2	3	4	5	6	7	8	Total
Marks									
Q No.	9	10	11	12	13	14	15	16	
Marks									
Q No.	17	18	19	20	21	22	23	24	
Marks									
Q No.	25	26	27	28	29	30	31	32	
Marks									
Q No.	33	34	35	36	37	38	39		
Marks									

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Question No: 1 (Marks: 1) - Please choose one

------------------- planes intersect at right angle to form three dimensional space.

- ► **Three**
- ► Four
- ► Eight
- ► Twelve

Question No: 2 (Marks: 1) - Please choose one

If the positive direction of x, y axes are known then ------------ the positive direction of z-axis.

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- ► Horizontal rightward direction is
- ► Vertical upward direction is
- ► **Left hand rule tells**
- ► Right hand rule tells

Question No: 3 (Marks: 1) - Please choose one

What is the distance between points (3, 2, 4) and (6, 10, -1)?

- ► $7\sqrt{2}$
- ► $2\sqrt{6}$
- ► $\sqrt{34}$
- ► $7\sqrt{3}$

Question No: 4 (Marks: 1) - Please choose one

The equation $ax+by+cz+d=0$, where a,b,c,d are real numbers, is the general equation of which of the following?

- ► Plane page # 12
- ► Line
- ► Curve
- ► Circle

Question No: 5 (Marks: 1) - Please choose one

The spherical co-ordinates of a point are $\left(\sqrt{3}, \frac{\pi}{3}, \frac{\pi}{2}\right)$. What are its cylindrical co-ordinates?

- ► $\left(\frac{\sqrt{3}}{2}, \frac{3}{2}, 0\right)$
- ► $\left(\sqrt{3}\cos\frac{\pi}{3}, \sqrt{3}\sin\frac{\pi}{3}, 0\right)$
- ► $\left(\sqrt{3}\sin\frac{\pi}{3}, \frac{\pi}{2}, \sqrt{3}\cos\frac{\pi}{3}\right)$
- ► $\left(\sqrt{3}, \frac{\pi}{3}, 0\right)$

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Question No: 6 (Marks: 1) - Please choose one

Domain of the function $f(x, y)=\sqrt{y-x^2}$ is

- ► $y<x^2$
- ► $y \geq x^2$
- ► $y \neq x^2$
- ► Entire space

Question No: 7 (Marks: 1) - Please choose one

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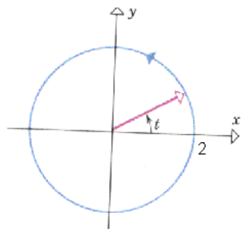
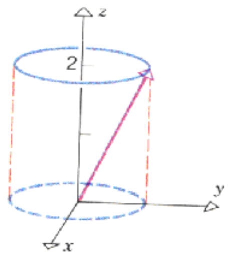
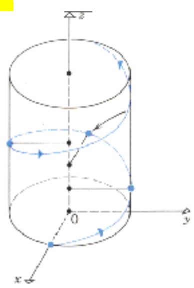
Match the following vector-valued function with its graph.

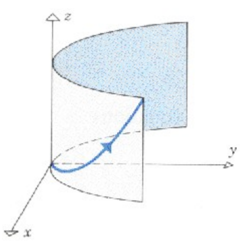
$r(t) = \cos t\,\hat{i} + \sin t\,\hat{j} + 2\,\hat{k}$ And $0 \le t \le 2\pi$

▶ 

▶ 

Correct

▶ 

▶ 

Question No: 8 (Marks: 1) - Please choose one

What are the parametric equations that correspond to the following vector equation?

$\vec{r}(t) = \sin^2 t\,\hat{i} + (1 - \cos 2t)\,\hat{j}$

- ▶ $x = \sin^2 t$, $y = 1 - \cos 2t$, $z = 0$
- ▶ $y = \sin^2 t$, $x = 1 - \cos 2t$, $z = 0$

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- $x=\sin^2 t$, $y=1-\cos 2t$, $z=1$
- $x=\sin^2 t$, $y=\cos 2t$, $z=1$

Question No: 9 (Marks: 1) - Please choose one

What are the parametric equations that correspond to the following vector equation?

$r(t)=(2t-1)\hat{i}-3\sqrt{t}\,\hat{j}+\sin 3t\,\hat{k}$

- $z=2t-1$, $x=-3\sqrt{t}$, $y=\sin 3t$
- $y=2t-1$, $x=-3\sqrt{t}$, $z=\sin 3t$
- $x=2t-1$, $z=-3\sqrt{t}$, $y=\sin 3t$
- $x=2t-1$, $y=-3\sqrt{t}$, $z=\sin 3t$

Question No: 10 (Marks: 1) - Please choose one

Is the following vector-valued function $\vec{r}(t)$ continuous at $t=1$? If not, why?

$\vec{r}(t)=\left(\frac{t+1}{t-1}, t^2 , 2t\right)$

- $\vec{r}(t)$ is continuous at $t=1$
- $\vec{r}(1)$ is not defined
- $\vec{r}(1)$ is defined but $\lim_{t\to 1}\vec{r}(t)$ does not exist
- $\vec{r}(1)$ **is defined and** $\lim_{t\to 1}\vec{r}(t)$ **exists but these two numbers are not equal.**

Question No: 11 (Marks: 1) - Please choose one

Which one of the following is correct Wallis Sine formula when n is even and $n\geq 2$?

▶ $\int_0^{\frac{\pi}{2}} \sin^n x\, dx = \frac{\pi}{2}\frac{(n-1)}{n}\frac{(n-3)}{(n-2)}\frac{(n-5)}{(n-4)}-------\frac{5}{6}\frac{3}{4}\frac{1}{2}$ page183

▶ $\int_0^{\frac{\pi}{2}} \sin^n x\, dx = \frac{(n-1)}{n}\frac{(n-3)}{(n-2)}\frac{(n-5)}{(n-4)}-------\frac{6}{7}\frac{4}{5}\frac{2}{3}$

▶ $\int_0^{\frac{\pi}{2}} \sin^n x\, dx = \frac{\pi}{2}\frac{(n)}{(n-1)}\frac{(n-2)}{(n-3)}\frac{(n-4)}{(n-5)}-------\frac{6}{5}\frac{4}{3}\frac{2}{1}$

▶ $\int_0^{\frac{\pi}{2}} \sin^n x\, dx = \frac{(n)}{(n-1)}\frac{(n-2)}{(n-3)}\frac{(n-4)}{(n-5)}-------\frac{6}{5}\frac{4}{3}\frac{2}{1}$

Question No: 12 (Marks: 1) - Please choose one

Which one of the following is correct Wallis Cosine formula when n is odd and $n \geq 3$?

▶ $\int_0^{\frac{\pi}{2}} \cos^n x\, dx = \frac{\pi}{2}\frac{(n-1)}{n}\frac{(n-3)}{(n-2)}\frac{(n-5)}{(n-4)}-------\frac{5}{6}\frac{3}{4}\frac{1}{2}$

▶ $\int_0^{\frac{\pi}{2}} \cos^n x\, dx = \frac{\pi}{2}\frac{(n)}{(n-1)}\frac{(n-2)}{(n-3)}\frac{(n-4)}{(n-5)}-------\frac{6}{5}\frac{4}{3}\frac{2}{1}$

▶ $\int_0^{\frac{\pi}{2}} \cos^n x\, dx = \frac{(n)}{(n-1)}\frac{(n-2)}{(n-3)}\frac{(n-4)}{(n-5)}-------\frac{6}{5}\frac{4}{3}\frac{2}{1}$

▶ $\int_0^{\frac{\pi}{2}} \cos^n x\, dx = \frac{(n-1)}{n}\frac{(n-3)}{(n-2)}\frac{(n-5)}{(n-4)}-------\frac{6}{7}\frac{4}{5}\frac{2}{3}$ page183

Question No: 13 (Marks: 1) - Please choose one

If the equation of a curve, in polar co-ordinates, remains unchanged after replacing (r, θ) by $(r, \pi-\theta)$ then the curve is said to be symmetric about which of the following?

► Initial line
► y-axis
► Pole

Question No: 14 (Marks: 1) - Please choose one

If $a > 0$, then the equation, in polar co-ordinates, of the form $r^2 = a^2 \cos 2\theta$ represent which of the following family of curves?

► Leminscate page 130
► Cardiods
► Rose curves
► Spiral

Question No: 15 (Marks: 1) - Please choose one

What is the period of a periodic function defined by $f(x) = \sin\frac{x}{2}$?

► $\frac{\pi}{2}$
► π
► $\frac{3\pi}{2}$
► 4π

Question No: 16 (Marks: 1) - Please choose one

What is the period of periodic function whose graph is as below?

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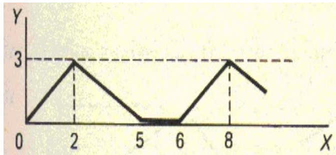
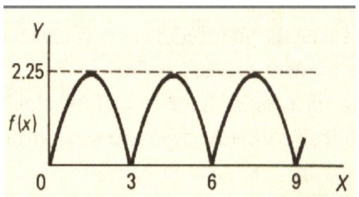


- ▶ 2
- ▶ 5
- ▶ 6
- ▶ 8

Question No: 17 (Marks: 1) - Please choose one

What is the period of periodic function whose graph is as below?



- ▶ 0
- ▶ 2.25
- ▶ 3
- ▶ 6

Question No: 18 (Marks: 1) - Please choose one

Let L **denotes the Laplace Transform.**

If $L\{F(t)\}=f(s)$ **where** s **is a constant, then which of the following equation holds?**

- ▶ $L\{t\ F(t)\}=-\frac{d}{ds}\{f(s)\}$
- ▶ $L\{t\ F(t)\}=f(s+t)$
- ▶ $L\{t\ F(t)\}=f(s)$
- ▶ $L\{t F(t)\}=\int_s^{\infty} f(s)\ ds$

Question No: 19 (Marks: 1) - Please choose one

The graph of an odd function is symmetrical about ----------------

- ► x-axis
- ► y-axis
- ► **origin**

Question No: 20 (Marks: 1) - Please choose one

Consider the function $f(x, y, z) = \sqrt{1 - x^2 - y^2 - z^2}$. What is the value of $f\left(0, \frac{1}{2}, \frac{1}{2}\right)$□

- ► $f\left(0, \frac{1}{2}, \frac{1}{2}\right) = \sqrt{\frac{1}{2}}$
- ► $f\left(0, \frac{1}{2}, \frac{1}{2}\right) = 2$
- ► $f\left(0, \frac{1}{2}, \frac{1}{2}\right) = \frac{1}{2}$
- ► $f\left(0, \frac{1}{2}, \frac{1}{2}\right) = 0$

Question No: 21 (Marks: 1) - Please choose one

The path of integration of a line integral must be -------------

- ► straight and single-valued
- ► **continuous and single-valued**
- ► straight and multiple-valued
- ► continuous and multiple-valued

Question No: 22 (Marks: 1) - Please choose one

Sign of line integral is reversed when -----------

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► path of integration is divided into parts.
► path of integration is parallel to y-axis.
► **direction of path of integration is reversed.**
► path of integration is parallel to x-axis.

Question No: 23 (Marks: 1) - Please choose one

Let the functions $P(x, y)$ and $Q(x, y)$ are finite and continuous inside and at the boundary of a closed curve C in the xy-plane.

If $(P\,dx + Q\,dy)$ is an exact differential then

$\oint_C (P\,dx + Q\,dy) =$

► **Zero**

► **One**

► Infinite

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Question No: 24 (Marks: 1) - Please choose one

What is the value of $L\{e^{5t}\}$ if L denotes laplace transform?

► $L\{e^{5t}\} = \dfrac{1}{s-5}$

► $L\{e^{5t}\} = \dfrac{s}{s^2+25}$

► $L\{e^{5t}\} = \dfrac{5}{s^2+25}$

► $L\{e^{5t}\} = \dfrac{5!}{s^6}$

Question No: 25 (Marks: 1) - Please choose one

What is laplace transform of the function $F(t)$ if $F(t) = \sin 3t$?

► $L\{\sin 3t\} = \dfrac{3}{s^2+9}$

► $L\{\sin 3t\} = \dfrac{s}{s^2+9}$

- $L\{\sin 3t\} = \frac{1}{s-3}$
- $L\{\sin 3t\} = \frac{3!}{s^4}$

Question No: 26 (Marks: 1) - Please choose one

If L denotes laplace transform then

$L\{te^{5t}\} =$

- $L\{te^{5t}\} = \frac{1}{s^2-5}$
- $L\{te^{5t}\} = \frac{1}{s^2+5}$
- $L\{te^{5t}\} = \frac{1}{(s+5)^2}$
- $L\{te^{5t}\} = \frac{1}{(s-5)^2}$

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Question No: 27 (Marks: 1) - Please choose one

Evaluate the line integral $\int_C (3x+2y)\,dx+(2x-y)\,dy$ **where C is the line segment from (0, 0) to (0, 2).**

- 1
- 0
- 2
- -2

Question No: 28 (Marks: 1) - Please choose one

Evaluate the line integral $\int_C (2x+y)\,dx+(x^2-y)\,dy$ where C is the line segment from (0, 0) to (0, 2).

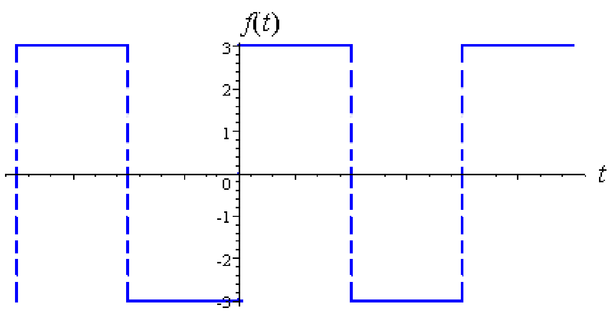
► -4
► **-2**
► 0
► 2

Question No: 29 (Marks: 1) - Please choose one

Divergence of a vector function is always a ------------

► Scalar
► **Vector**

Question No: 30 (Marks: 1) - Please choose one



Which of the following is true for a function whose graph is given above

► **An odd function**

► An even function

► Neither even nor odd

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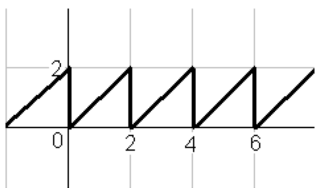
Question No: 31 (Marks: 2)

Does the following limit exist? If yes find its value, if no give reason

$$\lim_{t \to 0}\left[(e^{2t}+5)\hat{i}+(t^2+2t-3)\hat{j}+\left(\frac{1}{t}\right)\hat{k}\right]$$

Question No: 32 (Marks: 2)

Define the periodic function whose graph is shown below.



Question No: 33 (Marks: 2)

Find Laplace Transform of the function $F(t)$ if $F(t)=t^4$

Solution:

The Laplace transform of the given function will be:

$$f(t)=t^4$$
$$L\{t^4\}=\frac{4!}{s^5}$$

Question No: 34 (Marks: 3)

Determine whether the following differential is exact or not.

$$dz=(4x^3y+2xy^3)\,dx+(x^4+3x^2y^2)\,dy$$

Question No: 35 (Marks: 3)

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Use Wallis sine formula to evaluate $\int_0^{\frac{\pi}{2}} \left(\sin^3 x + \sin^5 x\right) dx$

Solution:

$$\int_0^{\frac{\pi}{2}} \sin^8 x dx = \frac{7}{8}.\frac{5}{6}.\frac{3}{4}.\frac{1}{2} - - - - - - - - - \frac{\pi}{2}$$

Question No: 36 (Marks: 3)

Find Laplace transform of the function $F(t)$ if

$F(t) = e^{2t} \sin 3t$

Solution:

Laplace transform will be

$L(t) = e^{2t}..............1$

$= \frac{1}{s-2}$

$L(t) = \sin 3t..........2$

$L(t) = \frac{a}{s^2 + 3^2}$

$L(t) = \frac{a}{s^2 + 9}$

$Combining,$

$L(t) = (\frac{1}{s-2})(\frac{a}{s^2 + 9})$

Question No: 37 (Marks: 5)

Using definite integral, find area of the region that is enclosed between the curves $y = x^2 \ \ and \ \ y = \sqrt{x}$

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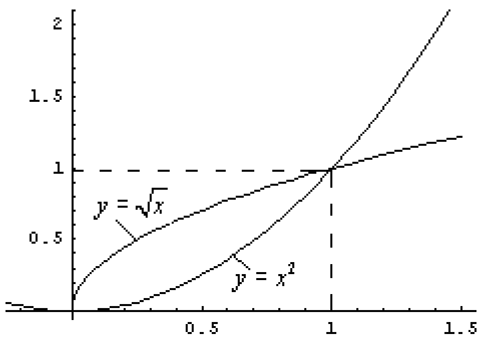

Question No: 38 (Marks: 5)

Determine the fourier co-efficient b_n of the following function.

$f(x) = x^2 \qquad 0 < x < 2\pi$

Question No: 39 (Marks: 5)

Determine whether the following vector field $\vec{F}$ is conservative or not.

$\vec{F}(x, y, z) = (4x - z)\hat{i} + (3y + z)\hat{j} + (y - x)\hat{k}$

ASSALAM O ALAIKUM

All Dearz fellows

ALL IN ONE MTH301

Final term PAPERS & MCQz

Created BY Farhan & Ali

BS (cs) 2nd sem

Hackers Group

From Mandi Bahauddin
Remember us in your prayers

Mindhacker124@gmail.com
Hearthacker124@gmail.com

FINALTERM EXAMINATION
Fall 2009
MTH301- Calculus II

Time: 120 min
Marks: 80

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Question No: 1 (Marks: 1) - Please choose one

π is an example of -----------

► **Irrational numbers**

► Rational numbers

► Integers

► Natural numbers

Question No: 2 (Marks: 1) - Please choose one

Straight line is a special kind of ----------------

► Surface

► **Curve**

► Plane

► Parabola

Question No: 3 (Marks: 1) - Please choose one

An ordered triple corresponds to ----------- in three dimensional space.

► ***A unique point***

► *A point in each octant*

► Three points

► Infinite number of points

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Question No: 4 (Marks: 1) - Please choose one

The angles which a line makes with positive x ,y and z-axis are known as ------------

- ► Direction cosines
- ► Direction ratios
- ► **Direction angles**

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Question No: 5 (Marks: 1) - Please choose one

Is the function $f(x, y)$ continuous at origin? If not, why?

$f(x, y) = 4xy + \sin 3x^2 y$

- ► $f(x, y)$ **is continuous at origin**
- ► $f(0, 0)$ is not defined
- ► $f(0, 0)$ is defined but $\lim_{(x,y)\to(0,0)} f(x, y)$ does not exist
- ► $f(0, 0)$ is defined and $\lim_{(x,y)\to(0,0)} f(x, y)$ exists but these two numbers are not equal.

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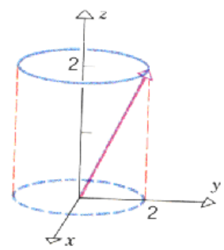
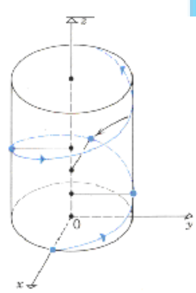
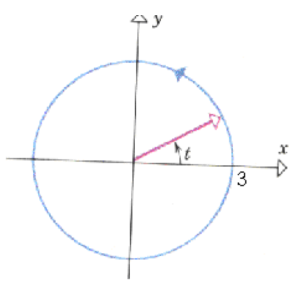
Question No: 6 (Marks: 1) - Please choose one

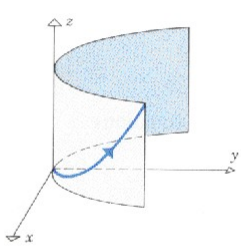
Match the following vector-valued function with its graph.

$r(t) = 3\cos t\,\hat{i} + 3\sin t\,\hat{j}$ And $0 \le t \le 2\pi$

► 

► 

► 

► 

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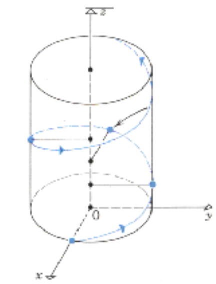
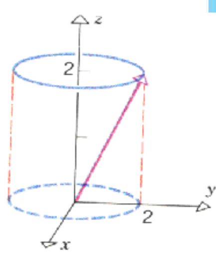
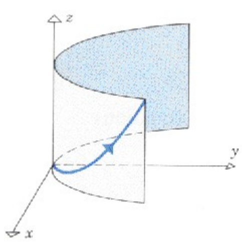
Question No: 7 (Marks: 1) - Please choose one

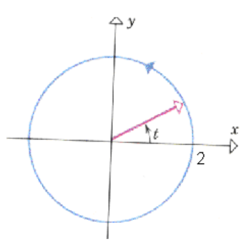
Match the following vector-valued function with its graph.

$r(t) = t\hat{i} + t^2\hat{j} + t^3\hat{k}$ and $t \geq 0$

► 

► 

► 

► 

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Question No: 8 (Marks: 1) - Please choose one

What are the parametric equations that correspond to the following vector equation?

$\vec{r}(t) = \sin^2 t\,\hat{i} + (1 - \cos 2t)\,\hat{j}$

- ► $x = \sin^2 t \quad , \quad y = 1 - \cos 2t \quad , \quad z = 0$
- ► $y = \sin^2 t \quad , \quad x = 1 - \cos 2t \quad , \quad z = 0$
- ► $x = \sin^2 t \quad , \quad y = 1 - \cos 2t \quad , \quad z = 1$
- ► $x = \sin^2 t \quad , \quad y = \cos 2t \quad , \quad z = 1$

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Question No: 9 (Marks: 1) - Please choose one

Is the following vector-valued function $\vec{r}(t)$ continuous at $t = 0$? If not, why?

$\vec{r}(t) = (4\cos t, \sqrt{t}\ , 4\sin t)$

- ► $\vec{r}(0)$ is not defined
- ► $\vec{r}(0)$ is defined but $\lim\limits_{t \to 0} \vec{r}(t)$ does not exist
- ► $\vec{r}(0)$ is defined and $\lim\limits_{t \to 0} \vec{r}(t)$ exists but these two numbers are not equal.

► $\vec{r}(t)$ is continuous at $t=0$

Question No: 10 (Marks: 1) - Please choose one

What is the derivative of following vector-valued function?

$\vec{r}(t) = (\cos 5t,\ \tan t\ ,\ 6\sin t)$

► $\vec{r}'(t) = \left(\frac{\sin 5t}{5},\ \sec t\ ,\ 6\cos t\right)$

► $\vec{r}'(t) = (\frac{-\sin 5t}{5},\ \sec t\ ,\ 6\cos t)$

► $\vec{r}'(t) = (-5\sin 5t,\ \sec^2 t\ ,\ 6\cos t)$

► $\vec{r}'(t) = (\sin 5t,\ \sec^2 t\ ,\ -6\cos t)$

Question No: 11 (Marks: 1) - Please choose one

The following differential is exact

$dz = (3x^2 y + 2)\ dx + (x^3 + y)\ dy$

- ► True

► False

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Question No: 12 (Marks: 1) - Please choose one

The following differential is exact

$dz = (3x^2 + 4xy)\ dx + (2x^2 + 2y)\ dy$

► True

► False

Question No: 13 (Marks: 1) - Please choose one

Which one of the following is correct Wallis Sine formula when n is odd and $n \geq 3$?

► $\int_0^{\frac{\pi}{2}} \sin^n x\ dx = \frac{\pi}{2}\frac{(n-1)}{n}\frac{(n-3)}{(n-2)}\frac{(n-5)}{(n-4)}-------\frac{5}{6}\frac{3}{4}\frac{1}{2}$

► $\int_0^{\frac{\pi}{2}} \sin^n x\ dx = \frac{\pi}{2}\frac{(n)}{(n-1)}\frac{(n-2)}{(n-3)}\frac{(n-4)}{(n-5)}-------\frac{6}{5}\frac{4}{3}\frac{2}{1}$

► $\int_0^{\frac{\pi}{2}} \sin^n x\ dx = \frac{(n-1)}{n}\frac{(n-3)}{(n-2)}\frac{(n-5)}{(n-4)}-------\frac{6}{7}\frac{4}{5}\frac{2}{3}$

► $\int_0^{\frac{\pi}{2}} \sin^n x\ dx = \frac{(n)}{(n-1)}\frac{(n-2)}{(n-3)}\frac{(n-4)}{(n-5)}-------\frac{6}{5}\frac{4}{3}\frac{2}{1}$

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Question No: 14 (Marks: 1) - Please choose one

Which of the following is correct?

► $\int_0^{\frac{\pi}{2}} \sin^4 x \, dx = \frac{3}{16}$

► $\int_0^{\frac{\pi}{2}} \sin^4 x \, dx = \frac{3\pi}{16}$

► $\int_0^{\frac{\pi}{2}} \sin^4 x \, dx = \frac{3}{8}$

► $\int_0^{\frac{\pi}{2}} \sin^4 x \, dx = \frac{2\pi}{3}$

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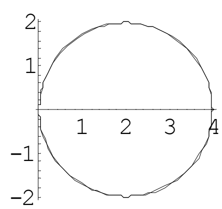
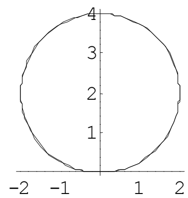
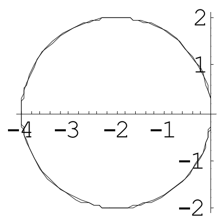
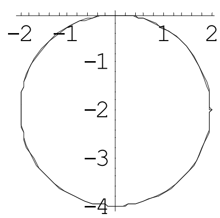
Question No: 15 (Marks: 1) - Please choose one

Match the following equation in polar co-ordinates with its graph.
$r = 4\sin\theta$

► 

► 

► 

► 

Question No: 16 (Marks: 1) - Please choose one

If the equation of a curve, in polar co-ordinates, remains unchanged after replacing (r, θ) by $(r, \pi - \theta)$ then the curve is said to be symmetric about which of the following?

► Initial line

► **y-axis**

► Pole

Question No: 17 (Marks: 1) - Please choose one

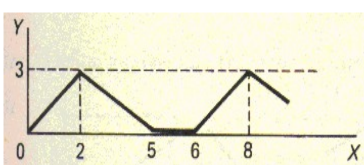
What is the period of a periodic function defined by $f(x) = \sin\frac{x}{2}$?

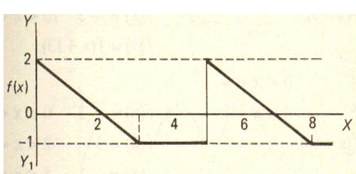
- ▶ $\frac{\pi}{2}$
- ▶ π
- ▶ $\frac{3\pi}{2}$
- ▶ 4π

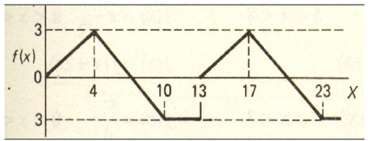
Question No: 18 (Marks: 1) - Please choose one

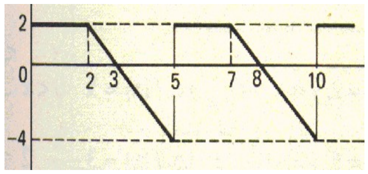
Match the following periodic function with its graph.

$$f(x) = \begin{cases} \frac{3}{4}x & 0 < x < 4 \\ 7 - x & 4 < x < 10 \\ -3 & 10 < x < 13 \end{cases}$$

▶ 

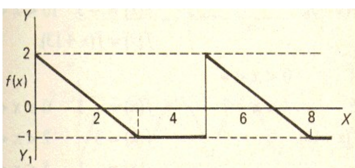
▶ 

► 

► 

Question No: 19 (Marks: 1) - Please choose one

What is the period of periodic function whose graph is as below?



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► 2

► 3

► 4

► 5

Question No: 20 (Marks: 1) - Please choose one

What is the period of periodic function whose graph is as below?

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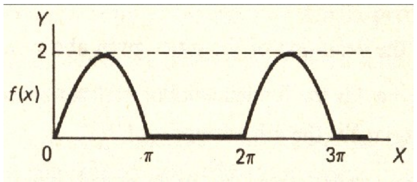

► 0

► 2

► π

► 2π not sure

Question No: 21 (Marks: 1) - Please choose one

Polar co-ordinates of a point are $\left(-2, \frac{-3\pi}{2}\right)$. Which of the following is another possible polar co-ordinates representation of this point?

► $\left(2, \frac{-\pi}{4}\right)$

► $\left(2, \frac{-\pi}{2}\right)$

► $\left(2, \frac{-\pi}{3}\right)$

► $\left(2, \frac{3\pi}{4}\right)$

Question No: 22 (Marks: 1) - Please choose one

The function $f(x) = x^3 e^x$ is -------------

► Even function

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► Odd function

► Neither even nor odd

Question No: 23 (Marks: 1) - Please choose one

The graph of an even function is symmetrical about ---------------

► x-axis

► **y-axis** **page 208**

► origin

Question No: 24 (Marks: 1) - Please choose one

At which point the vertex of parabola, represented by the equation $y = x^2 - 4x + 3$, occurs?

- $(0, 3)$
- $(2, -1)$
- $(-2, 15)$
- $(1, 0)$ pge 08

Question No: 25 (Marks: 1) - Please choose one

The equation $y = x^2 - 4x + 2$ represents a parabola. Find a point at which the vertex of given parabola occurs?

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- $(2, -2)$
- $(-4, 34)$
- $(0, 0)$
- $(-2, 14)$

Question No: 26 (Marks: 1) - Please choose one

Is the function $f(x, y)$ continuous at origin? If not, why?

$$f(x, y) = \frac{xy}{x^2 + y^2}$$

- $f(x, y)$ is continuous at origin

► $\lim_{(x,y)\to(0,0)} f(x, y)$ does not exist

► $f(0, 0)$ is defined and $\lim_{(x,y)\to(0,0)} f(x, y)$ exists but these two numbers are not equal.

Question No: 27 (Marks: 1) - Please choose one

Sign of line integral is reversed when -----------

► path of integration is divided into parts.

► path of integration is parallel to y-axis.

► **direction of path of integration is reversed.**

► path of integration is parallel to x-axis.

Question No: 28 (Marks: 1) - Please choose one

What is Laplace transform of a function F(t)?

(s is a constant)

► $\int_0^s e^{-st} F(t)\, dt$

► $\int_0^\infty e^{st} F(t)\, dt$

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► $\int_{-\infty}^{\infty} e^{-st}\ F(t)\ dt$

► $\int_{0}^{\infty} e^{-st}\ F(t)\ dt$

Question No: 29 (Marks: 1) - Please choose one

What is the value of $L\{e^{5t}\}$ if L denotes laplace transform?

► $L\{e^{5t}\} = \dfrac{1}{s-5}$

► $L\{e^{5t}\} = \dfrac{s}{s^2+25}$

► $L\{e^{5t}\} = \dfrac{5}{s^2+25}$

► $L\{e^{5t}\} = \dfrac{5!}{s^6}$

Question No: 30 (Marks: 1) - Please choose one

What is the Laplace Inverse Transform of $\dfrac{1}{s+1}$

► $L^{-1}\left\{\dfrac{1}{s+1}\right\} = t+1$

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► $L^{-1}\left\{\frac{1}{s+1}\right\} = e^{-t} + e^{t}$

► $L^{-1}\left\{\frac{1}{s+1}\right\} = e^{t}$

► $L^{-1}\left\{\frac{1}{s+1}\right\} = e^{-t}$

Question No: 31 (Marks: 1) - Please choose one

What is Laplace Inverse Transform of $\frac{5}{s^2+25}$

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► $L^{-1}\left\{\frac{5}{s^2+25}\right\} = \sin 5t$

► $L^{-1}\left\{\frac{5}{s^2+25}\right\} = \cos 5t$

► $L^{-1}\left\{\frac{5}{s^2+25}\right\} = \sin 25t$

► $L^{-1}\left\{\frac{5}{s^2+25}\right\} = \cos 25t$

Question No: 32 (Marks: 1) - Please choose one

What is $L\{-6\}$ if L denotes Laplace Transform?

- ► $L\{-6\} = \frac{1}{s+6}$
- ► $L\{-6\} = \frac{-6}{s}$
- ► $L\{-6\} = \frac{s}{s^2+36}$
- ► $L\{-6\} = \frac{-6}{s^2+36}$

Question No: 33 (Marks: 1) - Please choose one

Evaluate the line integral $\int_C (3x+2y)\,dx + (2x-y)\,dy$ where C is the line segment from (0, 0) to (2, 0).

- ► 6
- ► -6
- ► 0
- ► Do not exist

Question No: 34 (Marks: 1) - Please choose one

Evaluate the line integral $\int_C (2x+y)\,dx + (x^2 - y)\,dy$ where C is the line segment from (0, 0) to (0, 2).

► -4

► -2

► 0

► 2

Question No: 35 (Marks: 1) - Please choose one

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Plane is an example of --------------------

► Curve

► **Surface**

► Sphere

► Cone

Question No: 36 (Marks: 1) - Please choose one

$If\ R=\{(x,y)/0\le x\le 2\,and\,-1\le y\le 1\},then$

$\iint\limits_R (x+2y^2)dA=$

- ▶ $\int_{-1}^{1}\int_{0}^{2}(x+2y^2)dydx$
- ▶ $\int_{0}^{2}\int_{1}^{-1}(x+2y^2)dxdy$
- ▶ $\int_{-1}^{1}\int_{0}^{2}(x+2y^2)dxdy$
- ▶ $\int_{1}^{2}\int_{-1}^{0}(x+2y^2)dxdy$

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Question No: 37 (Marks: 1) - Please choose one

To evaluate the line integral, the integrand is expressed in terms of x, y, z with

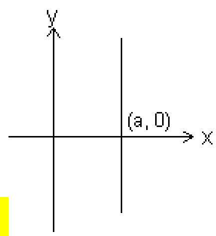
- ▶ $dr=dx\hat{i}+dy\hat{j}$
- ▶ $dr=dx\hat{i}+dy\hat{j}+dy\hat{k}$
- ▶ $dr=dx+dy+dz$
- ▶ $dr=dx+dy$

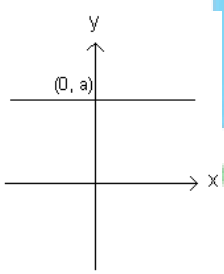
Question No: 38 (Marks: 1) - Please choose one

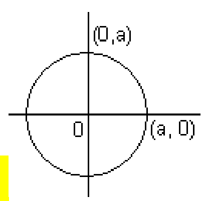
Match the following equation in polar co-ordinates with its graph.

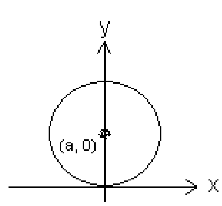
$r = a$

where a is an arbitrary constant.

▶ 

▶ 

▶ 

▶ 

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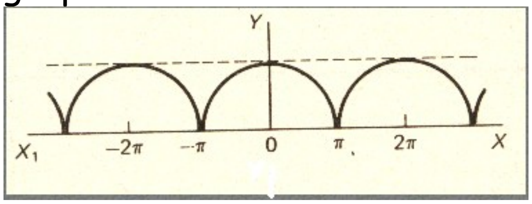
Question No: 39 (Marks: 1) - Please choose one

Which of the following is true for a periodic function whose graph is as below?



► Even function

► Odd function

► Neither even nor odd function

Question No: 40 (Marks: 1) - Please choose one

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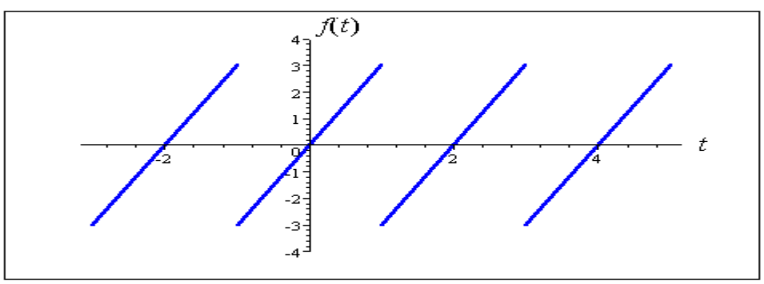

The graph of "saw tooth wave" given above is --------------

► An odd function

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► An even function

► Neither even nor odd   not sure

Question No: 1   (Marks: 2)   - Please choose one

Laplace transform of 't ' is

► $\frac{1}{s}$

► $\frac{1}{s^2}$

► e^{-s}

► s

Question No: 2   (Marks: 2)   - Please choose one

Symmetric equation for the line through (1,3,5) and (2,-2,3) is

► $x-2=-\frac{y+2}{3}=-\frac{z-3}{5}$

► $x+2=-\frac{y+3}{5}=-\frac{z+5}{2}$

► $x-1=-\frac{y-3}{5}=-\frac{z-5}{2}$

► $x+1=\frac{y+3}{5}=\frac{z-5}{5}$

Question No: 3 (Marks: 1) - Please choose one

The level curves of f(x, y) = y Cscx are parabolas.

► True.

► False.

Question No: 4 (Marks: 1) - Please choose one

The equation $z=r$ is written in

► Rectangular coordinates

► Cylindrical coordinates

► Spherical coordinates

► None of the above

FINALTERM EXAMINATION
Spring 2010
MTH301- Calculus II (Session - 4)

Ref No: Time: 90 min
Marks: 60

Student Info	
StudentID:	
Center:	OPKST
ExamDate:	07 Aug 2010

For Teacher's Use Only									
Q No.	1	2	3	4	5	6	7	8	Total
Marks									
Q No.	9	10	11	12	13	14	15	16	
Marks									
Q No.	17	18	19	20	21	22	23	24	
Marks									
Q No.	25	26	27	28	29	30	31	32	
Marks									
Q No.	33	34	35	36	37	38	39		
Marks									

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Question No: 1 (Marks: 1) - Please choose one

There is one-to-one correspondence between the set of points on co-ordinate line and ------------

- ► **Set of real numbers**
- ► Set of integers
- ► Set of natural numbers
- ► Set of rational numbers

Question No: 2 (Marks: 1) - Please choose one

Straight line is a special kind of ----------------

- ► Surface
- ► **Curve**
- ► Plane
- ► Parabola

Question No: 3 (Marks: 1) - Please choose one

$\lim_{(x,y)\to(0,0)} \frac{xy^2}{x^2+y^2} =$

- ► ∞
- ► 0
- ► 1
- ► 0.5

Question No: 4 (Marks: 1) - Please choose one

If $f(x, y) = x^2 y - y^3 + \ln x$

then $\frac{\partial^2 f}{\partial x^2} =$

▶ $2xy+\frac{1}{x^2}$

▶ $2y+\frac{1}{x^2}$

▶ $2xy-\frac{1}{x^2}$

▶ $2y-\frac{1}{x^2}$

Question No: 5 (Marks: 1) - Please choose one

Suppose $f(x, y)=xy-2y^2$ *where* $x=3t+1$ *and* $y=2t$. Which one of the following is true?

□□□□□□□□▶□ $\frac{df}{dt}=-4t+2$

□□□□□□□□▶□ $\frac{df}{dt}=-16t-t$

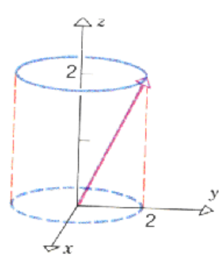
□□□□□□□□▶□ $\frac{df}{dt}=18t+2$

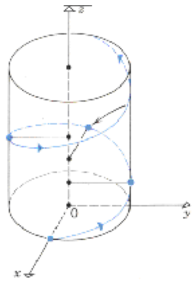
□□□□□□□□▶□ $\frac{df}{dt}=-10t^2+8t+1$

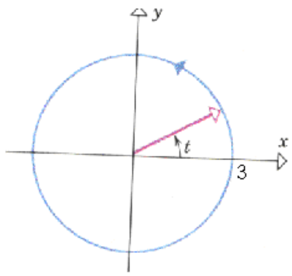
Question No: 6 (Marks: 1) - Please choose one

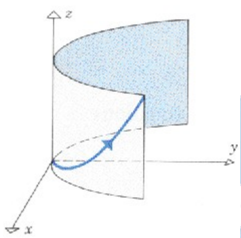
Match the following vector-valued function with its graph.

$r(t)=3\cos t\,\hat{i}+3\sin t\,\hat{j}$ and $0\le t\le 2\pi$

□□□□□□□□▶□ 

□□□□□□□□▶□ 

□□□□□□□□▶□ 

□□□□□□□□▶□ 

Question No: 7 (Marks: 1) - Please choose one

What are the parametric equations that correspond to the following vector equation?

$$\vec{r}(t) = \sin^2 t\,\hat{i} + (1-\cos 2t)\,\hat{j}$$

- ▶ $x = \sin^2 t$, $y = 1-\cos 2t$, $z = 0$
- ▶ $y = \sin^2 t$, $x = 1-\cos 2t$, $z = 0$
- ▶ $x = \sin^2 t$, $y = 1-\cos 2t$, $z = 1$
- ▶ $x = \sin^2 t$, $y = \cos 2t$, $z = 1$

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Question No: 8 (Marks: 1) - Please choose one

Is the following vector-valued function $\vec{r}(t)$ continuous at $t=\frac{\pi}{2}$? If not, why?

$\vec{r}(t)=(\tan t, \sin t^2 , \cos t)$

□□□□□□□▶□ $\vec{r}(t)$ is continuous at $t=\frac{\pi}{2}$

□□□□□□□▶□ $\vec{r}\left(\frac{\pi}{2}\right)$ is not defined

□□□□□□□▶□ $\vec{r}\left(\frac{\pi}{2}\right)$ is defined but $\lim_{t\to\frac{\pi}{2}} \vec{r}(t)$ does not exist

□□□□□□□▶□ $\vec{r}\left(\frac{\pi}{2}\right)$ is defined and $\lim_{t\to\frac{\pi}{2}} \vec{r}(t)$ exists but these two numbers are not equal. Not sure

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Question No: 9 (Marks: 1) - Please choose one

What is the derivative of following vector-valued function?

$\vec{r}(t)=\left(t^4, \sqrt{t+1} , \frac{3}{t^2}\right)$

□□□□□□□▶□ $\vec{r}'(t)=\left(4t^3, \frac{1}{\sqrt{t+1}} , \frac{-6}{t^3}\right)$

□□□□□□□▶□ $\vec{r}'(t)=\left(4t^3, \frac{1}{2\sqrt{t+1}} , \frac{6}{t^3}\right)$

□□□□□□□▶□ $\vec{r}'(t)=\left(4t^4, \frac{1}{2\sqrt{t+1}} , \frac{-6}{t^3}\right)$

□□□□□□□▶□ $\vec{r}'(t)=\left(4t^3, \frac{1}{2\sqrt{t+1}} , \frac{-6}{t^3}\right)$

Question No: 10 (Marks: 1) - Please choose one

The following differential is exact

$dz=(6xy+2y^2-5)\ dx+(3x^2+4xy-6)\ dy$

▶ True
▶ False

Question No: 11 (Marks: 1) - Please choose one

The following differential is exact
$dz = (x^2 y + y)\, dx - x\, dy$

▶ True
▶ **False**

Question No: 12 (Marks: 1) - Please choose one

Which one of the following is correct Wallis Sine formula when n is even and $n \ge 2$?

▶ $\int_0^{\frac{\pi}{2}} \sin^n x\, dx = \frac{\pi}{2} \frac{(n-1)}{n} \frac{(n-3)}{(n-2)} \frac{(n-5)}{(n-4)} -------- \frac{5}{6}\,\frac{3}{4}\,\frac{1}{2}$

▶ $\int_0^{\frac{\pi}{2}} \sin^n x\, dx = \frac{(n-1)}{n} \frac{(n-3)}{(n-2)} \frac{(n-5)}{(n-4)} -------- \frac{6}{7}\,\frac{4}{5}\,\frac{2}{3}$

▶ $\int_0^{\frac{\pi}{2}} \sin^n x\, dx = \frac{\pi}{2} \frac{(n)}{(n-1)} \frac{(n-2)}{(n-3)} \frac{(n-4)}{(n-5)} -------- \frac{6}{5}\,\frac{4}{3}\,\frac{2}{1}$

▶ $\int_0^{\frac{\pi}{2}} \sin^n x\, dx = \frac{(n)}{(n-1)} \frac{(n-2)}{(n-3)} \frac{(n-4)}{(n-5)} -------- \frac{6}{5}\,\frac{4}{3}\,\frac{2}{1}$

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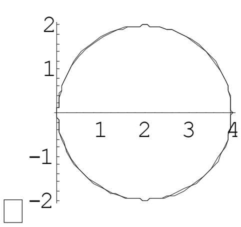
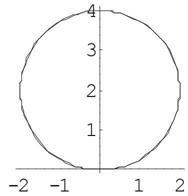
Question No: 13 (Marks: 1) - Please choose one

Match the following equation in polar co-ordinates with its graph.
$r = 4\sin\theta$

▶ 



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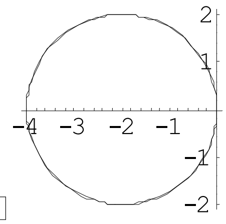

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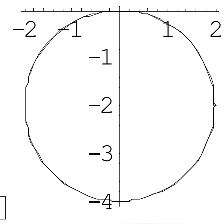

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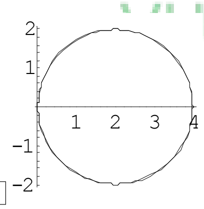
Question No: 14 (Marks: 1) - Please choose one

Match the following equation in polar co-ordinates with its graph.

$r = -4\sin\theta$



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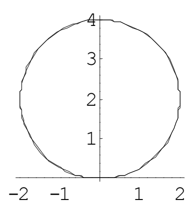

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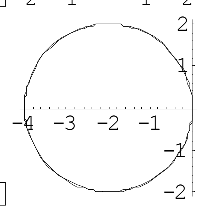

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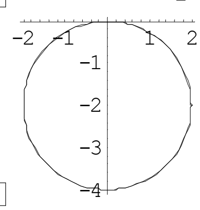

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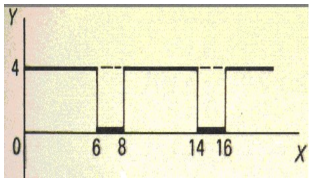
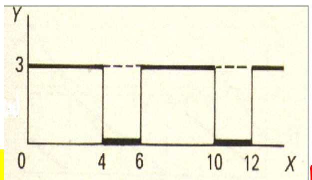
Question No: 15 (Marks: 1) - Please choose one

Match the following periodic function with its graph.

$$f(x)=\begin{cases}3 & 0<x<4\\ 0 & 4<x<6\end{cases}$$

□□□□□□□□▶□ 

□□□□□□□□▶□ 

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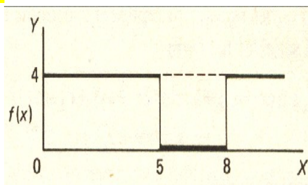
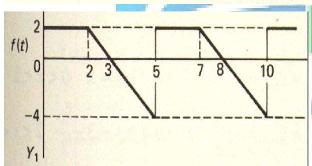
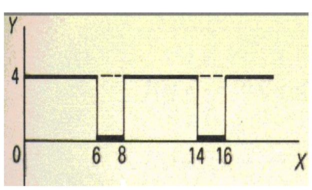
□□□□□□□□▶□ 

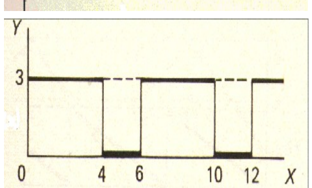
□□□□□□□□▶□ 

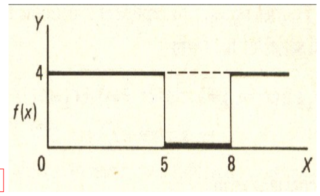
Question No: 16 (Marks: 1) - Please choose one

Match the following periodic function with its graph.

$$f(x)=\begin{cases}4 & 0<x<5\\ 0 & 5<x<8\end{cases}$$

□□□□□□□□▶□ 

□□□□□□□□▶□ 

□□□□□□□□► □ 

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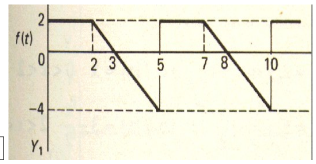
□□□□□□□□► □ 

Question No: 17 (Marks: 1) - Please choose one

Which of the following condition must be satisfied for a vector field $\vec{F}$ to be a conservative vector field?

□□□□□□□□

► □Line integral of $\vec{F}$ along a curve, depends only on the endpoints of that curve, not on the particular route taken.

□□□□□□□□

► □Divergence of $\vec{F}$ should be zero

□□□□□□□□

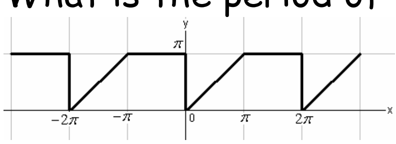
► □Gradient of $\vec{F}$ should be zero.

□□□□□□□□

► □ $\vec{F} = 0$

Question No: 18 (Marks: 1) - Please choose one

What is the period of periodic function whose graph is as below?



□□□□□□□□► □ π

□□□□□□□□► □ $-\pi$

□□□□□□□□► □ 2π

□□□□□□□□► □ -2π

Question No: 19 (Marks: 1) - Please choose one

Let L denotes the Laplace Transform.

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According to First Shift Theorem, if $L\{F(t)\}=f(s)$ then which of the following equation holds?

s *and* a are constants.

□□□□□□□□▶□ $L\{e^{-at}F(t)\}=f(s-a)$

□□□□□□□□▶□ $L\{e^{-at}F(t)\}=f(s+a)$

□□□□□□□□▶□ $L\{e^{-at}F(t)\}=f(s)$

□□□□□□□□▶□ $L\{e^{-at}F(t)\}=f(a)$

Question No: 20 (Marks: 1) - Please choose one

Polar co-ordinates of a point are $\left(7, \frac{-\pi}{4}\right)$. Which of the following is another possible polar co-ordinates representation of this point?

□□□□□□□□▶□ $\left(7, \frac{3\pi}{4}\right)$

□□□□□□□□▶□ $\left(-7, \frac{3\pi}{4}\right)$

□□□□□□□□▶□ $\left(-7, \frac{-\pi}{4}\right)$

□□□□□□□□▶□ $\left(7, \frac{-3\pi}{4}\right)$

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Question No: 21 (Marks: 1) - Please choose one

The graph of an even function is symmetrical about ---------------

□□□□□□□□▶□x-axis

□□□□□□□□▶□**y-axis**

□□□□□□□□▶□origin

Question No: 22 (Marks: 1) - Please choose one

Is the function $f(x, y)$ continuous at origin? If not, why?

$$f(x, y)=\frac{xy}{x^2+y^2}$$

□□□□□□□□▶□ $f(x, y)$ is continuous at origin

□□□□□□□□▶□ $\lim_{(x,y)\to(0,0)} f(x, y)$ does not exist

► $f(0,0)$ is defined and $\lim_{(x,y)\to(0,0)} f(x,y)$ exists but these two numbers are not equal.

Question No: 23 (Marks: 1) - Please choose one

Consider the function $f(x,y,z)=\sqrt{1-x^2-y^2-z^2}$. What is the value of $f\left(0,\frac{1}{2},\frac{1}{2}\right)$?

► $f\left(0,\frac{1}{2},\frac{1}{2}\right)=\sqrt{\frac{1}{2}}$

► $f\left(0,\frac{1}{2},\frac{1}{2}\right)=2$

► $f\left(0,\frac{1}{2},\frac{1}{2}\right)=\frac{1}{2}$

► $f\left(0,\frac{1}{2},\frac{1}{2}\right)=0$

Question No: 24 (Marks: 1) - Please choose one

Sign of line integral is reversed when -----------

► path of integration is divided into parts.

► path of integration is parallel to y-axis.

► **direction of path of integration is reversed.**

► path of integration is parallel to x-axis.

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Question No: 25 (Marks: 1) - Please choose one

Let the functions $P(x,y)$ and $Q(x,y)$ are finite and continuous inside and at the boundary of a closed curve C in the xy-plane. If $(P\,dx+Q\,dy)$ is an exact differential then

$\oint_C (P\,dx+Q\,dy)=$

► **Zero**

- One
- Infinite

Question No: 26 (Marks: 1) - Please choose one

What is laplace transform of the function $F(t)$ if $F(t) = \cos 2t$?

- $L\{\cos 2t\} = \dfrac{2}{s^2+4}$
- $L\{\cos 2t\} = \dfrac{1}{s-2}$
- $L\{\cos 2t\} = \dfrac{s}{s^2+4}$
- $L\{\cos 2t\} = \dfrac{2!}{s^3}$

Question No: 27 (Marks: 1) - Please choose one

What is $L\{-6\}$ if L denotes Laplace Transform?

- $L\{-6\} = \dfrac{1}{s+6}$
- $L\{-6\} = \dfrac{-6}{s}$
- $L\{-6\} = \dfrac{s}{s^2+36}$
- $L\{-6\} = \dfrac{-6}{s^2+36}$

Question No: 28 (Marks: 1) - Please choose one

Curl of vector function is always a --------------

- Scalar
- Vector

Question No: 29 (Marks: 1) - Please choose one

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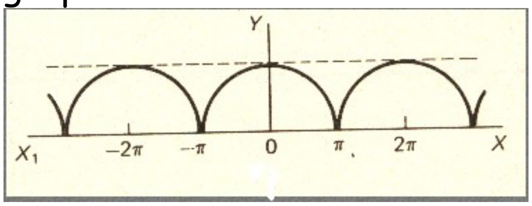
Which of the following is true for a periodic function whose graph is as below?

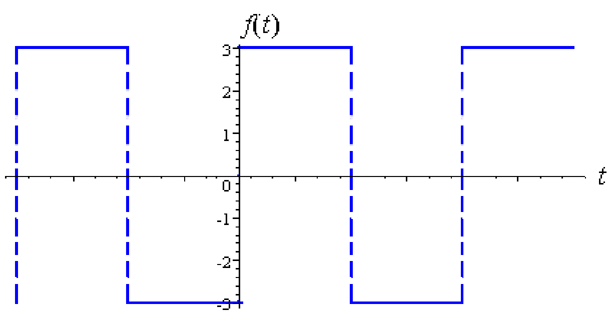


► Even function

► Odd function

► Neither even nor odd function

Question No: 30 (Marks: 1) - Please choose one



Which of the following is true for a function whose graph is given above

► *An odd function*

► An even function

► Neither even nor odd

Question No: 31 (Marks: 2)

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Evaluate the line integral $\int_C 2x\,dx$ where C is the line segment from $(0, 2)$ to $(2, 6)$

Question No: 32 (Marks: 2)

Use Wallis sine formula to evaluate $\int_0^{\frac{\pi}{2}} \sin^5 x\,dx$

Question No: 33 (Marks: 2)

Find Laplace Transform of the function $F(t)$ if $F(t) = \sin 2t$.

Question No: 34 (Marks: 3)

Find $div\,\vec{F}$, if $\vec{F} = (3x + y)\hat{i} + xy^2z\,\hat{j} + (xz^2)\hat{k}$

Question No: 35 (Marks: 3)

Determine whether the following differential is exact or not.

$dz = (4x^3y + 2xy^3)\,dx + (x^4 + 3x^2y^2)\,dy$

Question No: 36 (Marks: 3)

Prove whether the following function is even, odd or neither.

$f(x) = x^3e^x$

Question No: 37 (Marks: 5)

Evaluate the following line integral which is independent of path.

$$\int_{(2,-2)}^{(-1,0)} (2xy^3)\,dx + (3y^2x^2)\,dy$$

Question No: 38 (Marks: 5)

Determine the fourier co-efficient b_n of the following function.

$f(x) = x^2 \qquad 0 < x < 2\pi$

Question No: 39 (Marks: 5)

Determine whether the following vector field $\vec{F}$ is conservative or not.

$\vec{F}(x, y, z) = (3x + y)\hat{i} + xy^2z\,\hat{j} + xz^2\,\hat{k}$

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ALL IN ONE MTH301
Final term PAPERS &
MCQz
Created BY Farhan & Ali
BS (cs) 2nd sem
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From Mandi Bahauddin
Remember us in your prayers
Mindhacker124@gmail.com
Hearthacker124@gmail.com

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FINALTERM EXAMINATION

Spring 2010

MTH301- Calculus II

Time: 90 min

Marks: 60

Student Info	
Student ID:	
Center:	
Exam Date:	

For Teacher's Use Only

Q No.	1	2	3	4	5	6	7	8	Total
Marks									
Q No.	9	10	11	12	13	14	15	16	
Marks									
Q No.	17	18	19	20	21	22	23	24	
Marks									
Q No.	25	26	27	28	29	30	31	32	
Marks									
Q No.	33	34	35	36	37	38	39		
Marks									

Question No: 1 (Marks: 1) - Please choose one

Intersection of two straight lines is --------------

- ► Surface
- ► Curve
- ► Plane
- ► ***Point***

Question No: 2 (Marks: 1) - Please choose one

Plane is a --------------- surface.

- ► One-dimensional
- ► ***Two-dimensional***
- ► Three-dimensional
- ► Dimensionless

Question No: 3 (Marks: 1) - Please choose one

Let w = f(x, y, z) and x = g(r, s), y = h(r, s), z = t(r, s) then by chain rule

$$\frac{\partial w}{\partial r} =$$

► $\frac{\partial w}{\partial x}\frac{\partial x}{\partial r}+\frac{\partial w}{\partial y}\frac{\partial y}{\partial r}+\frac{\partial w}{\partial z}\frac{\partial z}{\partial r}$

► $\frac{\partial w}{\partial r}\frac{\partial x}{\partial r}+\frac{\partial w}{\partial r}\frac{\partial y}{\partial r}+\frac{\partial w}{\partial r}\frac{\partial z}{\partial r}$

► $\frac{\partial w}{\partial x}\frac{\partial x}{\partial r}\frac{\partial x}{\partial s}+\frac{\partial w}{\partial y}\frac{\partial y}{\partial r}\frac{\partial y}{\partial s}+\frac{\partial w}{\partial z}\frac{\partial z}{\partial r}\frac{\partial z}{\partial s}$

► $\frac{\partial w}{\partial r}\frac{\partial r}{\partial x}+\frac{\partial w}{\partial r}\frac{\partial r}{\partial y}+\frac{\partial w}{\partial r}\frac{\partial r}{\partial z}$

Question No: 4 (Marks: 1) - Please choose one

What are the parametric equations that correspond to the following vector equation?

$$\vec{r}(t) = \sin^2 t\,\hat{i} + (1-\cos 2t)\,\hat{j}$$

► $x = \sin^2 t \quad , \quad y = 1-\cos 2t \quad , \; z = 0$

► $y = \sin^2 t \quad , \quad x = 1-\cos 2t \quad , \; z = 0$

- ▶ $x = \sin^2 t$, $y = 1 - \cos 2t$, $z = 1$
- ▶ $x = \sin^2 t$, $y = \cos 2t$, $z = 1$

Question No: 5 (Marks: 1) - Please choose one

What are the parametric equations that correspond to the following vector equation?

$r(t) = (2t - 1)\hat{i} - 3\sqrt{t}\,\hat{j} + \sin 3t\,\hat{k}$

- ▶ $z = 2t - 1$, $x = -3\sqrt{t}$, $y = \sin 3t$
- ▶ $y = 2t - 1$, $x = -3\sqrt{t}$, $z = \sin 3t$
- ▶ $x = 2t - 1$, $z = -3\sqrt{t}$, $y = \sin 3t$
- ▶ $x = 2t - 1$, $y = -3\sqrt{t}$, $z = \sin 3t$

Question No: 6 (Marks: 1) - Please choose one

What is the derivative of following vector-valued function?

$\vec{r}(t) = (\cos 5t,\ \tan t\ ,\ 6\sin t)$

- ▶ $\vec{r}'(t) = \left(\frac{\sin 5t}{5},\ \sec t\ ,\ 6\cos t\right)$

► $\vec{r}'(t) = (\frac{-\sin 5t}{5}, \sec t\ , 6\cos t)$

► $\vec{r}'(t) = (-5\sin 5t, \sec^2 t\ , 6\cos t)$

► $\vec{r}'(t) = (\sin 5t, \sec^2 t\ , -6\cos t)$

Question No: 7 (Marks: 1) - Please choose one

What is the derivative of following vector-valued function?

$\vec{r}(t) = \left(t^4, \sqrt{t+1}\ , \frac{3}{t^2} \right)$

► $\vec{r}'(t) = \left(4t^3, \frac{1}{\sqrt{t+1}}, \frac{-6}{t^3} \right)$

► $\vec{r}'(t) = \left(4t^3, \frac{1}{2\sqrt{t+1}}, \frac{6}{t^3} \right)$

► $\vec{r}'(t) = \left(4t^4, \frac{1}{2\sqrt{t+1}}, \frac{-6}{t^3} \right)$

► $\vec{r}'(t) = \left(4t^3, \frac{1}{2\sqrt{t+1}}, \frac{-6}{t^3} \right)$

Question No: 8 (Marks: 1) - Please choose one

The following differential is exact

$dz = (x^2 y + y)\ dx - x\ dy$

► True

► False

Question No: 9 (Marks: 1) - Please choose one

Which one of the following is correct Wallis Sine formula when n is even and $n \geq 2$?

► $\int_0^{\frac{\pi}{2}} \sin^n x \, dx = \frac{\pi}{2} \frac{(n-1)}{n} \frac{(n-3)}{(n-2)} \frac{(n-5)}{(n-4)} -------- \frac{5}{6} \frac{3}{4} \frac{1}{2}$ page #182

► $\int_0^{\frac{\pi}{2}} \sin^n x \, dx = \frac{(n-1)}{n} \frac{(n-3)}{(n-2)} \frac{(n-5)}{(n-4)} -------- \frac{6}{7} \frac{4}{5} \frac{2}{3}$

► $\int_0^{\frac{\pi}{2}} \sin^n x \, dx = \frac{\pi}{2} \frac{(n)}{(n-1)} \frac{(n-2)}{(n-3)} \frac{(n-4)}{(n-5)} -------- \frac{6}{5} \frac{4}{3} \frac{2}{1}$

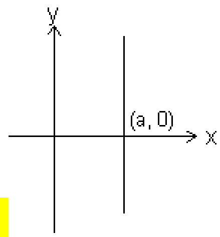
► $\int_0^{\frac{\pi}{2}} \sin^n x \, dx = \frac{(n)}{(n-1)} \frac{(n-2)}{(n-3)} \frac{(n-4)}{(n-5)} -------- \frac{6}{5} \frac{4}{3} \frac{2}{1}$

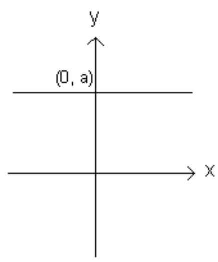
Question No: 10 (Marks: 1) - Please choose one

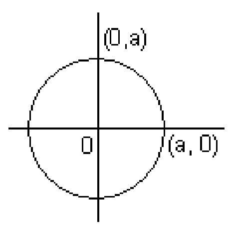
Match the following equation in polar co-ordinates with its graph.

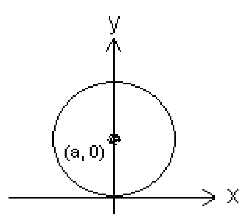
$r \cos\theta = a$

where a is an arbitrary constant

▶ 

▶ 

▶ 

▶ 

Question No: 11 (Marks: 1) - Please choose one

If the equation of a curve, in polar co-ordinates, remains unchanged after replacing (r, θ) by $(r, \pi - \theta)$ then the curve is said to be symmetric about which of the following?

► Initial line

► Y-axis

► Pole

Question No: 12 (Marks: 1) - Please choose one

If the equation of a curve, in polar co-ordinates, remains unchanged after replacing (r, θ) by $(-r, \theta)$ then the curve is said to be symmetric about which of the following?

► Initial line

► y-axis

► ***Pole***

Question No: 13 (Marks: 1) - Please choose one

What is the amplitude of a periodic function defined by $f(x) = \sin\frac{x}{3}$?

► **0**

► 1

► $\frac{1}{3}$

► Does not exist

Question No: 14 (Marks: 1) - Please choose one

What is the period of a periodic function defined by $f(x) = 4\cos 3x$**?**

► $\frac{\pi}{4}$

► $\frac{\pi}{3}$

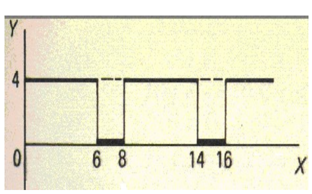
► $\frac{2\pi}{3}$

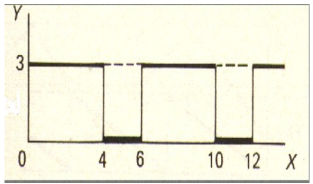
► π

Question No: 15 (Marks: 1) - Please choose one

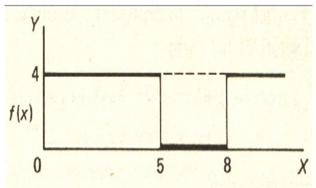
Match the following periodic function with its graph.

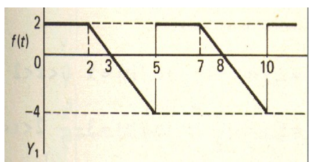
$$f(x) = \begin{cases} 3 & 0 < x < 4 \\ 0 & 4 < x < 6 \end{cases}$$

► 

► 

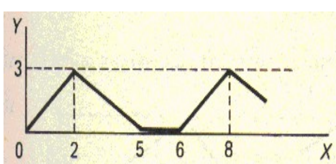
Page #198

► 

► 

Question No: 16 (Marks: 1) - Please choose one

What is the period of periodic function whose graph is as below?



► 2

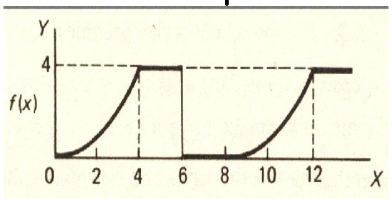
► 5

► **6**

► 8

Question No: 17 (Marks: 1) - Please choose one

What is the period of periodic function whose graph is as below?



► 0

► 4

► 6

► **8**

Question No: 18 (Marks: 1) - Please choose one

Let L denotes the Laplace Transform.

If $L\{F(t)\} = f(s)$ where s is a constant.and $\lim_{t \to 0}\left(\frac{F(t)}{t}\right)$ exists then which of the following equation holds?

► $L\left(\frac{F(t)}{t}\right) = f(s+a)$

► $L\left(\frac{F(t)}{t}\right) = f(s-a)$

► $L\left(\frac{F(t)}{t}\right)=\int_s^\infty f(s)\ ds$

► $L\left(\frac{F(t)}{t}\right)=-\frac{d}{ds}\{f(s)\}$

Question No: 19 (Marks: 1) - Please choose one

Which of the following is Laplace inverse transform of the function $f(s)$ defined by $f(s)=\frac{3}{s-2}-\frac{2}{s}$?

► $3te^{2t}-2$

► $3e^{2t}-2t$

► $3e^{2t}-2$

► None of these.

Question No: 20 (Marks: 1) - Please choose one

Let (x_1, y_1, z_1) and (x_2, y_2, z_2) be any two points in three dimensional space. What does the formula $\sqrt{(x_2-x_1)^2+(y_2-y_1)^2+(z_2-z_1)^2}$ calculates?

► Distance between these two points

► Midpoint of the line joining these two points

► Ratio between these two points

Question No: 21 (Marks: 1) - Please choose one

Let the functions $P(x, y)$ and $Q(x, y)$ are finite and continuous inside and at the boundary of a closed curve C in the xy-plane. If $(P\,dx + Q\,dy)$ is an exact differential then

$\oint_C (P\,dx + Q\,dy) =$

► **Zero**

► One

► Infinite

Question No: 22 (Marks: 1) - Please choose one

What is Laplace transform of the function $F(t)$ if $F(t) = t$?

► $L\{t\} = \frac{1}{s}$

► $L\{t\} = \frac{1}{s^2}$

► $L\{t\} = e^{-s}$

► $L\{t\} = s$

Question No: 23 (Marks: 1) - Please choose one

What is the value of $L\{e^{5t}\}$ if L denotes laplace transform?

► $L\{e^{5t}\} = \frac{1}{s-5}$

► $L\{e^{5t}\} = \frac{s}{s^2+25}$

► $L\{e^{5t}\} = \frac{5}{s^2+25}$

► $L\{e^{5t}\} = \frac{5!}{s^6}$

Question No: 24 (Marks: 1) - Please choose one

Evaluate the line integral $\int_C (3x+2y)\,dx + (2x-y)\,dy$ where *C* is the line segment from (0, 0) to (0, 2).

► 1

► 0

► **2**

► -2

Question No: 25 (Marks: 1) - Please choose one

Evaluate the line integral $\int_C (2x+y)\,dx+(x^2-y)\,dy$ where C is the line segment from (0, 0) to (2, 0).

► **0**

► -4

► 4

► Do not exist

Question No: 26 (Marks: 1) - Please choose one

Which of the following are direction ratios for the line joining the points $(1, 3, 5)$ and $(2, -1, 4)$?

► 3, 2 and 9

► 1, -4 and -1

► **2, -3 and 20**

► 0.5, -3 and 5/4

Question No: 27 (Marks: 1) - Please choose one

$If\ R = \{(x, y) / 0 \le x \le 2\ and\ 1 \le y \le 4\}, then$

$$\iint_R (6x^2 + 4xy^3) dA =$$

► $\int_1^4 \int_0^2 (6x^2 + 4xy^3) dydx$

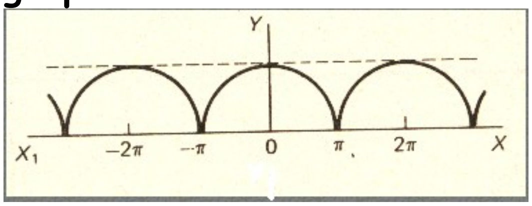
► $\int_0^2 \int_1^4 (6x^2 + 4xy^3) dxdy$

► $\int_1^4 \int_0^2 (6x^2 + 4xy^3) dxdy$

► $\int_2^4 \int_0^1 (6x^2 + 4xy^3) dxdy$

Question No: 28 (Marks: 1) - Please choose one

Which of the following is true for a periodic function whose graph is as below?

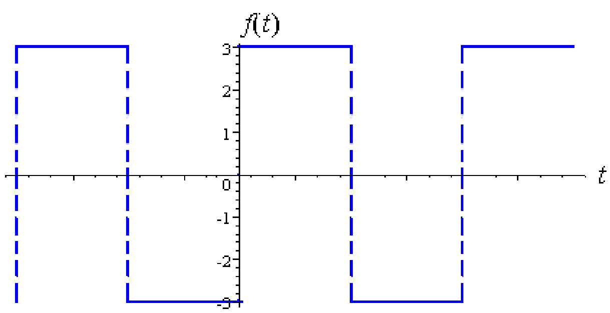


► Even function page209

► Odd function

► Neither even nor odd function

Question No: 29 (Marks: 1) - Please choose one



Which of the following is true for a function whose graph is given above.

► An odd function pg 212

► An even function

► Neither even nor odd

Question No: 30 (Marks: 1) - Please choose one

At each point of domain, the function ----------------

► **Is defined**

► Is continuous

► Is infinite

► Has a limit

Question No: 31 (Marks: 2)

Determine whether the following differential is exact or not.

$dz = 4x^3y^3\ dx + 3x^4y^2\ dy$

Solution:

$$dz = 4x^3y^3\ dx + 3x^4y^2\ dy$$
$$\frac{\partial p}{\partial y} = 12x^3y^2$$
$$\frac{\partial Q}{\partial X} = 12x^3y^2$$
$$\frac{\partial p}{\partial y} = \frac{\partial Q}{\partial X}$$
yes

Question No: 32 (Marks: 2)

Evaluate

$$\int_{-\pi}^{\pi} \sin\ nx\ dx$$

where n is an integer other than zero.

Solution:

$$\int_{-\pi}^{\pi} \sin\ nx\ dx$$
$$= \left[\frac{-\cos nx}{n}\right]_{-\pi}^{\pi}$$
$$= \left[\frac{-\cos n\pi}{n} + \frac{\cos n\pi}{n}\right]$$
$$= \frac{1}{n}(-\cos n\pi + \cos n\pi)$$
$$= 0$$

Question No: 33 (Marks: 2)

Find Laplce transform of the function $F(t)$ if $F(t) = e^{3t}$

Solution:

$$L(e^{3t}) = \int_0^\infty e^{3t} - e^{-st}$$

$$= \int_0^\infty e^{-(s-3)t}.dt$$

$$= \{\frac{e^{-(s-3)t}}{-(s-3)}\} \lim 0 - \infty$$

$$= \frac{-1}{s-3}(\frac{1}{e^{(s-3)t}})$$

$$= \frac{-1}{s-3}(0-1)$$

$$= \frac{1}{s-3} \text{......Ans}$$

Question No: 34 (Marks: 3)

Determine the Fourier co-efficient a_0 of the periodic function defined below:

$f(x) = 2x+1 \qquad 0 < x < 2$

Solution:

$$a_\circ = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)dx$$

$$f(x) = (2x+1)$$

$$(0,2)$$

$$= \int_0^2 (2x+1)dx$$

$$= \left[x^2 + x\right]_0^2$$

$$= 6$$

Question No: 35 (Marks: 3)

Determine whether the following differential is exact or not.

$$dz = (3x^2e^{2y} - 2y^2e^{3x})\ dx + (2x^3e^{2y} - 2ye^{3x})\,dy$$

Solution:

dz = Pdx + Qdy

Therefore,

For dz to be an exact differential it must satisfy $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$

But this test fails becuase $\frac{\partial P}{\partial y} \neq \frac{\partial Q}{\partial x}$

Not Exact

Question No: 36 (Marks: 3)

Use Wallis sine formula to evaluate $\int_0^{\frac{\pi}{2}} \left(\sin^3 x + \sin^5 x\right) dx$

Solution:

$$\int_0^{\frac{\pi}{2}} \sin^3 x dx$$
$$= \frac{n-1}{n}.$$
$$= \frac{3-1}{3}$$
$$= \frac{2}{3}$$
$$\int_0^{\frac{\pi}{2}} \sin^5 x dx$$
$$= \frac{n-1}{n}.\frac{n-3}{n-2}$$
$$= \frac{5-1}{5}.\frac{5-3}{5-2}$$
$$= \frac{4}{5}.\frac{2}{3}$$
$$\int_0^{\frac{\pi}{2}} \left(\sin^3 x + \sin^5 x\right) dx$$
$$= \frac{2}{3} + \frac{4}{5}.\frac{2}{3}$$

Question No: 37 (Marks: 5)

Evaluate the following line integral which is independent of path.

$$\int_{(0,0)}^{(3,2)} (2xe^y)\ dx + (x^2e^y)\ dy$$

Solution:

$$p=\frac{\partial z}{\partial x}=2e^{y} \qquad \int 2e^{y}dx$$

$$Q=\frac{\partial z}{\partial y}=x^{2}e^{y} \qquad \int x^{2}e^{y}dy$$

$$z=\int_{(0,0)}^{(3,2)} 2xe^{y}+x^{2}ye^{y}$$

$$z=6e^{2}+18e^{2}$$

$$z=24e^{2}$$

Question No: 38 (Marks: 5)

Determine the Fourier coefficients b_n for a periodic function $f(t)$ of period 2 defined by

$$f(t)=\begin{cases} 4\ (1+\mathrm{t}) & -1<\mathrm{t}<0 \\ 0 & 0<\mathrm{t}<1 \end{cases}$$

Solution:

$$bn=\frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nxdx$$

$$=\frac{1}{\pi}\int_{-1}^{1} 4(1+t)\sin nxdx$$

$$=\frac{1}{\pi}\left[\frac{-4(1+t)\cos nx}{n}\right]_{-1}^{1}$$

$$=\frac{-4(1+t)}{\pi n}\left[\cos n(1)-\cos n(-1)\right]$$

$$=\frac{-4(1+t)}{\pi n}(\cos n+\cos n)$$

Question No: 39 (Marks: 5)

Determine whether the following vector field $\vec{F}$ is conservative or not.

$$\vec{F}(x,\ y,\ z)=(4x-z)\hat{i}+(3y+z)\hat{j}+(y-x)\hat{k}$$

……………………………………

ASSALAM O ALAIKUM

All Dearz fellows

ALL IN ONE MTH301

Final term PAPERS & MCQz

Created BY Farhan & Ali

BS (cs) 2nd sem

Hackers Group

From Mandi Bahauddin

Remember us in your prayers

Mindhacker124@gmail.com

Hearthacker124@gmail.com

FINALTERM EXAMINATION

Spring 2010

MTH301- Calculus II (Session - 2)

Time: 90 min

Marks: 60

Student Info	
StudentID:	$$
Center:	OPKST
ExamDate:	19 Aug 2010

For Teacher's Use Only									
Q No.	1	2	3	4	5	6	7	8	Total
Marks									
Q No.	9	10	11	12	13	14	15	16	
Marks									
Q No.	17	18	19	20	21	22	23	24	
Marks									
Q No.	25	26	27	28	29	30	31	32	
Marks									
Q No.	33	34	35	36	37	38	39		
Marks									

Question No: 1 (Marks: 1) - Please choose one

------------------- planes intersect at right angle to form three dimensional space.

- ► **Three**
- ► Four
- ► Eight
- ► Twelve

Question No: 2 (Marks: 1) - Please choose one

If the positive direction of x, y axes are known then ------------ the positive direction of z-axis.

- ► Horizontal rightward direction is
- ► Vertical upward direction is
- ► **Left hand rule tells**
- ► Right hand rule tells

Question No: 3 (Marks: 1) - Please choose one

What is the distance between points (3, 2, 4) and (6, 10, -1)?

- ► $7\sqrt{2}$
- ► $2\sqrt{6}$
- ► $\sqrt{34}$
- ► $7\sqrt{3}$

Question No: 4 (Marks: 1) - Please choose one

The equation $ax + by + cz + d = 0$, where a,b,c,d are real numbers, is the general equation of which of the following?

- ► Plane page # 12
- ► Line
- ► Curve
- ► Circle

Question No: 5 (Marks: 1) - Please choose one

The spherical co-ordinates of a point are $\left(\sqrt{3}, \frac{\pi}{3}, \frac{\pi}{2}\right)$. What are its cylindrical co-ordinates?

- ► $\left(\frac{\sqrt{3}}{2}, \frac{3}{2}, 0\right)$
- ► $\left(\sqrt{3}\cos\frac{\pi}{3}, \sqrt{3}\sin\frac{\pi}{3}, 0\right)$
- ► $\left(\sqrt{3}\sin\frac{\pi}{3}, \frac{\pi}{2}, \sqrt{3}\cos\frac{\pi}{3}\right)$
- ► $\left(\sqrt{3}, \frac{\pi}{3}, 0\right)$

Question No: 6 (Marks: 1) - Please choose one

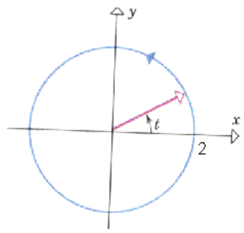
Domain of the function $f(x, y) = \sqrt{y - x^2}$ is

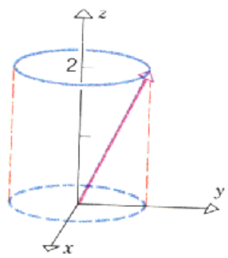
- ► $y < x^2$
- ► $y \geq x^2$
- ► $y \neq x^2$
- ► Entire space

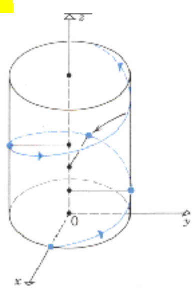
Question No: 7 (Marks: 1) - Please choose one

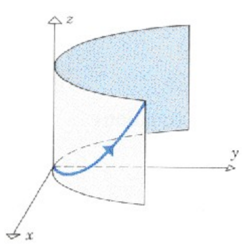
Match the following vector-valued function with its graph.

$r(t) = \cos t\,\hat{i} + \sin t\,\hat{j} + 2\,\hat{k}$ And $0 \le t \le 2\pi$

▶ 

▶  **Correct**

▶ 

▶ 

Question No: 8 (Marks: 1) - Please choose one

What are the parametric equations that correspond to the following vector equation?

$\vec{r}(t) = \sin^2 t\,\hat{i} + (1 - \cos 2t)\,\hat{j}$

▶ $x = \sin^2 t$, $y = 1 - \cos 2t$, $z = 0$

▶ $y = \sin^2 t$, $x = 1 - \cos 2t$, $z = 0$

- $x = \sin^2 t$, $y = 1 - \cos 2t$, $z = 1$
- $x = \sin^2 t$, $y = \cos 2t$, $z = 1$

Question No: 9 (Marks: 1) - Please choose one

What are the parametric equations that correspond to the following vector equation?

$r(t) = (2t-1)\hat{i} - 3\sqrt{t}\,\hat{j} + \sin 3t\,\hat{k}$

- $z = 2t-1$, $x = -3\sqrt{t}$, $y = \sin 3t$
- $y = 2t-1$, $x = -3\sqrt{t}$, $z = \sin 3t$
- $x = 2t-1$, $z = -3\sqrt{t}$, $y = \sin 3t$
- $x = 2t-1$, $y = -3\sqrt{t}$, $z = \sin 3t$

Question No: 10 (Marks: 1) - Please choose one

Is the following vector-valued function $\vec{r}(t)$ continuous at $t=1$? If not, why?

$\vec{r}(t) = \left(\frac{t+1}{t-1}, t^2 , 2t \right)$

- $\vec{r}(t)$ is continuous at $t=1$
- $\vec{r}(1)$ is not defined
- $\vec{r}(1)$ is defined but $\lim_{t \to 1} \vec{r}(t)$ does not exist
- **$\vec{r}(1)$ is defined and $\lim_{t \to 1} \vec{r}(t)$ exists but these two numbers are not equal.**

Question No: 11 (Marks: 1) - Please choose one

Which one of the following is correct Wallis Sine formula when n is even and $n \geq 2$?

▶ $\int_0^{\frac{\pi}{2}} \sin^n x\, dx = \frac{\pi}{2}\frac{(n-1)}{n}\frac{(n-3)}{(n-2)}\frac{(n-5)}{(n-4)}-------\frac{5}{6}\frac{3}{4}\frac{1}{2}$ **page183**

▶ $\int_0^{\frac{\pi}{2}} \sin^n x\, dx = \frac{(n-1)}{n}\frac{(n-3)}{(n-2)}\frac{(n-5)}{(n-4)}-------\frac{6}{7}\frac{4}{5}\frac{2}{3}$

▶ $\int_0^{\frac{\pi}{2}} \sin^n x\, dx = \frac{\pi}{2}\frac{(n)}{(n-1)}\frac{(n-2)}{(n-3)}\frac{(n-4)}{(n-5)}-------\frac{6}{5}\frac{4}{3}\frac{2}{1}$

▶ $\int_0^{\frac{\pi}{2}} \sin^n x\, dx = \frac{(n)}{(n-1)}\frac{(n-2)}{(n-3)}\frac{(n-4)}{(n-5)}-------\frac{6}{5}\frac{4}{3}\frac{2}{1}$

Question No: 12 (Marks: 1) - Please choose one

Which one of the following is correct Wallis Cosine formula when n is odd and $n \geq 3$?

▶ $\int_0^{\frac{\pi}{2}} \cos^n x\, dx = \frac{\pi}{2}\frac{(n-1)}{n}\frac{(n-3)}{(n-2)}\frac{(n-5)}{(n-4)}-------\frac{5}{6}\frac{3}{4}\frac{1}{2}$

▶ $\int_0^{\frac{\pi}{2}} \cos^n x\, dx = \frac{\pi}{2}\frac{(n)}{(n-1)}\frac{(n-2)}{(n-3)}\frac{(n-4)}{(n-5)}-------\frac{6}{5}\frac{4}{3}\frac{2}{1}$

▶ $\int_0^{\frac{\pi}{2}} \cos^n x\, dx = \frac{(n)}{(n-1)}\frac{(n-2)}{(n-3)}\frac{(n-4)}{(n-5)}-------\frac{6}{5}\frac{4}{3}\frac{2}{1}$

▶ $\int_0^{\frac{\pi}{2}} \cos^n x\, dx = \frac{(n-1)}{n}\frac{(n-3)}{(n-2)}\frac{(n-5)}{(n-4)}-------\frac{6}{7}\frac{4}{5}\frac{2}{3}$ **page183**

Question No: 13 (Marks: 1) - Please choose one

If the equation of a curve, in polar co-ordinates, remains unchanged after replacing (r,θ) by $(r,\pi-\theta)$ then the curve is said to be symmetric about which of the following?

► Initial line
► y-axis
► Pole

Question No: 14 (Marks: 1) - Please choose one

If $a > 0$, then the equation, in polar co-ordinates, of the form $r^2 = a^2 \cos 2\theta$ represent which of the following family of curves?

► Leminscate page 130
► Cardiods
► Rose curves
► Spiral

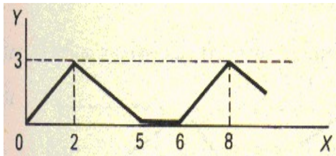
Question No: 15 (Marks: 1) - Please choose one

What is the period of a periodic function defined by $f(x) = \sin\frac{x}{2}$?

► $\frac{\pi}{2}$
► π
► $\frac{3\pi}{2}$
► 4π

Question No: 16 (Marks: 1) - Please choose one

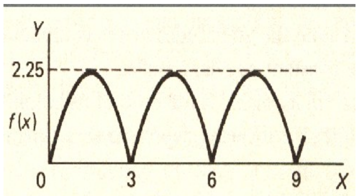
What is the period of periodic function whose graph is as below?



► 2
► 5
► 6
► 8

Question No: 17 (Marks: 1) - Please choose one

What is the period of periodic function whose graph is as below?



► 0
► 2.25
► 3
► 6

Question No: 18 (Marks: 1) - Please choose one

Let L denotes the Laplace Transform.
If $L\{F(t)\} = f(s)$ where s is a constant, then which of the following equation holds?

► $L\{t\ F(t)\} = -\frac{d}{ds}\{f(s)\}$
► $L\{t\ F(t)\} = f(s+t)$
► $L\{t\ F(t)\} = f(s)$
► $L\{t\,F(t)\} = \int_s^\infty f(s)\ ds$

Question No: 19 (Marks: 1) - Please choose one

The graph of an odd function is symmetrical about ----------------

- ► x-axis
- ► y-axis
- ► **origin**

Question No: 20 (Marks: 1) - Please choose one

Consider the function $f(x, y, z) = \sqrt{1 - x^2 - y^2 - z^2}$ **. What is the value of** $f\left(0, \frac{1}{2}, \frac{1}{2}\right)$

- ► $f\left(0, \frac{1}{2}, \frac{1}{2}\right) = \sqrt{\frac{1}{2}}$
- ► $f\left(0, \frac{1}{2}, \frac{1}{2}\right) = 2$
- ► $f\left(0, \frac{1}{2}, \frac{1}{2}\right) = \frac{1}{2}$
- ► $f\left(0, \frac{1}{2}, \frac{1}{2}\right) = 0$

Question No: 21 (Marks: 1) - Please choose one

The path of integration of a line integral must be -------------

- ► straight and single-valued
- ► **continuous and single-valued**
- ► straight and multiple-valued
- ► continuous and multiple-valued

Question No: 22 (Marks: 1) - Please choose one

Sign of line integral is reversed when -----------

► path of integration is divided into parts.
► path of integration is parallel to y-axis.
► **direction of path of integration is reversed.**
► path of integration is parallel to x-axis.

Question No: 23 (Marks: 1) - Please choose one

Let the functions $P(x, y)$ and $Q(x, y)$ are finite and continuous inside and at the boundary of a closed curve C in the xy-plane.

If $(P\,dx + Q\,dy)$ is an exact differential then

$$\oint_C (P\,dx + Q\,dy) =$$

► **Zero**
► **One**
► Infinite

Question No: 24 (Marks: 1) - Please choose one

What is the value of $L\{e^{5t}\}$ if L denotes laplace transform?

► $L\{e^{5t}\} = \dfrac{1}{s-5}$
► $L\{e^{5t}\} = \dfrac{s}{s^2+25}$
► $L\{e^{5t}\} = \dfrac{5}{s^2+25}$
► $L\{e^{5t}\} = \dfrac{5!}{s^6}$

Question No: 25 (Marks: 1) - Please choose one

What is laplace transform of the function $F(t)$ if $F(t) = \sin 3t$?

► $L\{\sin 3t\} = \dfrac{3}{s^2+9}$
► $L\{\sin 3t\} = \dfrac{s}{s^2+9}$

▶ $L\{\sin 3t\} = \dfrac{1}{s-3}$

▶ $L\{\sin 3t\} = \dfrac{3!}{s^4}$

Question No: 26 (Marks: 1) - Please choose one

If L denotes laplace transform then

$L\{te^{5t}\} =$

▶ $L\{te^{5t}\} = \dfrac{1}{s^2-5}$

▶ $L\{te^{5t}\} = \dfrac{1}{s^2+5}$

▶ $L\{te^{5t}\} = \dfrac{1}{(s+5)^2}$

▶ $L\{te^{5t}\} = \dfrac{1}{(s-5)^2}$

Question No: 27 (Marks: 1) - Please choose one

Evaluate the line integral $\int_C (3x+2y)\,dx+(2x-y)\,dy$ **where C is the line segment from (0, 0) to (0, 2).**

▶ 1

▶ 0

▶ 2

▶ -2

Question No: 28 (Marks: 1) - Please choose one

Evaluate the line integral $\int_C (2x+y)\,dx + (x^2-y)\,dy$ where C is the line segment from (0, 0) to (0, 2).

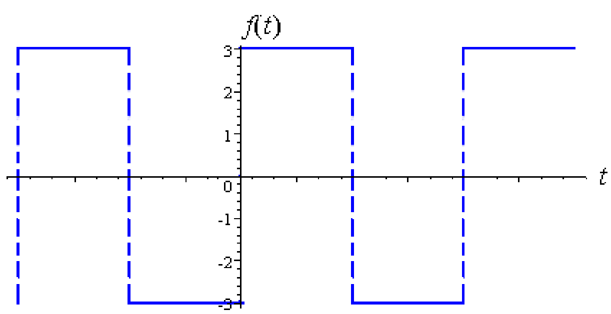
- ► -4
- ► -2
- ► 0
- ► 2

Question No: 29 (Marks: 1) - Please choose one

Divergence of a vector function is always a ------------

- ► Scalar
- ► Vector

Question No: 30 (Marks: 1) - Please choose one



Which of the following is true for a function whose graph is given above

- ► An odd function
- ► An even function
- ► Neither even nor odd

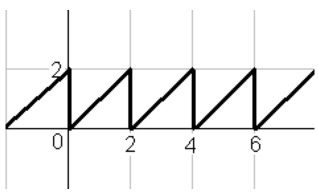
Question No: 31 (Marks: 2)

Does the following limit exist? If yes find its value, if no give reason

$$\lim_{t\to 0}\left[(e^{2t}+5)\hat{i}+(t^2+2t-3)\hat{j}+\left(\frac{1}{t}\right)\hat{k}\right]$$

Question No: 32 (Marks: 2)

Define the periodic function whose graph is shown below.



Question No: 33 (Marks: 2)

Find Laplace Transform of the function $F(t)$ if $F(t)=t^4$

Solution:

The Laplace transform of the given function will be:

$$f(t)=t^4$$
$$L\{t^4\}=\frac{4!}{s^5}$$

Question No: 34 (Marks: 3)

Determine whether the following differential is exact or not.

$$dz=(4x^3y+2xy^3)\,dx+(x^4+3x^2y^2)\,dy$$

Question No: 35 (Marks: 3)

Use Wallis sine formula to evaluate $\int_0^{\frac{\pi}{2}} \left(\sin^3 x + \sin^5 x\right) dx$

Solution:

$$\int_0^{\frac{\pi}{2}} \sin^8 x dx = \frac{7}{8} \cdot \frac{5}{6} \cdot \frac{3}{4} \cdot \frac{1}{2} --------- \frac{\pi}{2}$$

Question No: 36 (Marks: 3)

Find Laplace transform of the function $F(t)$ if

$F(t) = e^{2t} \sin 3t$

Solution:

Laplace transform will be

$$L(t) = e^{2t} ....1$$

$$= \frac{1}{s-2}$$

$$L(t) = \sin 3t ....2$$

$$L(t) = \frac{a}{s^2 + 3^2}$$

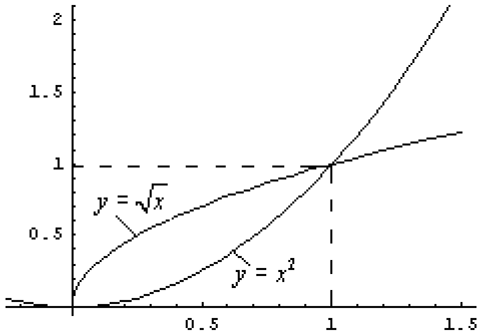
$$L(t) = \frac{a}{s^2 + 9}$$

Combining,

$$L(t) = (\frac{1}{s-2})(\frac{a}{s^2+9})$$

Question No: 37 (Marks: 5)

Using definite integral, find area of the region that is enclosed between the curves $y = x^2 \ \ and \ \ y = \sqrt{x}$



Question No: 38 (Marks: 5)

Determine the fourier co-efficient b_n of the following function.

$f(x) = x^2 \qquad 0 < x < 2\pi$

Question No: 39 (Marks: 5)

Determine whether the following vector field $\vec{F}$ is conservative or not.

$\vec{F}(x, y, z) = (4x - z)\hat{i} + (3y + z)\hat{j} + (y - x)\hat{k}$

ASSALAM O ALAIKUM

All Dearz fellows

ALL IN ONE MTH301

Final term PAPERS & MCQz

Created BY Farhan & Ali

BS (cs) 2nd sem

Hackers Group

From Mandi Bahauddin
Remember us in your prayers

Mindhacker124@gmail.com
Hearthacker124@gmail.com

FINALTERM EXAMINATION
Fall 2009
MTH301- Calculus II

Time: 120 min
Marks: 80

Question No: 1 (Marks: 1) - Please choose one

π is an example of -----------

► **Irrational numbers**

► Rational numbers

► Integers

► Natural numbers

Question No: 2 (Marks: 1) - Please choose one

Straight line is a special kind of ----------------

► Surface

► **Curve**

► Plane

► Parabola

Question No: 3 (Marks: 1) - Please choose one

An ordered triple corresponds to ----------- in three dimensional space.

► ***A unique point***

► *A point in each octant*

► Three points

► Infinite number of points

Question No: 4 (Marks: 1) - Please choose one

The angles which a line makes with positive x ,y and z-axis are known as ------------

- ► Direction cosines
- ► Direction ratios
- ► **Direction angles**

Question No: 5 (Marks: 1) - Please choose one

Is the function $f(x, y)$ continuous at origin? If not, why?

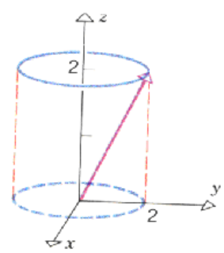
$f(x, y) = 4xy + \sin 3x^2 y$

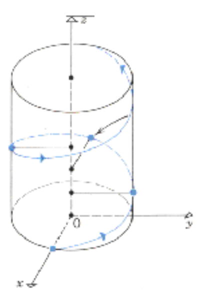
- ► $f(x, y)$ **is continuous at origin**
- ► $f(0, 0)$ is not defined
- ► $f(0, 0)$ is defined but $\lim_{(x,y)\to(0,0)} f(x, y)$ does not exist
- ► $f(0, 0)$ is defined and $\lim_{(x,y)\to(0,0)} f(x, y)$ exists but these two numbers are not equal.

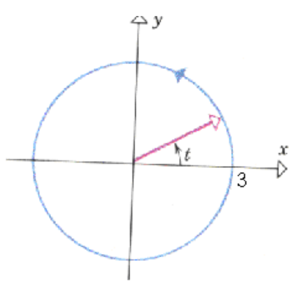
Question No: 6 (Marks: 1) - Please choose one

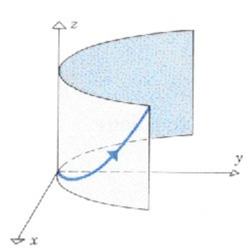
Match the following vector-valued function with its graph.

$r(t) = 3\cos t\,\hat{i} + 3\sin t\,\hat{j}$ And $0 \le t \le 2\pi$

► 

► 

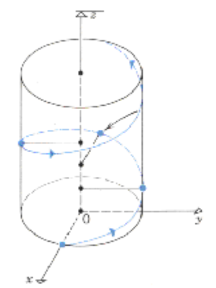
► 

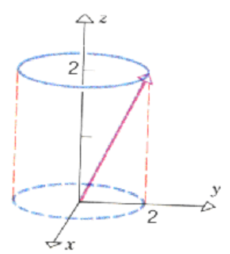
► 

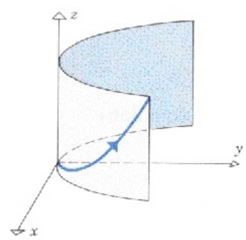
Question No: 7 (Marks: 1) - Please choose one

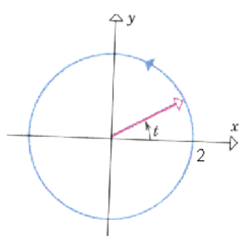
Match the following vector-valued function with its graph.

$r(t) = t\hat{i} + t^2\hat{j} + t^3\hat{k}$ and $t \geq 0$

► 

► 

► 

► 

Question No: 8 (Marks: 1) - Please choose one

What are the parametric equations that correspond to the following vector equation?

$\vec{r}(t) = \sin^2 t\,\hat{i} + (1-\cos 2t)\,\hat{j}$

- ► $x = \sin^2 t \quad , \quad y = 1-\cos 2t \quad , \ z = 0$
- ► $y = \sin^2 t \quad , \quad x = 1-\cos 2t \quad , \ z = 0$
- ► $x = \sin^2 t \quad , \quad y = 1-\cos 2t \quad , \ z = 1$
- ► $x = \sin^2 t \quad , \quad y = \cos 2t \quad , \ z = 1$

Question No: 9 (Marks: 1) - Please choose one

Is the following vector-valued function $\vec{r}(t)$ continuous at $t=0$? If not, why?

$\vec{r}(t) = (4\cos t, \sqrt{t}\ , 4\sin t)$

- ► $\vec{r}(0)$ is not defined
- ► $\vec{r}(0)$ is defined but $\lim_{t\to 0} \vec{r}(t)$ does not exist
- ► $\vec{r}(0)$ is defined and $\lim_{t\to 0} \vec{r}(t)$ exists but these two numbers are not equal.

► $\vec{r}(t)$ is continuous at $t=0$

Question No: 10 (Marks: 1) - Please choose one

What is the derivative of following vector-valued function?

$\vec{r}(t) = (\cos 5t,\ \tan t\ ,\ 6\sin t)$

► $\vec{r}'(t) = \left(\dfrac{\sin 5t}{5},\ \sec t\ ,\ 6\cos t\right)$

► $\vec{r}'(t) = (\dfrac{-\sin 5t}{5},\ \sec t\ ,\ 6\cos t)$

► $\vec{r}'(t) = (-5\sin 5t,\ \sec^2 t\ ,\ 6\cos t)$

► $\vec{r}'(t) = (\sin 5t,\ \sec^2 t\ ,\ -6\cos t)$

Question No: 11 (Marks: 1) - Please choose one

The following differential is exact

$dz = (3x^2 y + 2)\ dx + (x^3 + y)\ dy$

- ► True

► False

Question No: 12 (Marks: 1) - Please choose one

The following differential is exact

$dz = (3x^2 + 4xy)\ dx + (2x^2 + 2y)\ dy$

► True

► False

Question No: 13 (Marks: 1) - Please choose one

Which one of the following is correct Wallis Sine formula when n is odd and $n \geq 3$?

► $\int_0^{\frac{\pi}{2}} \sin^n x\ dx = \frac{\pi}{2} \frac{(n-1)}{n} \frac{(n-3)}{(n-2)} \frac{(n-5)}{(n-4)} -------- \frac{5}{6} \frac{3}{4} \frac{1}{2}$

► $\int_0^{\frac{\pi}{2}} \sin^n x\ dx = \frac{\pi}{2} \frac{(n)}{(n-1)} \frac{(n-2)}{(n-3)} \frac{(n-4)}{(n-5)} -------- \frac{6}{5} \frac{4}{3} \frac{2}{1}$

► $\int_0^{\frac{\pi}{2}} \sin^n x\ dx = \frac{(n-1)}{n} \frac{(n-3)}{(n-2)} \frac{(n-5)}{(n-4)} -------- \frac{6}{7} \frac{4}{5} \frac{2}{3}$

► $\int_0^{\frac{\pi}{2}} \sin^n x\ dx = \frac{(n)}{(n-1)} \frac{(n-2)}{(n-3)} \frac{(n-4)}{(n-5)} -------- \frac{6}{5} \frac{4}{3} \frac{2}{1}$

Question No: 14 (Marks: 1) - Please choose one

Which of the following is correct?

► $\int_0^{\frac{\pi}{2}} \sin^4 x \, dx = \frac{3}{16}$

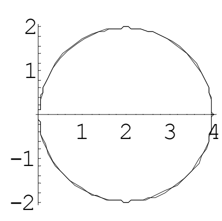
► $\int_0^{\frac{\pi}{2}} \sin^4 x \, dx = \frac{3\pi}{16}$

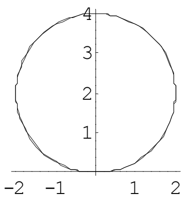
► $\int_0^{\frac{\pi}{2}} \sin^4 x \, dx = \frac{3}{8}$

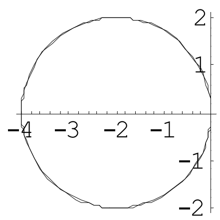
► $\int_0^{\frac{\pi}{2}} \sin^4 x \, dx = \frac{2\pi}{3}$

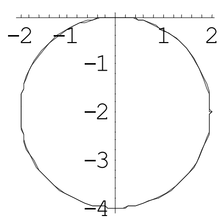
Question No: 15 (Marks: 1) - Please choose one

Match the following equation in polar co-ordinates with its graph.
$r = 4\sin\theta$

► 

► 

► 

► 

Question No: 16 (Marks: 1) - Please choose one

If the equation of a curve, in polar co-ordinates, remains unchanged after replacing (r, θ) by $(r, \pi - \theta)$ then the curve is said to be symmetric about which of the following?

► Initial line

► **y-axis**

► Pole

Question No: 17 (Marks: 1) - Please choose one

What is the period of a periodic function defined by $f(x) = \sin\frac{x}{2}$ **?**

► $\frac{\pi}{2}$

► π

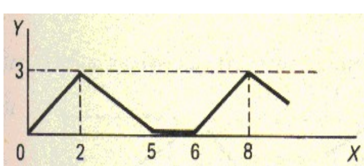
► $\frac{3\pi}{2}$

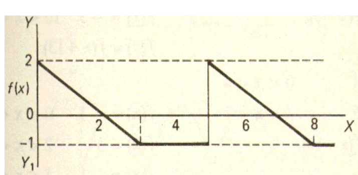
► 4π

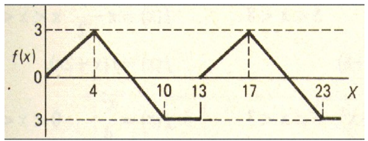
Question No: 18 (Marks: 1) - Please choose one

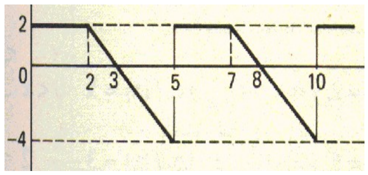
Match the following periodic function with its graph.

$$f(x) = \begin{cases} \frac{3}{4}x & 0 < x < 4 \\ 7 - x & 4 < x < 10 \\ -3 & 10 < x < 13 \end{cases}$$

► 

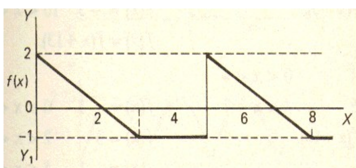
► 

► 

► 

Question No: 19 (Marks: 1) - Please choose one

What is the period of periodic function whose graph is as below?



► 2

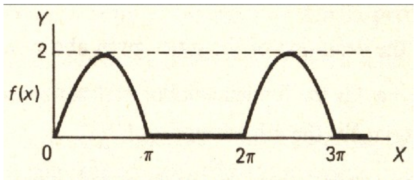
► 3

► 4

► 5

Question No: 20 (Marks: 1) - Please choose one

What is the period of periodic function whose graph is as below?



- ► 0
- ► 2
- ► π
- ► 2π not sure

Question No: 21 (Marks: 1) - Please choose one

Polar co-ordinates of a point are $\left(-2, \frac{-3\pi}{2}\right)$. Which of the following is another possible polar co-ordinates representation of this point?

- ► $\left(2, \frac{-\pi}{4}\right)$
- ► $\left(2, \frac{-\pi}{2}\right)$
- ► $\left(2, \frac{-\pi}{3}\right)$
- ► $\left(2, \frac{3\pi}{4}\right)$

Question No: 22 (Marks: 1) - Please choose one

The function $f(x) = x^3 e^x$ is -------------

► Even function

► Odd function

► Neither even nor odd

Question No: 23 (Marks: 1) - Please choose one

The graph of an even function is symmetrical about ---------------

► x-axis

► **y-axis** **page 208**

► origin

Question No: 24 (Marks: 1) - Please choose one

At which point the vertex of parabola, represented by the equation $y = x^2 - 4x + 3$, occurs?

► $(0, 3)$

► $(2, -1)$

► $(-2, 15)$

► $(1, 0)$ pge 08

Question No: 25 (Marks: 1) - Please choose one

The equation $y = x^2 - 4x + 2$ represents a parabola. Find a point at which the vertex of given parabola occurs?

► $(2, -2)$

► $(-4, 34)$

► $(0, 0)$

► $(-2, 14)$

Question No: 26 (Marks: 1) - Please choose one

Is the function $f(x, y)$ continuous at origin? If not, why?

$f(x, y) = \dfrac{xy}{x^2 + y^2}$

► $f(x, y)$ is continuous at origin

► $\lim_{(x,y)\to(0,0)} f(x,y)$ does not exist

► $f(0,0)$ is defined and $\lim_{(x,y)\to(0,0)} f(x,y)$ exists but these two numbers are not equal.

Question No: 27 (Marks: 1) - Please choose one

Sign of line integral is reversed when -----------

► path of integration is divided into parts.

► path of integration is parallel to y-axis.

► **direction of path of integration is reversed.**

► path of integration is parallel to x-axis.

Question No: 28 (Marks: 1) - Please choose one

What is Laplace transform of a function F(t)?

(s is a constant)

► $\int_0^s e^{-st} F(t)\, dt$

► $\int_0^\infty e^{st} F(t)\, dt$

► $\int_{-\infty}^{\infty} e^{-st}\ F(t)\ dt$

► $\int_{0}^{\infty} e^{-st}\ F(t)\ dt$

Question No: 29 (Marks: 1) - Please choose one

What is the value of $L\{e^{5t}\}$ if L denotes laplace transform?

► $L\{e^{5t}\} = \dfrac{1}{s-5}$

► $L\{e^{5t}\} = \dfrac{s}{s^2+25}$

► $L\{e^{5t}\} = \dfrac{5}{s^2+25}$

► $L\{e^{5t}\} = \dfrac{5!}{s^6}$

Question No: 30 (Marks: 1) - Please choose one

What is the Laplace Inverse Transform of $\dfrac{1}{s+1}$

► $L^{-1}\left\{\dfrac{1}{s+1}\right\} = t+1$

► $L^{-1}\left\{\frac{1}{s+1}\right\}=e^{-t}+e^{t}$

► $L^{-1}\left\{\frac{1}{s+1}\right\}=e^{t}$

► $L^{-1}\left\{\frac{1}{s+1}\right\}=e^{-t}$

Question No: 31 (Marks: 1) - Please choose one

What is Laplace Inverse Transform of $\frac{5}{s^2+25}$

► $L^{-1}\left\{\frac{5}{s^2+25}\right\}=\sin 5t$

► $L^{-1}\left\{\frac{5}{s^2+25}\right\}=\cos 5t$

► $L^{-1}\left\{\frac{5}{s^2+25}\right\}=\sin 25t$

► $L^{-1}\left\{\frac{5}{s^2+25}\right\}=\cos 25t$

Question No: 32 (Marks: 1) - Please choose one

What is $L\{-6\}$ if L denotes Laplace Transform?

► $L\{-6\} = \dfrac{1}{s+6}$

► $L\{-6\} = \dfrac{-6}{s}$

► $L\{-6\} = \dfrac{s}{s^2+36}$

► $L\{-6\} = \dfrac{-6}{s^2+36}$

Question No: 33 (Marks: 1) - Please choose one

Evaluate the line integral $\int_C (3x+2y)\, dx + (2x-y)\, dy$ where C is the line segment from (0, 0) to (2, 0).

► 6

► -6

► 0

► Do not exist

Question No: 34 (Marks: 1) - Please choose one

Evaluate the line integral $\int_C (2x+y)\,dx+(x^2-y)\,dy$ where C is the line segment from (0, 0) to (0, 2).

► -4

► -2

► 0

► 2

Question No: 35 (Marks: 1) - Please choose one

Plane is an example of --------------------

► Curve

► **Surface**

► Sphere

► Cone

Question No: 36 (Marks: 1) - Please choose one

$If\ R = \{(x,y)/0 \le x \le 2\ and -1 \le y \le 1\},then$

$\iint\limits_R (x+2y^2)dA =$

- ▶ $\int_{-1}^{1}\int_{0}^{2} (x+2y^2)dydx$
- ▶ $\int_{0}^{2}\int_{1}^{-1} (x+2y^2)dxdy$
- ▶ $\int_{-1}^{1}\int_{0}^{2} (x+2y^2)dxdy$
- ▶ $\int_{1}^{2}\int_{-1}^{0} (x+2y^2)dxdy$

Question No: 37 (Marks: 1) - Please choose one

To evaluate the line integral, the integrand is expressed in terms of x, y, z with

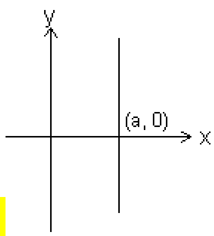
- ▶ $dr = dx\hat{i} + dy\,\hat{j}$
- ▶ $dr = dx\hat{i} + dy\,\hat{j} + dy\,\hat{k}$
- ▶ $dr = dx + dy + dz$
- ▶ $dr = dx + dy$

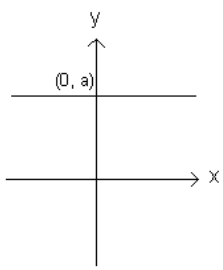
Question No: 38 (Marks: 1) - Please choose one

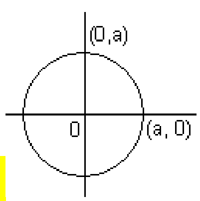
Match the following equation in polar co-ordinates with its graph.

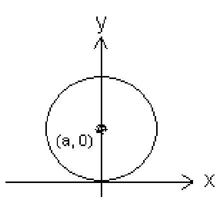
$r = a$

where a is an arbitrary constant.

▶ 

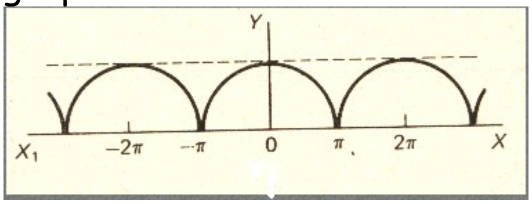
▶ 

▶ 

▶ 

Question No: 39 (Marks: 1) - Please choose one

Which of the following is true for a periodic function whose graph is as below?

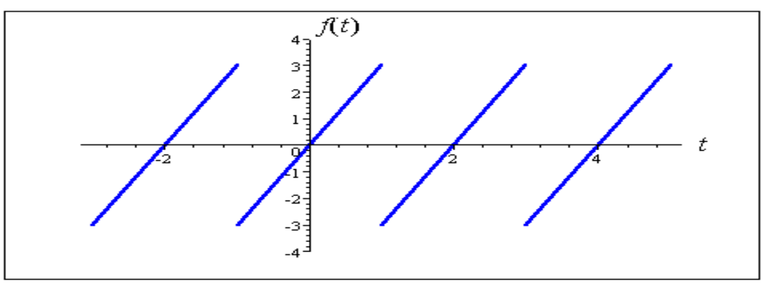


► Even function

► Odd function

► Neither even nor odd function

Question No: 40 (Marks: 1) - Please choose one



The graph of "saw tooth wave" given above is -------------

► An odd function

► An even function

► Neither even nor odd not sure

Question No: 1 (Marks: 2) - Please choose one
Laplace transform of 't' is ► $\frac{1}{s}$ ► $\frac{1}{s^2}$ ► e^{-s} ► s

Question No: 2 (Marks: 2) - Please choose one
Symmetric equation for the line through (1,3,5) and (2,-2,3) is ► $x-2=-\frac{y+2}{3}=-\frac{z-3}{5}$ ► $x+2=-\frac{y+3}{5}=-\frac{z+5}{2}$

► $x-1=-\frac{y-3}{5}=-\frac{z-5}{2}$

► $x+1=\frac{y+3}{5}=\frac{z-5}{5}$

Question No: 3 (Marks: 1) - Please choose one

The level curves of f(x, y) = y Cscx are parabolas.

► True.

► False.

Question No: 4 (Marks: 1) - Please choose one

The equation $z=r$ is written in

► Rectangular coordinates

► Cylindrical coordinates

► Spherical coordinates

► None of the above

FINALTERM EXAMINATION
Spring 2010
MTH301- Calculus II (Session - 4)

Ref No: Time: 90 min
Marks: 60

Student Info	
StudentID:	
Center:	OPKST
ExamDate:	07 Aug 2010

For Teacher's Use Only									
Q No.	1	2	3	4	5	6	7	8	Total
Marks									
Q No.	9	10	11	12	13	14	15	16	
Marks									
Q No.	17	18	19	20	21	22	23	24	
Marks									
Q No.	25	26	27	28	29	30	31	32	
Marks									
Q No.	33	34	35	36	37	38	39		
Marks									

Question No: 1 (Marks: 1) - Please choose one

There is one-to-one correspondence between the set of points on co-ordinate line and ------------

- ► **Set of real numbers**
- ► Set of integers
- ► Set of natural numbers
- ► Set of rational numbers

Question No: 2 (Marks: 1) - Please choose one

Straight line is a special kind of ---------------

- ► Surface
- ► **Curve**
- ► Plane
- ► Parabola

Question No: 3 (Marks: 1) - Please choose one

$$\lim_{(x,\,y)\to(0,\,0)} \frac{xy^2}{x^2+y^2} =$$

- ► ∞
- ► 0
- ► 1
- ► 0.5

Question No: 4 (Marks: 1) - Please choose one

If $f(x, y) = x^2 y - y^3 + \ln x$

then $\frac{\partial^2 f}{\partial x^2}$ =

▶ $2xy+\frac{1}{x^2}$

▶ $2y+\frac{1}{x^2}$

▶ $2xy-\frac{1}{x^2}$

▶ $2y-\frac{1}{x^2}$

Question No: 5 (Marks: 1) - Please choose one

Suppose $f(x, y)=xy-2y^2$ *where* $x=3t+1$ *and* $y=2t$. Which one of the following is true?

□□□□□□□▶□ $\frac{df}{dt}=-4t+2$

□□□□□□□▶□ $\frac{df}{dt}=-16t-t$

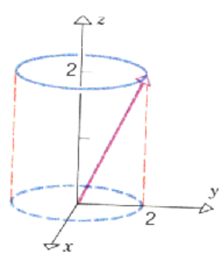
□□□□□□□▶□ $\frac{df}{dt}=18t+2$

□□□□□□□▶□ $\frac{df}{dt}=-10t^2+8t+1$

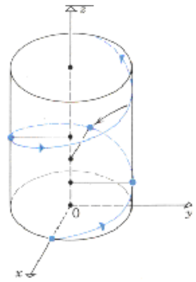
Question No: 6 (Marks: 1) - Please choose one

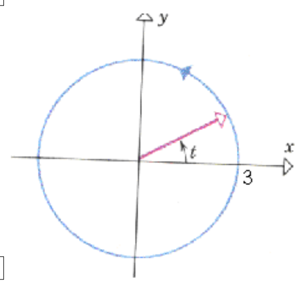
Match the following vector-valued function with its graph.

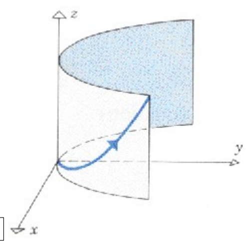
$r(t)=3\cos t\,\hat{i}+3\sin t\,\hat{j}$ and $0\le t\le 2\pi$



□□□□□□□▶□

□□□□□□□▶□ 

□□□□□□□▶□ 

□□□□□□□▶□ 

Question No: 7 (Marks: 1) - Please choose one

What are the parametric equations that correspond to the following vector equation?

$\vec{r}(t) = \sin^2 t\,\hat{i} + (1-\cos 2t)\,\hat{j}$

- ▶ $x = \sin^2 t$, $y = 1-\cos 2t$, $z = 0$
- ▶ $y = \sin^2 t$, $x = 1-\cos 2t$, $z = 0$
- ▶ $x = \sin^2 t$, $y = 1-\cos 2t$, $z = 1$
- ▶ $x = \sin^2 t$, $y = \cos 2t$, $z = 1$

Question No: 8 (Marks: 1) - Please choose one

Is the following vector-valued function $\vec{r}(t)$ continuous at $t = \frac{\pi}{2}$? If not, why?

$\vec{r}(t) = (\tan t, \sin t^2 , \cos t)$

□□□□□□□□▶□ $\vec{r}(t)$ is continuous at $t = \frac{\pi}{2}$

□□□□□□□□▶□ $\vec{r}\left(\frac{\pi}{2}\right)$ is not defined

□□□□□□□□▶□ $\vec{r}\left(\frac{\pi}{2}\right)$ is defined but $\lim_{t \to \frac{\pi}{2}} \vec{r}(t)$ does not exist

□□□□□□□□▶□ $\vec{r}\left(\frac{\pi}{2}\right)$ is defined and $\lim_{t \to \frac{\pi}{2}} \vec{r}(t)$ exists but these two numbers are not equal. Not sure

Question No: 9 (Marks: 1) - Please choose one

What is the derivative of following vector-valued function?

$\vec{r}(t) = \left(t^4, \sqrt{t+1} , \frac{3}{t^2}\right)$

□□□□□□□□▶□ $\vec{r}'(t) = \left(4t^3, \frac{1}{\sqrt{t+1}} , \frac{-6}{t^3}\right)$

□□□□□□□□▶□ $\vec{r}'(t) = \left(4t^3, \frac{1}{2\sqrt{t+1}} , \frac{6}{t^3}\right)$

□□□□□□□□▶□ $\vec{r}'(t) = \left(4t^4, \frac{1}{2\sqrt{t+1}} , \frac{-6}{t^3}\right)$

□□□□□□□□▶□ $\vec{r}'(t) = \left(4t^3, \frac{1}{2\sqrt{t+1}} , \frac{-6}{t^3}\right)$

Question No: 10 (Marks: 1) - Please choose one

The following differential is exact

$dz = (6xy + 2y^2 - 5)\, dx + (3x^2 + 4xy - 6)\, dy$

□□□□□□□□▶ True
□□□□□□□□▶ False

Question No: 11 (Marks: 1) - Please choose one

The following differential is exact

$dz = (x^2 y + y)\, dx - x\, dy$

□□□□□□□□▶ True
□□□□□□□□▶ **False**

Question No: 12 (Marks: 1) - Please choose one

Which one of the following is correct Wallis Sine formula when n is even and $n \geq 2$?

□□□□□□□□▶ □ $\int_0^{\frac{\pi}{2}} \sin^n x\, dx = \frac{\pi}{2} \frac{(n-1)}{n} \frac{(n-3)}{(n-2)} \frac{(n-5)}{(n-4)} -------- \frac{5}{6} \frac{3}{4} \frac{1}{2}$

□□□□□□□□▶ □ $\int_0^{\frac{\pi}{2}} \sin^n x\, dx = \frac{(n-1)}{n} \frac{(n-3)}{(n-2)} \frac{(n-5)}{(n-4)} -------- \frac{6}{7} \frac{4}{5} \frac{2}{3}$

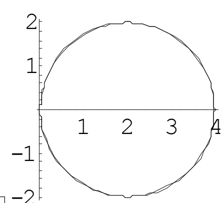
□□□□□□□□▶ □ $\int_0^{\frac{\pi}{2}} \sin^n x\, dx = \frac{\pi}{2} \frac{(n)}{(n-1)} \frac{(n-2)}{(n-3)} \frac{(n-4)}{(n-5)} -------- \frac{6}{5} \frac{4}{3} \frac{2}{1}$

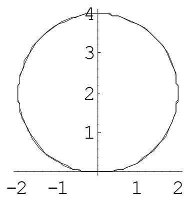
□□□□□□□□▶ □ $\int_0^{\frac{\pi}{2}} \sin^n x\, dx = \frac{(n)}{(n-1)} \frac{(n-2)}{(n-3)} \frac{(n-4)}{(n-5)} -------- \frac{6}{5} \frac{4}{3} \frac{2}{1}$

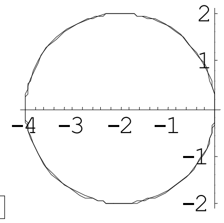
Question No: 13 (Marks: 1) - Please choose one

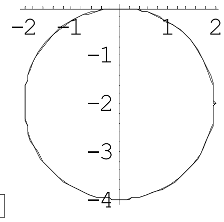
Match the following equation in polar co-ordinates with its graph.

$r = 4\sin\theta$

□□□□□□□□▶ □ 

□□□□□□□▶□ 

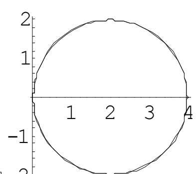
□□□□□□□▶□ 

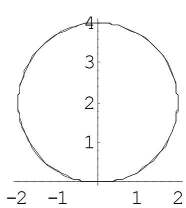
□□□□□□□▶□ 

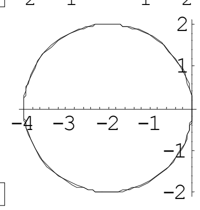
Question No: 14 (Marks: 1) - Please choose one

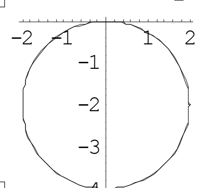
Match the following equation in polar co-ordinates with its graph.

$r = -4\sin\theta$

□□□□□□□▶□ 

□□□□□□□▶□ 

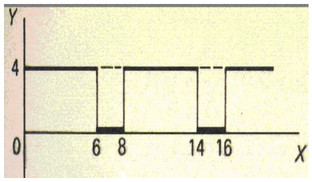
□□□□□□□▶□ 

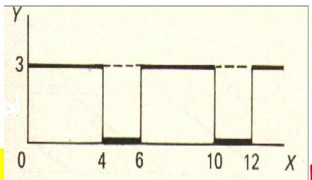
□□□□□□□▶□ 

Question No: 15 (Marks: 1) - Please choose one

Match the following periodic function with its graph.

$$f(x)=\begin{cases}3 & 0<x<4\\ 0 & 4<x<6\end{cases}$$

□□□□□□□□▶□ 

□□□□□□□□▶□ 

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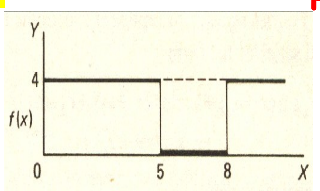
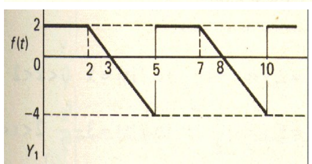
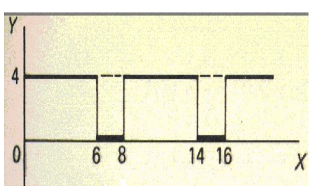
□□□□□□□□▶□ 

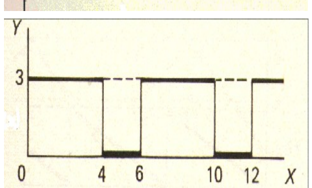
□□□□□□□□▶□ 

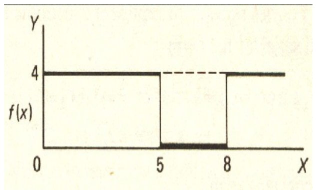
Question No: 16 (Marks: 1) - Please choose one

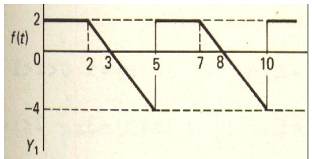
Match the following periodic function with its graph.

$$f(x)=\begin{cases}4 & 0<x<5\\ 0 & 5<x<8\end{cases}$$

□□□□□□□□▶□ 

□□□□□□□□▶□ 

▶  pg199

▶ 

Question No: 17 (Marks: 1) - Please choose one

Which of the following condition must be satisfied for a vector field $\vec{F}$ to be a conservative vector field?

▶ Line integral of $\vec{F}$ along a curve, depends only on the endpoints of that curve, not on the particular route taken.

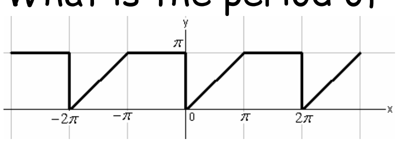
▶ Divergence of $\vec{F}$ should be zero

▶ Gradient of $\vec{F}$ should be zero.

▶ $\vec{F} = 0$

Question No: 18 (Marks: 1) - Please choose one

What is the period of periodic function whose graph is as below?



▶ π

▶ $-\pi$

▶ 2π

▶ -2π

Question No: 19 (Marks: 1) - Please choose one

Let L denotes the Laplace Transform.

According to First Shift Theorem, if $L\{F(t)\} = f(s)$ then which of the following equation holds?
s *and* a **are constants.**

□□□□□□□▶□ $L\{e^{-at}F(t)\} = f(s-a)$
□□□□□□□▶□ $L\{e^{-at}F(t)\} = f(s+a)$
□□□□□□□▶□ $L\{e^{-at}F(t)\} = f(s)$
□□□□□□□▶□ $L\{e^{-at}F(t)\} = f(a)$

Question No: 20 (Marks: 1) - Please choose one

Polar co-ordinates of a point are $\left(7, \frac{-\pi}{4}\right)$. Which of the following is another possible polar co-ordinates representation of this point?

□□□□□□□▶□ $\left(7, \frac{3\pi}{4}\right)$
□□□□□□□▶□ $\left(-7, \frac{3\pi}{4}\right)$
□□□□□□□▶□ $\left(-7, \frac{-\pi}{4}\right)$
□□□□□□□▶□ $\left(7, \frac{-3\pi}{4}\right)$

Question No: 21 (Marks: 1) - Please choose one

The graph of an even function is symmetrical about ---------------

□□□□□□□▶□x-axis
□□□□□□□▶□**y-axis**
□□□□□□□▶□origin

Question No: 22 (Marks: 1) - Please choose one

Is the function $f(x, y)$ continuous at origin? If not, why?

$$f(x, y) = \frac{xy}{x^2 + y^2}$$

□□□□□□□▶□ $f(x, y)$ is continuous at origin
□□□□□□□▶□ $\lim_{(x,y)\to(0,0)} f(x, y)$ does not exist

□□□□□□□► □ $f(0, 0)$ is defined and $\lim_{(x,y)\to(0,0)} f(x, y)$ exists but these two numbers are not equal.

Question No: 23 (Marks: 1) - Please choose one

Consider the function $f(x, y, z) = \sqrt{1-x^2-y^2-z^2}$. What is the value of $f\left(0, \frac{1}{2}, \frac{1}{2}\right)$?

□□□□□□□► □ $f\left(0, \frac{1}{2}, \frac{1}{2}\right) = \sqrt{\frac{1}{2}}$

□□□□□□□► □ $f\left(0, \frac{1}{2}, \frac{1}{2}\right) = 2$

□□□□□□□► □ $f\left(0, \frac{1}{2}, \frac{1}{2}\right) = \frac{1}{2}$

□□□□□□□► □ $f\left(0, \frac{1}{2}, \frac{1}{2}\right) = 0$

Question No: 24 (Marks: 1) - Please choose one

Sign of line integral is reversed when -----------

□□□□□□□► □path of integration is divided into parts.

□□□□□□□► □path of integration is parallel to y-axis.

□□□□□□□► □**direction of path of integration is reversed.**

□□□□□□□► □path of integration is parallel to x-axis.

Question No: 25 (Marks: 1) - Please choose one

Let the functions $P(x, y)$ and $Q(x, y)$ are finite and continuous inside and at the boundary of a closed curve C in the xy-plane.

If $(P\,dx + Q\,dy)$ is an exact differential then

$$\oint_C (P\,dx + Q\,dy) =$$

► **Zero**

► One
► Infinite

Question No: 26 (Marks: 1) - Please choose one

What is laplace transform of the function $F(t)$ if $F(t)=\cos 2t$?

► $L\{\cos 2t\} = \dfrac{2}{s^2+4}$

► $L\{\cos 2t\} = \dfrac{1}{s-2}$

► $L\{\cos 2t\} = \dfrac{s}{s^2+4}$

► $L\{\cos 2t\} = \dfrac{2!}{s^3}$

Question No: 27 (Marks: 1) - Please choose one

What is $L\{-6\}$ if L denotes Laplace Transform?

► $L\{-6\} = \dfrac{1}{s+6}$

► $L\{-6\} = \dfrac{-6}{s}$

► $L\{-6\} = \dfrac{s}{s^2+36}$

► $L\{-6\} = \dfrac{-6}{s^2+36}$

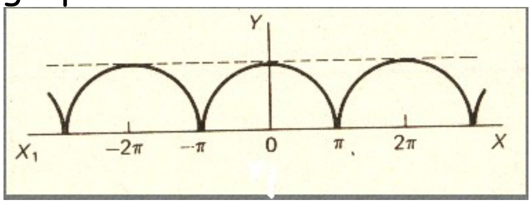
Question No: 28 (Marks: 1) - Please choose one

Curl of vector function is always a --------------

► Scalar
► Vector

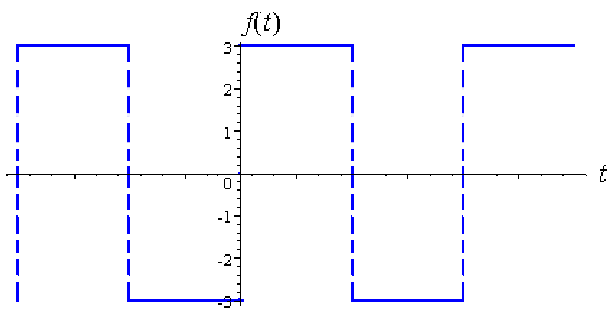
Question No: 29 (Marks: 1) - Please choose one

Which of the following is true for a periodic function whose graph is as below?



- ► **Even function**
- ► Odd function
- ► Neither even nor odd function

Question No: 30 (Marks: 1) - Please choose one



Which of the following is true for a function whose graph is given above

- ► ***An odd function***
- ► An even function
- ► Neither even nor odd

Question No: 31 (Marks: 2)

Evaluate the line integral $\int_C 2x\,dx$ where C is the line segment from $(0,2)$ to $(2,6)$

Question No: 32 (Marks: 2)

Use Wallis sine formula to evaluate $\int_0^{\frac{\pi}{2}} \sin^5 x\,dx$

Question No: 33 (Marks: 2)

Find Laplace Transform of the function $F(t)$ if $F(t) = \sin 2t$.

Question No: 34 (Marks: 3)

Find $div\,\vec{F}$, if $\vec{F} = (3x+y)\hat{i} + xy^2z\,\hat{j} + (xz^2)\hat{k}$

Question No: 35 (Marks: 3)

Determine whether the following differential is exact or not.

$dz = (4x^3y + 2xy^3)\,dx + (x^4 + 3x^2y^2)\,dy$

Question No: 36 (Marks: 3)

Prove whether the following function is even, odd or neither.

$f(x) = x^3e^x$

Question No: 37 (Marks: 5)

Evaluate the following line integral which is independent of path.

$$\int_{(2,-2)}^{(-1,0)} (2xy^3)\,dx + (3y^2x^2)\,dy$$

Question No: 38 (Marks: 5)

Determine the fourier co-efficient b_n of the following function.

$f(x) = x^2 \qquad 0 < x < 2\pi$

Question No: 39 (Marks: 5)

Determine whether the following vector field $\vec{F}$ is conservative or not.

$\vec{F}(x, y, z) = (3x + y)\hat{i} + xy^2z\hat{j} + xz^2\hat{k}$

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All Dearz fellows

ALL IN ONE MTH301

Final term PAPERS &

MCQz

Created BY Farhan & Ali

BS (cs) 2nd sem

Hackers Group

From Mandi Bahauddin

Remember us in your prayers

Mindhacker124@gmail.com

Hearthacker124@gmail.com

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ALL IN ONE MTH301

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Created BY Farhan & Ali

BS (cs) 2nd sem

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Mindhacker124@gmail.com

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FINALTERM EXAMINATION

Spring 2010

MTH301- Calculus II

Time: 90 min

Marks: 60

Student Info	
Student ID:	
Center:	
Exam Date:	

For Teacher's Use Only

Q No.	1	2	3	4	5	6	7	8	Total
Marks									
Q No.	9	10	11	12	13	14	15	16	
Marks									
Q No.	17	18	19	20	21	22	23	24	
Marks									
Q No.	25	26	27	28	29	30	31	32	
Marks									
Q No.	33	34	35	36	37	38	39		
Marks									

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Question No: 1 (Marks: 1) - Please choose one

Intersection of two straight lines is -------------

► Surface

► Curve

► Plane

► ***Point***

Question No: 2 (Marks: 1) - Please choose one

Plane is a --------------- surface.

► One-dimensional

► ***Two-dimensional***

► Three-dimensional

► Dimensionless

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Question No: 3 (Marks: 1) - Please choose one

Let w = f(x, y, z) and x = g(r, s), y = h(r, s), z = t(r, s) then by chain rule

$\frac{\partial w}{\partial r} =$

► $\frac{\partial w}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial r} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial r}$

► $\frac{\partial w}{\partial r}\frac{\partial x}{\partial r} + \frac{\partial w}{\partial r}\frac{\partial y}{\partial r} + \frac{\partial w}{\partial r}\frac{\partial z}{\partial r}$

► $\frac{\partial w}{\partial x}\frac{\partial x}{\partial r}\frac{\partial x}{\partial s} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial r}\frac{\partial y}{\partial s} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial r}\frac{\partial z}{\partial s}$

► $\frac{\partial w}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial w}{\partial r}\frac{\partial r}{\partial y} + \frac{\partial w}{\partial r}\frac{\partial r}{\partial z}$

Question No: 4 (Marks: 1) - Please choose one

What are the parametric equations that correspond to the following vector equation?

$\vec{r}(t) = \sin^2 t\,\hat{i} + (1-\cos 2t)\,\hat{j}$

► $x = \sin^2 t \quad , \quad y = 1-\cos 2t \quad , \quad z = 0$

► $y = \sin^2 t \quad , \quad x = 1-\cos 2t \quad , \quad z = 0$

- $x = \sin^2 t$, $y = 1 - \cos 2t$, $z = 1$
- $x = \sin^2 t$, $y = \cos 2t$, $z = 1$

Question No: 5 (Marks: 1) - Please choose one

What are the parametric equations that correspond to the following vector equation?

$r(t) = (2t-1)\hat{i} - 3\sqrt{t}\,\hat{j} + \sin 3t\,\hat{k}$

- $z = 2t - 1$, $x = -3\sqrt{t}$, $y = \sin 3t$
- $y = 2t - 1$, $x = -3\sqrt{t}$, $z = \sin 3t$
- $x = 2t - 1$, $z = -3\sqrt{t}$, $y = \sin 3t$
- $x = 2t - 1$, $y = -3\sqrt{t}$, $z = \sin 3t$

Question No: 6 (Marks: 1) - Please choose one

What is the derivative of following vector-valued function?

$\vec{r}(t) = (\cos 5t, \tan t , 6\sin t)$

- $\vec{r}'(t) = \left(\frac{\sin 5t}{5}, \sec t , 6\cos t\right)$

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▶ $\vec{r}'(t) = (\frac{-\sin 5t}{5}, \sec t\ , 6\cos t)$

▶ $\vec{r}'(t) = (-5\sin 5t, \sec^2 t\ , 6\cos t)$

▶ $\vec{r}'(t) = (\sin 5t, \sec^2 t\ , -6\cos t)$

Question No: 7 (Marks: 1) - Please choose one

What is the derivative of following vector-valued function?

$\vec{r}(t) = \left(t^4, \sqrt{t+1}\ , \frac{3}{t^2} \right)$

▶ $\vec{r}'(t) = \left(4t^3, \frac{1}{\sqrt{t+1}}, \frac{-6}{t^3} \right)$

▶ $\vec{r}'(t) = \left(4t^3, \frac{1}{2\sqrt{t+1}}, \frac{6}{t^3} \right)$

▶ $\vec{r}'(t) = \left(4t^4, \frac{1}{2\sqrt{t+1}}, \frac{-6}{t^3} \right)$

▶ $\vec{r}'(t) = \left(4t^3, \frac{1}{2\sqrt{t+1}}, \frac{-6}{t^3} \right)$

Question No: 8 (Marks: 1) - Please choose one

The following differential is exact

$dz = (x^2 y + y)\ dx - x\ dy$

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► True

► False

Question No: 9 (Marks: 1) - Please choose one

Which one of the following is correct Wallis Sine formula when n is even and $n \geq 2$?

► $\int_0^{\frac{\pi}{2}} \sin^n x \, dx = \frac{\pi}{2} \frac{(n-1)}{n} \frac{(n-3)}{(n-2)} \frac{(n-5)}{(n-4)} ------- \frac{5}{6} \frac{3}{4} \frac{1}{2}$ page #182

► $\int_0^{\frac{\pi}{2}} \sin^n x \, dx = \frac{(n-1)}{n} \frac{(n-3)}{(n-2)} \frac{(n-5)}{(n-4)} ------- \frac{6}{7} \frac{4}{5} \frac{2}{3}$

► $\int_0^{\frac{\pi}{2}} \sin^n x \, dx = \frac{\pi}{2} \frac{(n)}{(n-1)} \frac{(n-2)}{(n-3)} \frac{(n-4)}{(n-5)} ------- \frac{6}{5} \frac{4}{3} \frac{2}{1}$

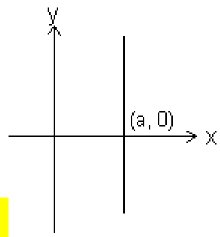
► $\int_0^{\frac{\pi}{2}} \sin^n x \, dx = \frac{(n)}{(n-1)} \frac{(n-2)}{(n-3)} \frac{(n-4)}{(n-5)} ------- \frac{6}{5} \frac{4}{3} \frac{2}{1}$

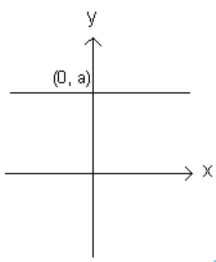
Question No: 10 (Marks: 1) - Please choose one

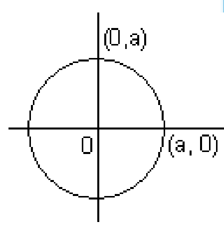
Match the following equation in polar co-ordinates with its graph.

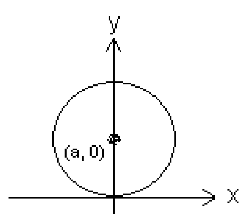
$r\cos\theta = a$

where a is an arbitrary constant

▶ 

▶ 

▶ 

▶ 

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Question No: 11 (Marks: 1) - Please choose one

If the equation of a curve, in polar co-ordinates, remains unchanged after replacing (r,θ) by $(r,\pi-\theta)$ then the curve is said to be symmetric about which of the following?

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► Initial line

► Y-axis

► Pole

Question No: 12 (Marks: 1) - Please choose one

If the equation of a curve, in polar co-ordinates, remains unchanged after replacing (r, θ) by $(-r, \theta)$ then the curve is said to be symmetric about which of the following?

► Initial line

► y-axis

► *Pole*

Question No: 13 (Marks: 1) - Please choose one

What is the amplitude of a periodic function defined by $f(x) = \sin\frac{x}{3}$?

► 0

► 1

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► $\frac{1}{3}$

► Does not exist

Question No: 14 (Marks: 1) - Please choose one

What is the period of a periodic function defined by $f(x) = 4\cos 3x$ **?**

► $\frac{\pi}{4}$

► $\frac{\pi}{3}$

► $\frac{2\pi}{3}$

► π

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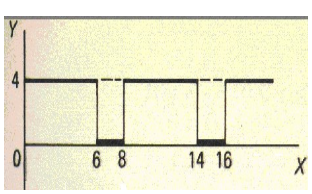
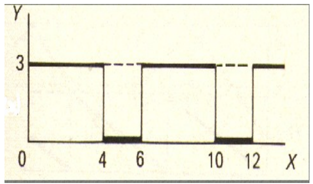
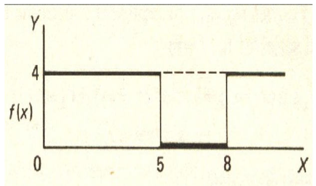
Question No: 15 (Marks: 1) - Please choose one

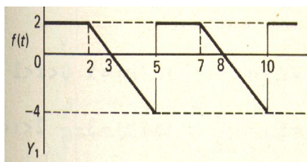
Match the following periodic function with its graph.

$$f(x) = \begin{cases} 3 & 0 < x < 4 \\ 0 & 4 < x < 6 \end{cases}$$

► 

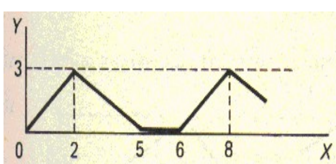
►  Page #198

► 

► 

Question No: 16 (Marks: 1) - Please choose one

What is the period of periodic function whose graph is as below?



► 2

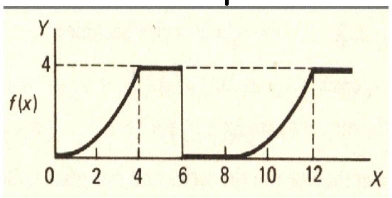
► 5

► 6

► 8

Question No: 17 (Marks: 1) - Please choose one

What is the period of periodic function whose graph is as below?



► 0

► 4

► 6

► **8**

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Question No: 18 (Marks: 1) - Please choose one

Let L **denotes the Laplace Transform.**

If $L\{F(t)\} = f(s)$ **where** s **is a constant.and** $\lim_{t \to 0}\left(\frac{F(t)}{t}\right)$ **exists then which of the following equation holds?**

► $L\left(\frac{F(t)}{t}\right) = f(s+a)$

► $L\left(\frac{F(t)}{t}\right) = f(s-a)$

► $L\left(\frac{F(t)}{t}\right) = \int_{s}^{\infty} f(s)\, ds$

► $L\left(\frac{F(t)}{t}\right) = -\frac{d}{ds}\{f(s)\}$

Question No: 19 (Marks: 1) - Please choose one

Which of the following is Laplace inverse transform of the function $f(s)$ defined by $f(s) = \frac{3}{s-2} - \frac{2}{s}$?

► $3te^{2t} - 2$

► $3e^{2t} - 2t$

► $3e^{2t} - 2$

► None of these.

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Question No: 20 (Marks: 1) - Please choose one

Let (x_1, y_1, z_1) and (x_2, y_2, z_2) be any two points in three dimensional space. What does the formula $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$ calculates?

► Distance between these two points

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► Midpoint of the line joining these two points

► Ratio between these two points

Question No: 21 (Marks: 1) - Please choose one

Let the functions $P(x, y)$ and $Q(x, y)$ are finite and continuous inside and at the boundary of a closed curve C in the xy-plane. If $(P\,dx + Q\,dy)$ is an exact differential then

$$\oint_C (P\,dx + Q\,dy) =$$

► Zero

► One

► Infinite

Question No: 22 (Marks: 1) - Please choose one

What is Laplace transform of the function $F(t)$ if $F(t) = t$?

► $L\{t\} = \dfrac{1}{s}$

► $L\{t\} = \dfrac{1}{s^2}$

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► $L\{t\} = e^{-s}$

► $L\{t\} = s$

Question No: 23 (Marks: 1) - Please choose one

What is the value of $L\{e^{5t}\}$ if L denotes laplace transform?

► $L\{e^{5t}\} = \dfrac{1}{s-5}$

► $L\{e^{5t}\} = \dfrac{s}{s^2+25}$

► $L\{e^{5t}\} = \dfrac{5}{s^2+25}$

► $L\{e^{5t}\} = \dfrac{5!}{s^6}$

Question No: 24 (Marks: 1) - Please choose one

Evaluate the line integral $\int_C (3x+2y)\,dx+(2x-y)\,dy$ where C is the line segment from (0, 0) to (0, 2).

► 1

► 0

► 2

► -2

Question No: 25 (Marks: 1) - Please choose one

Evaluate the line integral $\int_C (2x+y)\,dx + (x^2-y)\,dy$ **where C is the line segment from (0, 0) to (2, 0).**

► 0

► -4

► 4

► Do not exist

Question No: 26 (Marks: 1) - Please choose one

Which of the following are direction ratios for the line joining the points $(1, 3, 5)$ and $(2, -1, 4)$?

► 3, 2 and 9

► 1, -4 and -1

► 2, -3 and 20

► 0.5, -3 and 5/4

Question No: 27 (Marks: 1) - Please choose one

$If\ R = \{(x,y)/0 \le x \le 2\ and\ 1 \le y \le 4\}, then$

$$\iint_R (6x^2 + 4xy^3)dA =$$

► $\int_1^4 \int_0^2 (6x^2 + 4xy^3)dydx$

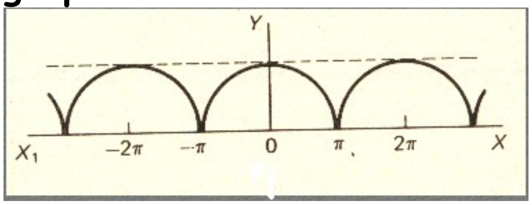
► $\int_0^2 \int_1^4 (6x^2 + 4xy^3)dxdy$

► $\int_1^4 \int_0^2 (6x^2 + 4xy^3)dxdy$

► $\int_2^4 \int_0^1 (6x^2 + 4xy^3)dxdy$

Question No: 28 (Marks: 1) - Please choose one

Which of the following is true for a periodic function whose graph is as below?

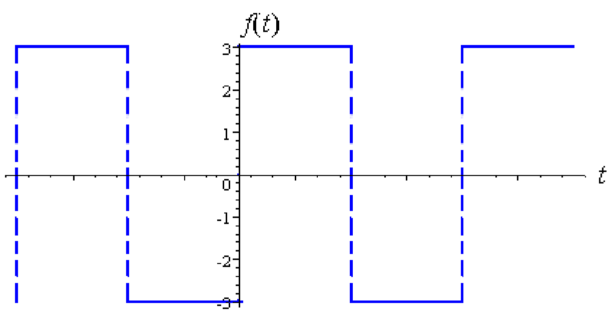


► Even function page209

► Odd function

► Neither even nor odd function

Question No: 29 (Marks: 1) - Please choose one



Which of the following is true for a function whose graph is given above.

► An odd function pg 212

► An even function

► Neither even nor odd

Question No: 30 (Marks: 1) - Please choose one

At each point of domain, the function ----------------

► **Is defined**

► Is continuous

► Is infinite

► Has a limit

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Question No: 31 (Marks: 2)

Determine whether the following differential is exact or not.

$dz = 4x^3y^3\ dx + 3x^4y^2\ dy$

Solution:

$$dz = 4x^3y^3\ dx + 3x^4y^2\ dy$$

$$\frac{\partial p}{\partial y} = 12x^3y^2$$

$$\frac{\partial Q}{\partial X} = 12x^3y^2$$

$$\frac{\partial p}{\partial y} = \frac{\partial Q}{\partial X}$$

yes

Question No: 32 (Marks: 2)

Evaluate

$$\int_{-\pi}^{\pi} \sin\ nx\ dx$$

where n is an integer other than zero.

Solution:

$$\int_{-\pi}^{\pi} \sin\ nx\ dx$$

$$= \left[\frac{-\cos nx}{n}\right]_{-\pi}^{\pi}$$

$$= \left[\frac{-\cos n\pi}{n} + \frac{\cos n\pi}{n}\right]$$

$$= \frac{1}{n}(-\cos n\pi + \cos n\pi)$$

$$= 0$$

Question No: 33 (Marks: 2)

Find Laplce transform of the function $F(t)$ if $F(t) = e^{3t}$

Solution:

$$L(e^{3t}) = \int_0^\infty e^{3t} - e^{-st}$$

$$= \int_0^\infty e^{-(s-3)t}.dt$$

$$= \{\frac{e^{-(s-3)t}}{-(s-3)}\} \lim 0 - \infty$$

$$= \frac{-1}{s-3}(\frac{1}{e^{(s-3)t}})$$

$$= \frac{-1}{s-3}(0-1)$$

$$= \frac{1}{s-3} \text{......Ans}$$

Question No: 34 (Marks: 3)

Determine the Fourier co-efficient a_0 of the periodic function defined below:

$f(x) = 2x+1 \qquad 0 < x < 2$

Solution:

$$a_\circ = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)dx$$

$f(x) = (2x+1)$

$(0,2)$

$$= \int_0^2 (2x+1)dx$$

$$= \left[x^2 + x\right]_0^2$$

$$= 6$$

Question No: 35 (Marks: 3)

Determine whether the following differential is exact or not.

$$dz = (3x^2e^{2y} - 2y^2e^{3x})\ dx + (2x^3e^{2y} - 2ye^{3x})\,dy$$

Solution:

dz = Pdx + Qdy

Therefore,

For dz to be an exact differential it must satisfy $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$

But this test fails becuase $\frac{\partial P}{\partial y} \neq \frac{\partial Q}{\partial x}$

Not Exact

Question No: 36 (Marks: 3)

Use Wallis sine formula to evaluate $\int_0^{\frac{\pi}{2}} \left(\sin^3 x + \sin^5 x\right) dx$

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Solution:

$$\int_0^{\frac{\pi}{2}} \sin^3 x dx$$

$$= \frac{n-1}{n}.$$

$$= \frac{3-1}{3}$$

$$= \frac{2}{3}$$

$$\int_0^{\frac{\pi}{2}} \sin^5 x dx$$

$$= \frac{n-1}{n}.\frac{n-3}{n-2}$$

$$= \frac{5-1}{5}.\frac{5-3}{5-2}$$

$$= \frac{4}{5}.\frac{2}{3}$$

$$\int_0^{\frac{\pi}{2}} \left(\sin^3 x + \sin^5 x\right) dx$$

$$= \frac{2}{3} + \frac{4}{5}.\frac{2}{3}$$

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Question No: 37 (Marks: 5)

Evaluate the following line integral which is independent of path.

$$\int_{(0,0)}^{(3,2)} (2xe^y)\, dx + (x^2 e^y)\, dy$$

Solution:

$$p = \frac{\partial z}{\partial x} = 2e^y \qquad \int 2e^y\,dx$$

$$Q = \frac{\partial z}{\partial y} = x^2 e^y \qquad \int x^2 e^y\,dy$$

$$z = \int_{(0,0)}^{(3,2)} 2xe^y + x^2 ye^y$$

$$z = 6e^2 + 18e^2$$

$$z = 24e^2$$

Question No: 38 (Marks: 5)

Determine the Fourier coefficients b_n for a periodic function $f(t)$ of period 2 defined by

$$f(t) = \begin{cases} 4\,(1+t) & -1 < t < 0 \\ 0 & 0 < t < 1 \end{cases}$$

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Solution:

$$bn = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx dx$$

$$= \frac{1}{\pi}\int_{-1}^{1} 4(1+t)\sin nx dx$$

$$= \frac{1}{\pi}\left[\frac{-4(1+t)\cos nx}{n}\right]_{-1}^{1}$$

$$= \frac{-4(1+t)}{\pi n}\left[\cos n(1) - \cos n(-1)\right]$$

$$= \frac{-4(1+t)}{\pi n}(\cos n + \cos n)$$

Question No: 39 (Marks: 5)

Determine whether the following vector field $\vec{F}$ is conservative or not.

$$\vec{F}(x,\ y,\ z) = (4x - z)\hat{i} + (3y + z)\hat{j} + (y - x)\hat{k}$$

……………………………………

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ALL IN ONE MTH301
Final term PAPERS & MCQz
Created BY Farhan & Ali
BS (cs) 2nd sem
Hackers Group
From Mandi Bahauddin
Remember us in your prayers

Mindhacker124@gmail.com
Hearthacker124@gmail.com

FINALTERM EXAMINATION
Spring 2010
MTH301- Calculus II (Session - 2)

Time: 90 min
Marks: 60

Student Info	
StudentID:	$$
Center:	OPKST
ExamDate:	19 Aug 2010

For Teacher's Use Only									
Q No.	1	2	3	4	5	6	7	8	Total
Marks									
Q No.	9	10	11	12	13	14	15	16	
Marks									
Q No.	17	18	19	20	21	22	23	24	
Marks									
Q No.	25	26	27	28	29	30	31	32	
Marks									
Q No.	33	34	35	36	37	38	39		
Marks									

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Question No: 1 (Marks: 1) - Please choose one

------------------- planes intersect at right angle to form three dimensional space.

- ► **Three**
- ► Four
- ► Eight
- ► Twelve

Question No: 2 (Marks: 1) - Please choose one

If the positive direction of x, y axes are known then ------------ the positive direction of z-axis.

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- ► Horizontal rightward direction is
- ► Vertical upward direction is
- ► **Left hand rule tells**
- ► Right hand rule tells

Question No: 3 (Marks: 1) - Please choose one

What is the distance between points (3, 2, 4) and (6, 10, -1)?

- ► $7\sqrt{2}$
- ► $2\sqrt{6}$
- ► $\sqrt{34}$
- ► $7\sqrt{3}$

Question No: 4 (Marks: 1) - Please choose one

The equation $ax+by+cz+d=0$, where a,b,c,d are real numbers, is the general equation of which of the following?

- ► Plane page # 12
- ► Line
- ► Curve
- ► Circle

Question No: 5 (Marks: 1) - Please choose one

The spherical co-ordinates of a point are $\left(\sqrt{3}, \frac{\pi}{3}, \frac{\pi}{2}\right)$. What are its cylindrical co-ordinates?

- ► $\left(\frac{\sqrt{3}}{2}, \frac{3}{2}, 0\right)$
- ► $\left(\sqrt{3}\cos\frac{\pi}{3}, \sqrt{3}\sin\frac{\pi}{3}, 0\right)$
- ► $\left(\sqrt{3}\sin\frac{\pi}{3}, \frac{\pi}{2}, \sqrt{3}\cos\frac{\pi}{3}\right)$
- ► $\left(\sqrt{3}, \frac{\pi}{3}, 0\right)$

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Question No: 6 (Marks: 1) - Please choose one

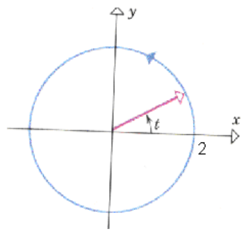
Domain of the function $f(x, y)=\sqrt{y-x^2}$ is

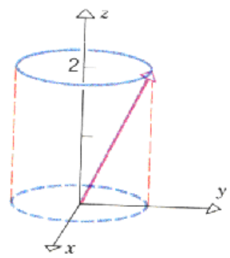
- ► $y < x^2$
- ► $y \geq x^2$
- ► $y \neq x^2$
- ► Entire space

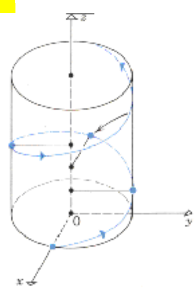
Question No: 7 (Marks: 1) - Please choose one

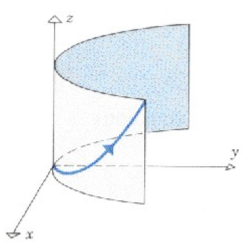
Match the following vector-valued function with its graph.

$r(t) = \cos t\,\hat{i} + \sin t\,\hat{j} + 2\hat{k}$ And $0 \le t \le 2\pi$

- 

- 

Correct
- 

- 

Question No: 8 (Marks: 1) - Please choose one

What are the parametric equations that correspond to the following vector equation?

$\vec{r}(t) = \sin^2 t\,\hat{i} + (1 - \cos 2t)\,\hat{j}$

- $x = \sin^2 t$, $y = 1 - \cos 2t$, $z = 0$
- $y = \sin^2 t$, $x = 1 - \cos 2t$, $z = 0$

- ▶ $x=\sin^2 t$, $y=1-\cos 2t$, $z=1$
- ▶ $x=\sin^2 t$, $y=\cos 2t$, $z=1$

Question No: 9 (Marks: 1) - Please choose one

What are the parametric equations that correspond to the following vector equation?

$r(t)=(2t-1)\hat{i}-3\sqrt{t}\,\hat{j}+\sin 3t\,\hat{k}$

- ▶ $z=2t-1$, $x=-3\sqrt{t}$, $y=\sin 3t$
- ▶ $y=2t-1$, $x=-3\sqrt{t}$, $z=\sin 3t$
- ▶ $x=2t-1$, $z=-3\sqrt{t}$, $y=\sin 3t$
- ▶ $x=2t-1$, $y=-3\sqrt{t}$, $z=\sin 3t$

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Question No: 10 (Marks: 1) - Please choose one

Is the following vector-valued function $\vec{r}(t)$ continuous at $t=1$? If not, why?

$\vec{r}(t)=\left(\dfrac{t+1}{t-1}, t^2, 2t\right)$

- ▶ $\vec{r}(t)$ is continuous at $t=1$
- ▶ $\vec{r}(1)$ is not defined
- ▶ $\vec{r}(1)$ is defined but $\lim\limits_{t\to 1}\vec{r}(t)$ does not exist
- ▶ **$\vec{r}(1)$ is defined and $\lim\limits_{t\to 1}\vec{r}(t)$ exists but these two numbers are not equal.**

Question No: 11 (Marks: 1) - Please choose one

Which one of the following is correct Wallis Sine formula when n is even and $n\geq 2$?

► $\int_0^{\frac{\pi}{2}} \sin^n x \, dx = \frac{\pi}{2} \frac{(n-1)}{n} \frac{(n-3)}{(n-2)} \frac{(n-5)}{(n-4)} ------- \frac{5}{6} \frac{3}{4} \frac{1}{2}$ page183

► $\int_0^{\frac{\pi}{2}} \sin^n x \, dx = \frac{(n-1)}{n} \frac{(n-3)}{(n-2)} \frac{(n-5)}{(n-4)} ------- \frac{6}{7} \frac{4}{5} \frac{2}{3}$

► $\int_0^{\frac{\pi}{2}} \sin^n x \, dx = \frac{\pi}{2} \frac{(n)}{(n-1)} \frac{(n-2)}{(n-3)} \frac{(n-4)}{(n-5)} ------- \frac{6}{5} \frac{4}{3} \frac{2}{1}$

► $\int_0^{\frac{\pi}{2}} \sin^n x \, dx = \frac{(n)}{(n-1)} \frac{(n-2)}{(n-3)} \frac{(n-4)}{(n-5)} ------- \frac{6}{5} \frac{4}{3} \frac{2}{1}$

Question No: 12 (Marks: 1) - Please choose one

Which one of the following is correct Wallis Cosine formula when n is odd and $n \geq 3$?

► $\int_0^{\frac{\pi}{2}} \cos^n x \, dx = \frac{\pi}{2} \frac{(n-1)}{n} \frac{(n-3)}{(n-2)} \frac{(n-5)}{(n-4)} ------- \frac{5}{6} \frac{3}{4} \frac{1}{2}$

► $\int_0^{\frac{\pi}{2}} \cos^n x \, dx = \frac{\pi}{2} \frac{(n)}{(n-1)} \frac{(n-2)}{(n-3)} \frac{(n-4)}{(n-5)} ------- \frac{6}{5} \frac{4}{3} \frac{2}{1}$

► $\int_0^{\frac{\pi}{2}} \cos^n x \, dx = \frac{(n)}{(n-1)} \frac{(n-2)}{(n-3)} \frac{(n-4)}{(n-5)} ------- \frac{6}{5} \frac{4}{3} \frac{2}{1}$

► $\int_0^{\frac{\pi}{2}} \cos^n x \, dx = \frac{(n-1)}{n} \frac{(n-3)}{(n-2)} \frac{(n-5)}{(n-4)} ------- \frac{6}{7} \frac{4}{5} \frac{2}{3}$ page183

Question No: 13 (Marks: 1) - Please choose one

If the equation of a curve, in polar co-ordinates, remains unchanged after replacing (r, θ) by $(r, \pi - \theta)$ then the curve is said to be symmetric about which of the following?

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► Initial line
► y-axis
► Pole

Question No: 14 (Marks: 1) - Please choose one

If $a > 0$, then the equation, in polar co-ordinates, of the form $r^2 = a^2 \cos 2\theta$ represent which of the following family of curves?

► Leminscate page 130
► Cardiods
► Rose curves
► Spiral

Question No: 15 (Marks: 1) - Please choose one

What is the period of a periodic function defined by $f(x) = \sin\frac{x}{2}$?

► $\frac{\pi}{2}$
► π
► $\frac{3\pi}{2}$
► 4π

Question No: 16 (Marks: 1) - Please choose one

What is the period of periodic function whose graph is as below?

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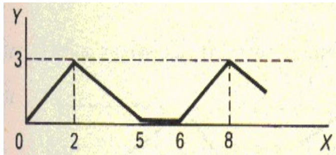
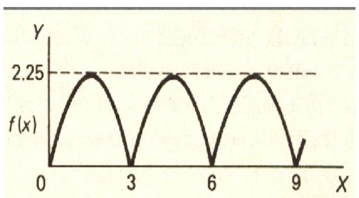


► 2
► 5
► 6
► 8

Question No: 17 (Marks: 1) - Please choose one

What is the period of periodic function whose graph is as below?



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► 0
► 2.25
► 3
► 6

Question No: 18 (Marks: 1) - Please choose one

Let L **denotes the Laplace Transform.**
If $L\{F(t)\} = f(s)$ **where** s **is a constant, then which of the following equation holds?**

► $L\{t\ F(t)\} = -\frac{d}{ds}\{f(s)\}$
► $L\{t\ F(t)\} = f(s+t)$
► $L\{t\ F(t)\} = f(s)$
► $L\{t F(t)\} = \int_{s}^{\infty} f(s)\ ds$

Question No: 19 (Marks: 1) - Please choose one

The graph of an odd function is symmetrical about ----------------

- ► x-axis
- ► y-axis
- ► **origin**

Question No: 20 (Marks: 1) - Please choose one

Consider the function $f(x, y, z)=\sqrt{1-x^2-y^2-z^2}$ **. What is the value of** $f\left(0, \frac{1}{2}, \frac{1}{2}\right)$□

- ► $f\left(0, \frac{1}{2}, \frac{1}{2}\right)=\sqrt{\frac{1}{2}}$
- ► $f\left(0, \frac{1}{2}, \frac{1}{2}\right)=2$
- ► $f\left(0, \frac{1}{2}, \frac{1}{2}\right)=\frac{1}{2}$
- ► $f\left(0, \frac{1}{2}, \frac{1}{2}\right)=0$

Question No: 21 (Marks: 1) - Please choose one

The path of integration of a line integral must be -------------

- ► straight and single-valued
- ► **continuous and single-valued**
- ► straight and multiple-valued
- ► continuous and multiple-valued

Question No: 22 (Marks: 1) - Please choose one

Sign of line integral is reversed when -----------

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► path of integration is divided into parts.
► path of integration is parallel to y-axis.
► **direction of path of integration is reversed.**
► path of integration is parallel to x-axis.

Question No: 23 (Marks: 1) - Please choose one

Let the functions $P(x, y)$ and $Q(x, y)$ are finite and continuous inside and at the boundary of a closed curve C in the xy-plane.

If $(P\,dx + Q\,dy)$ is an exact differential then

$\oint_C (P\,dx + Q\,dy) =$

► **Zero**

► One

► Infinite

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Question No: 24 (Marks: 1) - Please choose one

What is the value of $L\{e^{5t}\}$ if L denotes laplace transform?

► $L\{e^{5t}\} = \dfrac{1}{s-5}$

► $L\{e^{5t}\} = \dfrac{s}{s^2+25}$

► $L\{e^{5t}\} = \dfrac{5}{s^2+25}$

► $L\{e^{5t}\} = \dfrac{5!}{s^6}$

Question No: 25 (Marks: 1) - Please choose one

What is laplace transform of the function $F(t)$ if $F(t) = \sin 3t$?

► $L\{\sin 3t\} = \dfrac{3}{s^2+9}$

► $L\{\sin 3t\} = \dfrac{s}{s^2+9}$

► $L\{\sin 3t\} = \frac{1}{s-3}$

► $L\{\sin 3t\} = \frac{3!}{s^4}$

Question No: 26 (Marks: 1) - Please choose one

If L denotes laplace transform then
$L\{te^{5t}\} =$

► $L\{te^{5t}\} = \frac{1}{s^2-5}$

► $L\{te^{5t}\} = \frac{1}{s^2+5}$

► $L\{te^{5t}\} = \frac{1}{(s+5)^2}$

► $L\{te^{5t}\} = \frac{1}{(s-5)^2}$

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Question No: 27 (Marks: 1) - Please choose one

Evaluate the line integral $\int_C (3x+2y)\,dx+(2x-y)\,dy$ **where C is the line segment from (0, 0) to (0, 2).**

► 1

► 0

► 2

► -2

Question No: 28 (Marks: 1) - Please choose one

Evaluate the line integral $\int_C (2x+y)\,dx + (x^2 - y)\,dy$ where C is the line segment from (0, 0) to (0, 2).

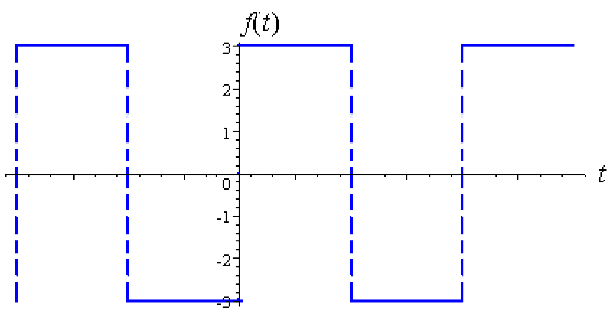
- ► -4
- ► **-2**
- ► 0
- ► 2

Question No: 29 (Marks: 1) - Please choose one

Divergence of a vector function is always a ------------

- ► Scalar
- ► **Vector**

Question No: 30 (Marks: 1) - Please choose one



Which of the following is true for a function whose graph is given above

- ► **An odd function**
- ► An even function
- ► Neither even nor odd

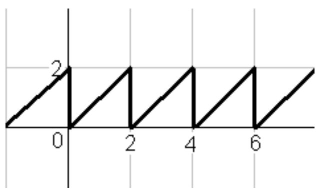
Question No: 31 (Marks: 2)

Does the following limit exist? If yes find its value, if no give reason

$$\lim_{t\to 0}\left[(e^{2t}+5)\hat{i}+(t^2+2t-3)\hat{j}+\left(\frac{1}{t}\right)\hat{k}\right]$$

Question No: 32 (Marks: 2)

Define the periodic function whose graph is shown below.



Question No: 33 (Marks: 2)

Find Laplace Transform of the function $F(t)$ if $F(t)=t^4$

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Solution:

The Laplace transform of the given function will be:

$$f(t)=t^4$$

$$L\{t^4\}=\frac{4!}{s^5}$$

Question No: 34 (Marks: 3)

Determine whether the following differential is exact or not.

$$dz=(4x^3y+2xy^3)\ dx+(x^4+3x^2y^2)\ dy$$

Question No: 35 (Marks: 3)

Use Wallis sine formula to evaluate $\int_0^{\frac{\pi}{2}} \left(\sin^3 x + \sin^5 x\right) dx$

Solution:

$$\int_0^{\frac{\pi}{2}} \sin^8 x dx = \frac{7}{8} \cdot \frac{5}{6} \cdot \frac{3}{4} \cdot \frac{1}{2} - - - - - - - - - \frac{\pi}{2}$$

Question No: 36 (Marks: 3)

Find Laplace transform of the function $F(t)$ if

$F(t) = e^{2t} \sin 3t$

Solution:

Laplace transform will be

$$L(t) = e^{2t} ....1$$

$$= \frac{1}{s-2}$$

$$L(t) = \sin 3t...........2$$

$$L(t) = \frac{a}{s^2 + 3^2}$$

$$L(t) = \frac{a}{s^2 + 9}$$

Combining,

$$L(t) = (\frac{1}{s-2})(\frac{a}{s^2+9})$$

Question No: 37 (Marks: 5)

Using definite integral, find area of the region that is enclosed between the curves $y = x^2$ *and* $y = \sqrt{x}$

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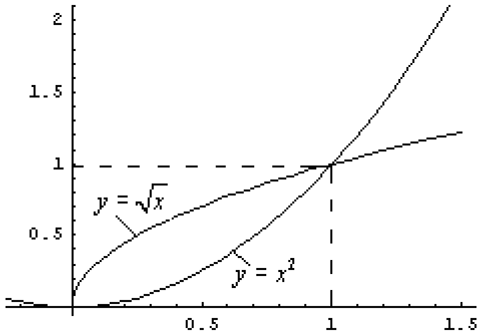

Question No: 38 (Marks: 5)

Determine the fourier co-efficient b_n of the following function.

$f(x) = x^2 \qquad 0 < x < 2\pi$

Question No: 39 (Marks: 5)

Determine whether the following vector field $\vec{F}$ is conservative or not.

$\vec{F}(x, y, z) = (4x - z)\hat{i} + (3y + z)\hat{j} + (y - x)\hat{k}$

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FINALTERM EXAMINATION
Fall 2009
MTH301- Calculus II

Time: 120 min
Marks: 80

Question No: 1 (Marks: 1) - Please choose one

π is an example of -----------

► **Irrational numbers**

► Rational numbers

► Integers

► Natural numbers

Question No: 2 (Marks: 1) - Please choose one

Straight line is a special kind of ----------------

► Surface

► **Curve**

► Plane

► Parabola

Question No: 3 (Marks: 1) - Please choose one

An ordered triple corresponds to ----------- in three dimensional space.

► ***A unique point***

► A point in each octant

► Three points

► Infinite number of points

Question No: 4 (Marks: 1) - Please choose one

The angles which a line makes with positive x ,y and z-axis are known as ------------

► Direction cosines

► Direction ratios

► **Direction angles**

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Question No: 5 (Marks: 1) - Please choose one

Is the function $f(x, y)$ continuous at origin? If not, why?

$f(x, y) = 4xy + \sin 3x^2 y$

► $f(x, y)$ **is continuous at origin**

► $f(0, 0)$ is not defined

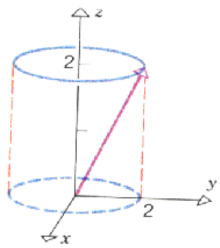
► $f(0, 0)$ is defined but $\lim_{(x,y)\to(0,0)} f(x, y)$ does not exist

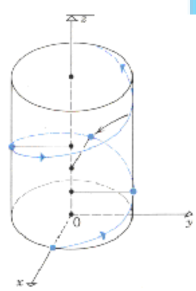
► $f(0, 0)$ is defined and $\lim_{(x,y)\to(0,0)} f(x, y)$ exists but these two numbers are not equal.

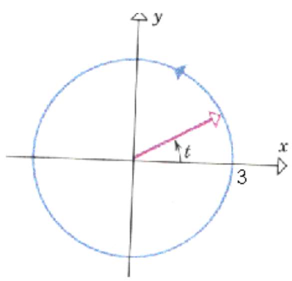
Question No: 6 (Marks: 1) - Please choose one

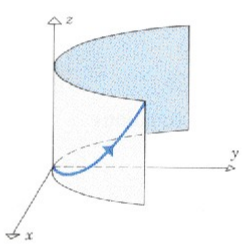
Match the following vector-valued function with its graph.

$r(t) = 3\cos t\,\hat{i} + 3\sin t\,\hat{j}$ And $0 \le t \le 2\pi$

► 

► 

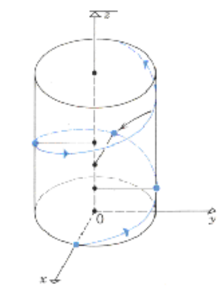
► 

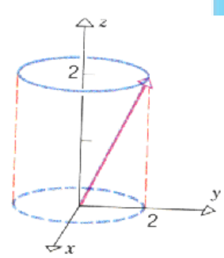
► 

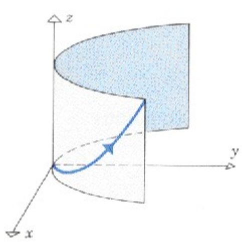
Question No: 7 (Marks: 1) - Please choose one

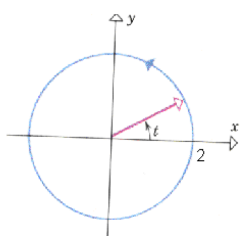
Match the following vector-valued function with its graph.

$r(t) = t\hat{i} + t^2\hat{j} + t^3\hat{k}$ and $t \geq 0$

► 

► 

► 

► 

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Question No: 8 (Marks: 1) - Please choose one

What are the parametric equations that correspond to the following vector equation?

$\vec{r}(t) = \sin^2 t\,\hat{i} + (1 - \cos 2t)\,\hat{j}$

- ► $x = \sin^2 t$, $y = 1 - \cos 2t$, $z = 0$
- ► $y = \sin^2 t$, $x = 1 - \cos 2t$, $z = 0$
- ► $x = \sin^2 t$, $y = 1 - \cos 2t$, $z = 1$
- ► $x = \sin^2 t$, $y = \cos 2t$, $z = 1$

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Question No: 9 (Marks: 1) - Please choose one

Is the following vector-valued function $\vec{r}(t)$ continuous at $t = 0$? If not, why?

$\vec{r}(t) = (4\cos t, \sqrt{t}\ , 4\sin t)$

- ► $\vec{r}(0)$ is not defined
- ► $\vec{r}(0)$ is defined but $\lim_{t\to 0} \vec{r}(t)$ does not exist
- ► $\vec{r}(0)$ is defined and $\lim_{t\to 0} \vec{r}(t)$ exists but these two numbers are not equal.

► $\vec{r}(t)$ is continuous at $t = 0$

Question No: 10 (Marks: 1) - Please choose one

What is the derivative of following vector-valued function?

$\vec{r}(t) = (\cos 5t,\ \tan t\ ,\ 6\sin t)$

► $\vec{r}'(t) = \left(\dfrac{\sin 5t}{5},\ \sec t\ ,\ 6\cos t\right)$

► $\vec{r}'(t) = (\dfrac{-\sin 5t}{5},\ \sec t\ ,\ 6\cos t)$

► $\vec{r}'(t) = (-5\sin 5t,\ \sec^2 t\ ,\ 6\cos t)$

► $\vec{r}'(t) = (\sin 5t,\ \sec^2 t\ ,\ -6\cos t)$

Question No: 11 (Marks: 1) - Please choose one

The following differential is exact

$dz = (3x^2 y + 2)\ dx + (x^3 + y)\ dy$

- ► True

► False

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Question No: 12 (Marks: 1) - Please choose one

The following differential is exact

$dz = (3x^2 + 4xy)\ dx + (2x^2 + 2y)\ dy$

► True

► False

Question No: 13 (Marks: 1) - Please choose one

Which one of the following is correct Wallis Sine formula when n is odd and $n \geq 3$?

► $\int_0^{\frac{\pi}{2}} \sin^n x\ dx = \frac{\pi}{2} \frac{(n-1)}{n} \frac{(n-3)}{(n-2)} \frac{(n-5)}{(n-4)} \text{--------} \frac{5}{6} \frac{3}{4} \frac{1}{2}$

► $\int_0^{\frac{\pi}{2}} \sin^n x\ dx = \frac{\pi}{2} \frac{(n)}{(n-1)} \frac{(n-2)}{(n-3)} \frac{(n-4)}{(n-5)} \text{--------} \frac{6}{5} \frac{4}{3} \frac{2}{1}$

► $\int_0^{\frac{\pi}{2}} \sin^n x\ dx = \frac{(n-1)}{n} \frac{(n-3)}{(n-2)} \frac{(n-5)}{(n-4)} \text{--------} \frac{6}{7} \frac{4}{5} \frac{2}{3}$

► $\int_0^{\frac{\pi}{2}} \sin^n x\ dx = \frac{(n)}{(n-1)} \frac{(n-2)}{(n-3)} \frac{(n-4)}{(n-5)} \text{--------} \frac{6}{5} \frac{4}{3} \frac{2}{1}$

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Question No: 14 (Marks: 1) - Please choose one

Which of the following is correct?

► $\int_0^{\frac{\pi}{2}} \sin^4 x \, dx = \frac{3}{16}$

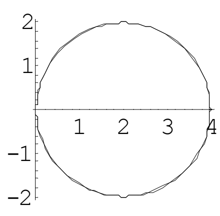
► $\int_0^{\frac{\pi}{2}} \sin^4 x \, dx = \frac{3\pi}{16}$

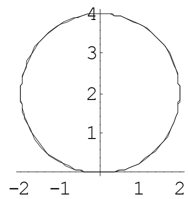
► $\int_0^{\frac{\pi}{2}} \sin^4 x \, dx = \frac{3}{8}$

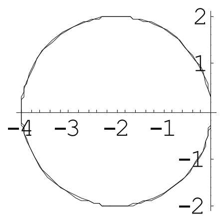
► $\int_0^{\frac{\pi}{2}} \sin^4 x \, dx = \frac{2\pi}{3}$

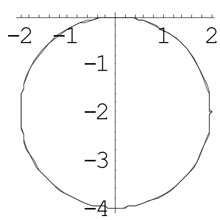
Question No: 15 (Marks: 1) - Please choose one

Match the following equation in polar co-ordinates with its graph.
$r = 4\sin\theta$

► 

► 

► 

► 

Question No: 16 (Marks: 1) - Please choose one

If the equation of a curve, in polar co-ordinates, remains unchanged after replacing (r, θ) by $(r, \pi - \theta)$ then the curve is said to be symmetric about which of the following?

► Initial line

► **y-axis**

► Pole

Question No: 17 (Marks: 1) - Please choose one

What is the period of a periodic function defined by $f(x) = \sin\frac{x}{2}$?

- ▶ $\frac{\pi}{2}$
- ▶ π
- ▶ $\frac{3\pi}{2}$
- ▶ 4π

Question No: 18 (Marks: 1) - Please choose one

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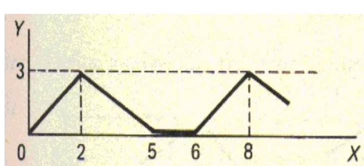
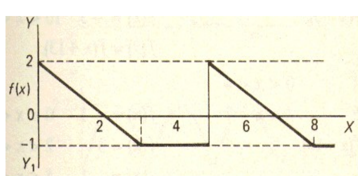
Match the following periodic function with its graph.

$$f(x) = \begin{cases} \frac{3}{4}x & 0 < x < 4 \\ 7 - x & 4 < x < 10 \\ -3 & 10 < x < 13 \end{cases}$$

▶ 

▶ 

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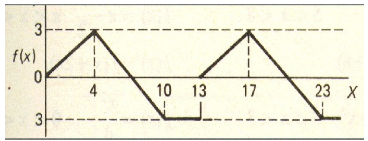
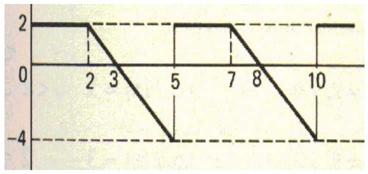
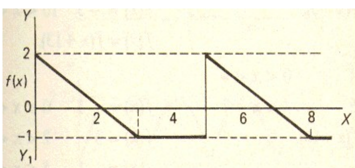
► 

► 

Question No: 19 (Marks: 1) - Please choose one

What is the period of periodic function whose graph is as below?



► 2

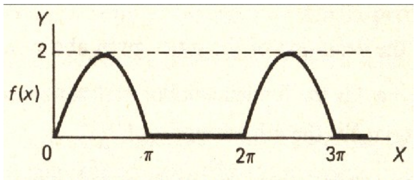
► 3

► 4

► 5

Question No: 20 (Marks: 1) - Please choose one

What is the period of periodic function whose graph is as below?



► 0

► 2

► π

► 2π not sure

Question No: 21 (Marks: 1) - Please choose one

Polar co-ordinates of a point are $\left(-2, \frac{-3\pi}{2}\right)$. Which of the following is another possible polar co-ordinates representation of this point?

► $\left(2, \frac{-\pi}{4}\right)$

► $\left(2, \frac{-\pi}{2}\right)$

► $\left(2, \frac{-\pi}{3}\right)$

► $\left(2, \frac{3\pi}{4}\right)$

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Question No: 22 (Marks: 1) - Please choose one

The function $f(x) = x^3 e^x$ is -------------

► Even function

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► Odd function

► Neither even nor odd

Question No: 23 (Marks: 1) - Please choose one

The graph of an even function is symmetrical about ---------------

► x-axis

► **y-axis** **page 208**

► origin

Question No: 24 (Marks: 1) - Please choose one

At which point the vertex of parabola, represented by the equation $y = x^2 - 4x + 3$, occurs?

► $(0, 3)$

► $(2, -1)$

► $(-2, 15)$

► $(1, 0)$ pge 08

Question No: 25 (Marks: 1) - Please choose one

The equation $y = x^2 - 4x + 2$ represents a parabola. Find a point at which the vertex of given parabola occurs?

► $(2, -2)$

► $(-4, 34)$

► $(0, 0)$

► $(-2, 14)$

Question No: 26 (Marks: 1) - Please choose one

Is the function $f(x, y)$ continuous at origin? If not, why?

$$f(x, y) = \frac{xy}{x^2 + y^2}$$

► $f(x, y)$ is continuous at origin

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► $\lim_{(x,y)\to(0,0)} f(x,y)$ does not exist

► $f(0,0)$ is defined and $\lim_{(x,y)\to(0,0)} f(x,y)$ exists but these two numbers are not equal.

Question No: 27 (Marks: 1) - Please choose one

Sign of line integral is reversed when -----------

► path of integration is divided into parts.

► path of integration is parallel to y-axis.

► **direction of path of integration is reversed.**

► path of integration is parallel to x-axis.

Question No: 28 (Marks: 1) - Please choose one

What is Laplace transform of a function F(t)?

(s is a constant)

► $\int_0^s e^{-st} F(t)\, dt$

► $\int_0^\infty e^{st} F(t)\, dt$

► $\int_{-\infty}^{\infty} e^{-st} F(t)\, dt$

► $\int_{0}^{\infty} e^{-st} F(t)\, dt$

Question No: 29 (Marks: 1) - Please choose one

What is the value of $L\{e^{5t}\}$ if L denotes laplace transform?

► $L\{e^{5t}\} = \frac{1}{s-5}$

► $L\{e^{5t}\} = \frac{s}{s^2+25}$

► $L\{e^{5t}\} = \frac{5}{s^2+25}$

► $L\{e^{5t}\} = \frac{5!}{s^6}$

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Question No: 30 (Marks: 1) - Please choose one

What is the Laplace Inverse Transform of $\frac{1}{s+1}$

► $L^{-1}\left\{\frac{1}{s+1}\right\} = t+1$

► $L^{-1}\left\{\frac{1}{s+1}\right\} = e^{-t} + e^{t}$

► $L^{-1}\left\{\frac{1}{s+1}\right\} = e^{t}$

► $L^{-1}\left\{\frac{1}{s+1}\right\} = e^{-t}$

Question No: 31 (Marks: 1) - Please choose one

What is Laplace Inverse Transform of $\frac{5}{s^2+25}$

► $L^{-1}\left\{\frac{5}{s^2+25}\right\} = \sin 5t$

► $L^{-1}\left\{\frac{5}{s^2+25}\right\} = \cos 5t$

► $L^{-1}\left\{\frac{5}{s^2+25}\right\} = \sin 25t$

► $L^{-1}\left\{\frac{5}{s^2+25}\right\} = \cos 25t$

Question No: 32 (Marks: 1) - Please choose one

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What is $L\{-6\}$ if L denotes Laplace Transform?

► $L\{-6\} = \frac{1}{s+6}$

► $L\{-6\} = \frac{-6}{s}$

► $L\{-6\} = \frac{s}{s^2+36}$

► $L\{-6\} = \frac{-6}{s^2+36}$

Question No: 33 (Marks: 1) - Please choose one

Evaluate the line integral $\int_C (3x+2y)\,dx+(2x-y)\,dy$ where C is the line segment from (0, 0) to (2, 0).

► 6

► -6

► 0

► Do not exist

Question No: 34 (Marks: 1) - Please choose one

Evaluate the line integral $\int_C (2x+y)\, dx + (x^2 - y)\, dy$ where C is the line segment from (0, 0) to (0, 2).

► -4

► -2

► 0

► 2

Question No: 35 (Marks: 1) - Please choose one

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Plane is an example of --------------------

► Curve

► **Surface**

► Sphere

► Cone

Question No: 36 (Marks: 1) - Please choose one

$If\ R = \{(x,y) / 0 \le x \le 2\, and -1 \le y \le 1\}, then$

$\iint_R (x+2y^2)dA =$

► $\int_{-1}^{1}\int_{0}^{2} (x+2y^2)dydx$

► $\int_{0}^{2}\int_{1}^{-1} (x+2y^2)dxdy$

► $\int_{-1}^{1}\int_{0}^{2} (x+2y^2)dxdy$

► $\int_{1}^{2}\int_{-1}^{0} (x+2y^2)dxdy$

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Question No: 37 (Marks: 1) - Please choose one

To evaluate the line integral, the integrand is expressed in terms of x, y, z with

► $dr = dx\hat{i} + dy\,\hat{j}$

► $dr = dx\hat{i} + dy\,\hat{j} + dy\,\hat{k}$

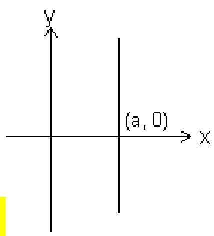
► $dr = dx + dy + dz$

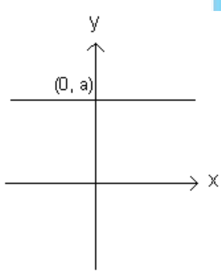
► $dr = dx + dy$

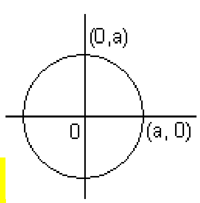
Question No: 38 (Marks: 1) - Please choose one

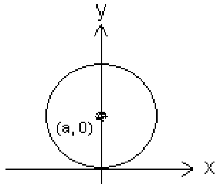
Match the following equation in polar co-ordinates with its graph.
$r = a$

where a is an arbitrary constant.

▶ 

▶ 

▶ 

▶ 

Question No: 39 (Marks: 1) - Please choose one

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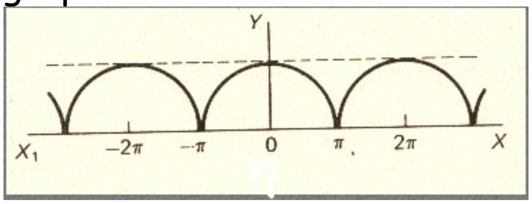
Which of the following is true for a periodic function whose graph is as below?



► **Even function**

► Odd function

► Neither even nor odd function

Question No: 40 (Marks: 1) - Please choose one

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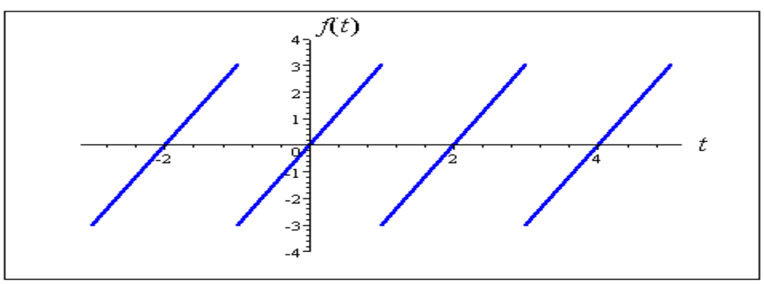

The graph of "saw tooth wave" given above is -------------

► An odd function

► An even function

► Neither even nor odd not sure

Question No: 1 (Marks: 2) - Please choose one
Laplace transform of 't ' is
► $\frac{1}{s}$
► $\frac{1}{s^2}$
► e^{-s}
► s

Question No: 2 (Marks: 2) - Please choose one
Symmetric equation for the line through (1,3,5) and (2,-2,3) is
► $x-2=-\frac{y+2}{3}=-\frac{z-3}{5}$
► $x+2=-\frac{y+3}{5}=-\frac{z+5}{2}$

► $x-1=-\dfrac{y-3}{5}=-\dfrac{z-5}{2}$

► $x+1=\dfrac{y+3}{5}=\dfrac{z-5}{5}$

Question No: 3 (Marks: 1) - Please choose one

The level curves of f(x, y) = y Cscx are parabolas.

► True.

► False.

Question No: 4 (Marks: 1) - Please choose one

The equation $z=r$ is written in

► Rectangular coordinates

► Cylindrical coordinates

► Spherical coordinates

► None of the above

FINALTERM EXAMINATION
Spring 2010
MTH301- Calculus II (Session - 4)

Ref No: Time: 90 min
Marks: 60

Student Info	
StudentID:	
Center:	OPKST
ExamDate:	07 Aug 2010

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Q No.	1	2	3	4	5	6	7	8	Total
Marks									
Q No.	9	10	11	12	13	14	15	16	
Marks									
Q No.	17	18	19	20	21	22	23	24	
Marks									
Q No.	25	26	27	28	29	30	31	32	
Marks									
Q No.	33	34	35	36	37	38	39		
Marks									

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Question No: 1 (Marks: 1) - Please choose one

There is one-to-one correspondence between the set of points on co-ordinate line and ------------

- ► **Set of real numbers**
- ► Set of integers
- ► Set of natural numbers
- ► Set of rational numbers

Question No: 2 (Marks: 1) - Please choose one

Straight line is a special kind of ----------------

- ► Surface
- ► **Curve**
- ► Plane
- ► Parabola

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Question No: 3 (Marks: 1) - Please choose one

$$\lim_{(x,\, y)\to(0,\, 0)} \frac{xy^2}{x^2+y^2} =$$

- ► ∞
- ► 0
- ► 1
- ► 0.5

Question No: 4 (Marks: 1) - Please choose one

If $f(x, y) = x^2 y - y^3 + \ln x$

then $\frac{\partial^2 f}{\partial x^2}$ =

▶ $2xy+\frac{1}{x^2}$

▶ $2y+\frac{1}{x^2}$

▶ $2xy-\frac{1}{x^2}$

▶ $2y-\frac{1}{x^2}$

Question No: 5 (Marks: 1) - Please choose one

Suppose $f(x, y) = xy - 2y^2$ *where* $x = 3t+1$ *and* $y = 2t$. Which one of the following is true?

▶ $\frac{df}{dt} = -4t+2$

▶ $\frac{df}{dt} = -16t - t$

▶ $\frac{df}{dt} = 18t+2$

▶ $\frac{df}{dt} = -10t^2+8t+1$

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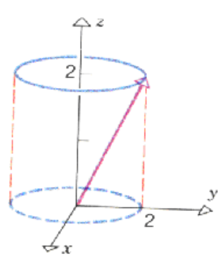
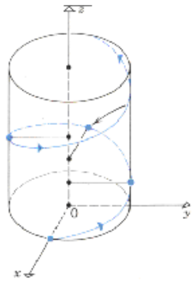
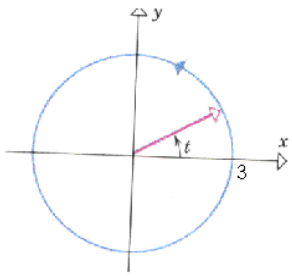
Question No: 6 (Marks: 1) - Please choose one

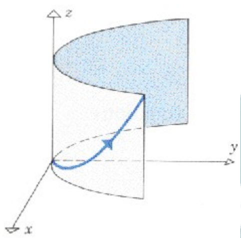
Match the following vector-valued function with its graph.

$r(t) = 3\cos t\,\hat{i} + 3\sin t\,\hat{j}$ and $0 \le t \le 2\pi$

▶ 

□□□□□□□▶□ 

□□□□□□□▶□ 

□□□□□□□▶□ 

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Question No: 7 (Marks: 1) - Please choose one

What are the parametric equations that correspond to the following vector equation?

$$\vec{r}(t) = \sin^2 t\,\hat{i} + (1 - \cos 2t)\,\hat{j}$$

- ▶ $x = \sin^2 t$, $y = 1 - \cos 2t$, $z = 0$
- ▶ $y = \sin^2 t$, $x = 1 - \cos 2t$, $z = 0$
- ▶ $x = \sin^2 t$, $y = 1 - \cos 2t$, $z = 1$
- ▶ $x = \sin^2 t$, $y = \cos 2t$, $z = 1$

Question No: 8 (Marks: 1) - Please choose one

Is the following vector-valued function $\vec{r}(t)$ continuous at $t=\frac{\pi}{2}$? If not, why?

$\vec{r}(t)=(\tan t, \sin t^2 \ , \cos t)$

□□□□□□□▶□ $\vec{r}(t)$ is continuous at $t=\frac{\pi}{2}$

□□□□□□□▶□ $\vec{r}\left(\frac{\pi}{2}\right)$ is not defined

□□□□□□□▶□ $\vec{r}\left(\frac{\pi}{2}\right)$ is defined but $\lim\limits_{t\to\frac{\pi}{2}} \vec{r}(t)$ does not exist

□□□□□□□▶□ $\vec{r}\left(\frac{\pi}{2}\right)$ is defined and $\lim\limits_{t\to\frac{\pi}{2}} \vec{r}(t)$ exists but these two numbers are not equal. Not sure

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Question No: 9 (Marks: 1) - Please choose one

What is the derivative of following vector-valued function?

$\vec{r}(t)=\left(t^4, \sqrt{t+1} \ , \frac{3}{t^2}\right)$

□□□□□□□▶□ $\vec{r}'(t)=\left(4t^3, \frac{1}{\sqrt{t+1}} \ , \frac{-6}{t^3}\right)$

□□□□□□□▶□ $\vec{r}'(t)=\left(4t^3, \frac{1}{2\sqrt{t+1}} \ , \frac{6}{t^3}\right)$

□□□□□□□▶□ $\vec{r}'(t)=\left(4t^4, \frac{1}{2\sqrt{t+1}} \ , \frac{-6}{t^3}\right)$

□□□□□□□▶□ $\vec{r}'(t)=\left(4t^3, \frac{1}{2\sqrt{t+1}} \ , \frac{-6}{t^3}\right)$

Question No: 10 (Marks: 1) - Please choose one

The following differential is exact

$dz=(6xy+2y^2-5)\ dx+(3x^2+4xy-6)\ dy$

□□□□□□□□▶ True
□□□□□□□□▶ False

Question No: 11 (Marks: 1) - Please choose one

The following differential is exact

$dz = (x^2 y + y)\ dx - x\ dy$

□□□□□□□□▶ True
□□□□□□□□▶ **False**

Question No: 12 (Marks: 1) - Please choose one

Which one of the following is correct Wallis Sine formula when n is even and $n \geq 2$?

□□□□□□□□▶ □ $\int_0^{\frac{\pi}{2}} \sin^n x\ dx = \frac{\pi}{2} \frac{(n-1)}{n} \frac{(n-3)}{(n-2)} \frac{(n-5)}{(n-4)} \text{--------} \frac{5}{6} \frac{3}{4} \frac{1}{2}$

□□□□□□□□▶ □ $\int_0^{\frac{\pi}{2}} \sin^n x\ dx = \frac{(n-1)}{n} \frac{(n-3)}{(n-2)} \frac{(n-5)}{(n-4)} \text{--------} \frac{6}{7} \frac{4}{5} \frac{2}{3}$

□□□□□□□□▶ □ $\int_0^{\frac{\pi}{2}} \sin^n x\ dx = \frac{\pi}{2} \frac{(n)}{(n-1)} \frac{(n-2)}{(n-3)} \frac{(n-4)}{(n-5)} \text{--------} \frac{6}{5} \frac{4}{3} \frac{2}{1}$

□□□□□□□□▶ □ $\int_0^{\frac{\pi}{2}} \sin^n x\ dx = \frac{(n)}{(n-1)} \frac{(n-2)}{(n-3)} \frac{(n-4)}{(n-5)} \text{--------} \frac{6}{5} \frac{4}{3} \frac{2}{1}$

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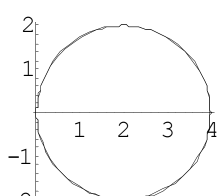
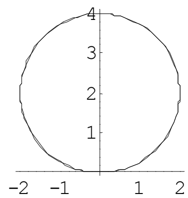
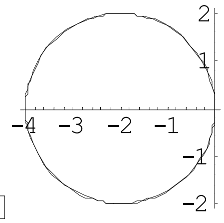
Question No: 13 (Marks: 1) - Please choose one

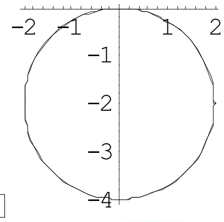
Match the following equation in polar co-ordinates with its graph.

$r = 4\sin\theta$

□□□□□□□□▶ □ 

□□□□□□□▶□ 

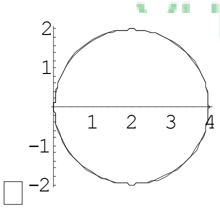
□□□□□□□▶□ 

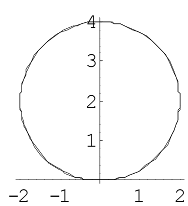
□□□□□□□▶□ 

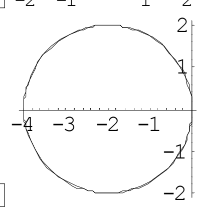
Question No: 14 (Marks: 1) - Please choose one

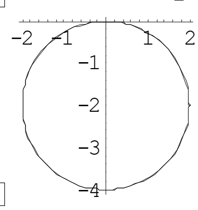
Match the following equation in polar co-ordinates with its graph.

$r = -4\sin\theta$

□□□□□□□▶□ 

□□□□□□□▶□ 

□□□□□□□▶□ 

□□□□□□□▶□ 

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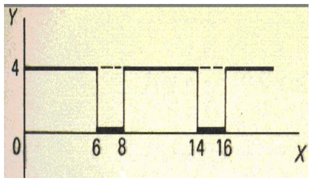
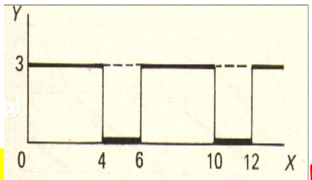
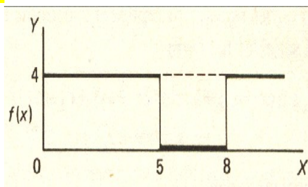
Question No: 15 (Marks: 1) - Please choose one

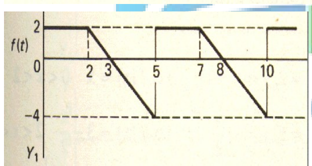
Match the following periodic function with its graph.

$$f(x)=\begin{cases}3 & 0<x<4\\ 0 & 4<x<6\end{cases}$$

□□□□□□□□▶□ 

□□□□□□□□▶□  **page 198**

□□□□□□□□▶□ 

□□□□□□□□▶□ 

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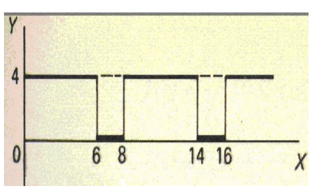
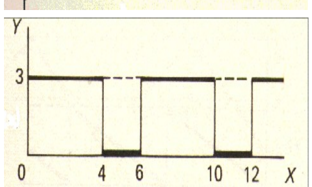
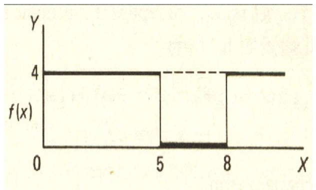
Question No: 16 (Marks: 1) - Please choose one

Match the following periodic function with its graph.

$$f(x)=\begin{cases}4 & 0<x<5\\ 0 & 5<x<8\end{cases}$$

□□□□□□□□▶□ 

□□□□□□□□▶□ 

► 

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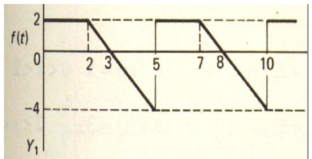
► 

Question No: 17 (Marks: 1) - Please choose one

Which of the following condition must be satisfied for a vector field $\vec{F}$ to be a conservative vector field?

► Line integral of $\vec{F}$ along a curve, depends only on the endpoints of that curve, not on the particular route taken.

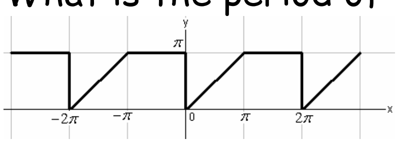
► Divergence of $\vec{F}$ should be zero

► Gradient of $\vec{F}$ should be zero.

► $\vec{F} = 0$

Question No: 18 (Marks: 1) - Please choose one

What is the period of periodic function whose graph is as below?



► π

► $-\pi$

► 2π

► -2π

Question No: 19 (Marks: 1) - Please choose one

Let L denotes the Laplace Transform.

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According to First Shift Theorem, if $L\{F(t)\} = f(s)$ then which of the following equation holds?
s *and* a are constants.

□□□□□□□□▶□ $L\{e^{-at}F(t)\} = f(s-a)$

□□□□□□□□▶□ $L\{e^{-at}F(t)\} = f(s+a)$

□□□□□□□□▶□ $L\{e^{-at}F(t)\} = f(s)$

□□□□□□□□▶□ $L\{e^{-at}F(t)\} = f(a)$

Question No: 20 (Marks: 1) - Please choose one

Polar co-ordinates of a point are $\left(7, \frac{-\pi}{4}\right)$. Which of the following is another possible polar co-ordinates representation of this point?

□□□□□□□□▶□ $\left(7, \frac{3\pi}{4}\right)$

□□□□□□□□▶□ $\left(-7, \frac{3\pi}{4}\right)$

□□□□□□□□▶□ $\left(-7, \frac{-\pi}{4}\right)$

□□□□□□□□▶□ $\left(7, \frac{-3\pi}{4}\right)$

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Question No: 21 (Marks: 1) - Please choose one

The graph of an even function is symmetrical about ---------------

□□□□□□□□▶□x-axis

□□□□□□□□▶□**y-axis**

□□□□□□□□▶□origin

Question No: 22 (Marks: 1) - Please choose one

Is the function $f(x, y)$ continuous at origin? If not, why?

$$f(x, y) = \frac{xy}{x^2 + y^2}$$

□□□□□□□□▶□ $f(x, y)$ is continuous at origin

□□□□□□□□▶□ $\lim_{(x,y)\to(0,0)} f(x, y)$ does not exist

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▶ $f(0, 0)$ is defined and $\lim_{(x, y) \to (0, 0)} f(x, y)$ exists but these two numbers are not equal.

Question No: 23 (Marks: 1) - Please choose one

Consider the function $f(x, y, z) = \sqrt{1 - x^2 - y^2 - z^2}$. What is the value of $f\left(0, \frac{1}{2}, \frac{1}{2}\right)$

▶ $f\left(0, \frac{1}{2}, \frac{1}{2}\right) = \sqrt{\frac{1}{2}}$

▶ $f\left(0, \frac{1}{2}, \frac{1}{2}\right) = 2$

▶ $f\left(0, \frac{1}{2}, \frac{1}{2}\right) = \frac{1}{2}$

▶ $f\left(0, \frac{1}{2}, \frac{1}{2}\right) = 0$

Question No: 24 (Marks: 1) - Please choose one

Sign of line integral is reversed when ------------

▶ path of integration is divided into parts.

▶ path of integration is parallel to y-axis.

▶ **direction of path of integration is reversed.**

▶ path of integration is parallel to x-axis.

Question No: 25 (Marks: 1) - Please choose one

Let the functions $P(x, y)$ and $Q(x, y)$ are finite and continuous inside and at the boundary of a closed curve C in the xy-plane.

If $(P\,dx + Q\,dy)$ is an exact differential then

$$\oint_C (P\,dx + Q\,dy) =$$

▶ **Zero**

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► One
► Infinite

Question No: 26 (Marks: 1) - Please choose one

What is laplace transform of the function $F(t)$ if $F(t) = \cos 2t$?

► $L\{\cos 2t\} = \dfrac{2}{s^2+4}$

► $L\{\cos 2t\} = \dfrac{1}{s-2}$

► $L\{\cos 2t\} = \dfrac{s}{s^2+4}$

► $L\{\cos 2t\} = \dfrac{2!}{s^3}$

Question No: 27 (Marks: 1) - Please choose one

What is $L\{-6\}$ if L denotes Laplace Transform?

► $L\{-6\} = \dfrac{1}{s+6}$

► $L\{-6\} = \dfrac{-6}{s}$

► $L\{-6\} = \dfrac{s}{s^2+36}$

► $L\{-6\} = \dfrac{-6}{s^2+36}$

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Question No: 28 (Marks: 1) - Please choose one

Curl of vector function is always a --------------

► Scalar
► Vector

Question No: 29 (Marks: 1) - Please choose one

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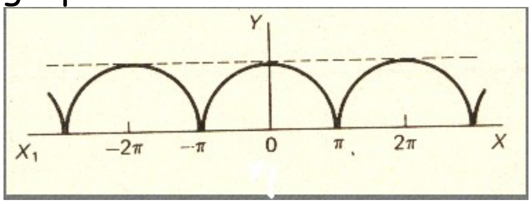
Which of the following is true for a periodic function whose graph is as below?

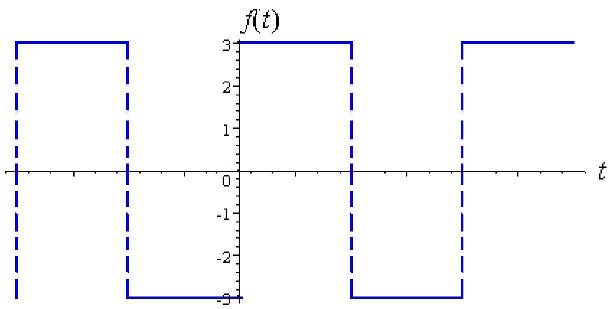


► **Even function**

► Odd function

► Neither even nor odd function

Question No: 30 (Marks: 1) - Please choose one



Which of the following is true for a function whose graph is given above

► ***An odd function***

► An even function

► Neither even nor odd

Question No: 31 (Marks: 2)

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Evaluate the line integral $\int_C 2x\ dx$ where C is the line segment from $(0, 2)$ to $(2, 6)$

Question No: 32 (Marks: 2)

Use Wallis sine formula to evaluate $\int_0^{\frac{\pi}{2}} \sin^5 x\ dx$

Question No: 33 (Marks: 2)

Find Laplace Transform of the function $F(t)$ if $F(t) = \sin 2t$.

Question No: 34 (Marks: 3)

Find $div\ \vec{F}$, if $\vec{F} = (3x+y)\hat{i} + xy^2 z\hat{j} + (xz^2)\hat{k}$

Question No: 35 (Marks: 3)

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Determine whether the following differential is exact or not.

$dz = (4x^3 y + 2xy^3)\ dx + (x^4 + 3x^2 y^2)\ dy$

Question No: 36 (Marks: 3)

Prove whether the following function is even, odd or neither.
$f(x) = x^3 e^x$

Question No: 37 (Marks: 5)

Evaluate the following line integral which is independent of path.

$$\int_{(2,-2)}^{(-1,0)} (2xy^3)\ dx + (3y^2 x^2)\ dy$$

Question No: 38 (Marks: 5)

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Determine the fourier co-efficient b_n of the following function.

$f(x) = x^2 \qquad 0 < x < 2\pi$

Question No: 39 (Marks: 5)

Determine whether the following vector field $\vec{F}$ is conservative or not.

$\vec{F}(x, y, z) = (3x + y)\hat{i} + xy^2 z\,\hat{j} + xz^2\,\hat{k}$

ASSALAM O ALAIKUM

All Dearz fellows

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Final term PAPERS &

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Created BY Farhan & Ali

BS (cs) 2nd sem

Hackers Group

From Mandi Bahauddin

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Mindhacker124@gmail.com

Hearthacker124@gmail.com

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